# Swarm Search and Rescue with Differing Scales of Autonomy

**Abstract**

[To write. Draft facts for it: 33 operators, two 8-minute sessions each (Strategic Heavy and Tactical Heavy scenarios, counterbalanced), a heterogeneous 33-drone fleet, two optional assistants at two decision tiers. Headline results: reliance is a stable personal style; hand-over rises with mission size on both tiers; momentary load does not push operators onto the assistants; the Tactical Assistant is trusted more; a visible flaw on the strategic tier and an invisible one on the tactical tier.]

## 1 Introduction

The deployment of large numbers of robots has advanced from speculation from practice in a short time frame. Present applications of drones and similar uncrewed vehicles are increasingly requiring either large numbers of well-organised operators [23], or else provide early demonstrations of semi-autonomous command-and-control software [3]. There is significant appetite for the incorporation of autonomy for reasons including but not limited to the impracticalities of supporting one pilot for each asset, and a desire for a single operator to maintain a complete picture of the entire operation [29]. As swarms scale further, the one-to-one or many-to-one models of human control must give way to a one-to-many paradigm, and some degree of autonomy becomes not merely desirable but unavoidable.

How that autonomy is best integrated, however, remains an open question. Autonomy is far from new; defence operations have long been organised around autonomy at different levels of a hierarchy. Auftragstaktik (mission command) requires strategic command to communicate high-level goals to subordinates and grants them latitude over low-level execution, and indeed further delegation. A similar logic can be applied to robot swarms: tactical decisions may be delegated to individual assets or sub-groups, while strategic decisions remain with a human commander. Yet it is not obvious that the best configuration places the tactical under full autonomy while leaving the strategic entirely manual. Lower-level interventions may benefit heavily from human oversight; robots still struggle with physical execution, while autonomous planners are near-perfect at many allocation problems. Of course, higher-level decisions carry greater consequence and arguably warrant closer human control; it is natural to prefer an autonomy-at-tactical approach, however the best possible combination of autonomy levels may not be so simple. Each of these judgements also depends on operator workload and situational context. The need for interfaces and control paradigms which allow autonomy is therefore high [29], but the question of *where* in a command hierarchy that autonomy should sit, and whether this should be fixed or operator-configurable, has received little attention. Previous works have explored many forms of autonomy [18], but these have typically been treated at the level of a single decision layer.

This work represents a first step toward a longer-term vision: We aim to understand how autonomy effects both user and mission when deployed at each level of a multi-layered command hierarchy, and move towards a system where each level of concern from strategic down to tactical details, can be automated to varying degrees. Finally, we propose a "co-pilot" that, instead of attempting to fully automate the mission, identifies where the user is most needed, and suggest to them an autonomy profile which adapts on-the-fly.

We develop a simulated search-and-rescue scenario using a heterogeneous drone swarm, in which a two-tier assistive system (a strategic tier and a tactical tier) is available to the operator throughout.

[Step 4 note (kept from the old draft): this is the natural place to make the *ORCHID lineage* explicit — name `jennings2014human` (human-agent collectives / flexible autonomy) and `ramchurn2021trustworthy` alongside the already-cited `ramchurn2016disaster`. See cluster C1 in the Related Work notes box. Not drafted.] Neither assistant has errors injected: the manipulation is not the assistants' reliability but *where the operational demand sits*. In a within-subjects design each participant plays two 8-minute sessions — **Strategic Heavy** (frequent, small

missions, placing demand on cross-mission allocation) and **Tactical Heavy** (fewer, larger missions, placing demand on within-mission planning) — with the fleet and the assistants identical in both. Using either assistant is always optional, so the study observes when operators choose to hand a decision over, at which tier, and how that relates to the mission, the moment, the person and their trust.

[RQ block — REVISED to match what the analysis found (the old block assumed a perfect-agent "value of intervention" RQ4 that the data cannot answer). Candidate framings, each mapped to a results section; merge or cut as you like:

- **RQ1 (who).** How much of an operator's use of the assistants is a stable personal style, and is it predicted by attitudes measured beforehand? → Section 7.2.
- **RQ2 (when).** What makes an operator hand a decision over — the mission (its size), the moment (load, the queue), or practice — and does the answer differ between the strategic and tactical tiers? → Sections 7.3–7.7.
- **RQ3 (flaws).** How do operators respond to an assistant's flaw that is visible (the illogical strategy card) versus one that is not (the tactical assistant's wrong refusal)? → Section 7.8.
- **RQ4 (trust by tier).** Is one tier's assistant trusted more, where does the difference come from, and does stated trust match use? → Section 7.9.

Note: "failure" in this study means hardware drone failure during execution, plus the two side effects of the assistants' rules (Section 4.3) — not injected agent error. Operator-built deadlocks were auto-repaired and never surfaced.]

## 2 Related Work

Essential to any autonomous system is the notion of trust. Trust in this context is defined by Lee and See as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" [11]. Such qualities are abundant in swarm contexts, where the positions, motions, and goals of the robots are rife with uncertainty, and there is a significant risk of failure stemming from the attritable nature of many such assets and the high levels of risk in their environment [27]. It is an important consideration, as a system which is not trusted will likely not be used by the operator, and a system which is trusted too much will be relied on without appropriate checking and oversight. A system which balances under- and over-trust is said to be well calibrated [13].

It has been observed that once a user falls into the trap of under- or over-trust, they have a tendency to remain there unless actively coaxed back into a well-calibrated relationship [20]. The standard tool for this is explanations, however these are not always given due attention [17, 31], and often-times users are biased in favour of action over inaction so accept suggestions readily [15]. In the domain of HSI the notion of neglect benevolence describes the observed advantage from leaving the swarm to resolve its own behaviour instead of intervening overly often [16], additionally supporting the notion of autonomy which manages itself under human supervision instead of constant interaction with the recommender. Equally important to explanations, then, may be a design pattern where users work in tandem with autonomous tools and can elect to deploy them selectively, with varying levels of autonomy on varying scales of a swarm or sub-swarm.

Traditionally swarm robotic systems are characterised by autonomy, however with the importance of human oversight a question arises as to what level of autonomy should be given, and how the human's actions interface with some level of autonomy. Traditional scales such as Sheridan and Verplank's LOAs are a common metric [26], as well as more swarm-focussed scales of a similar form [7]. An important note in more developed robotic systems is that the degree of autonomy can vary at different parts of the robot's role (for example information acquisition may be more autonomous than decision-making) [21].

An important note is that the terms "automation" and "autonomy" are often used interchangeably but are in fact different things, with the former describing well-defined processes of tasks deferred to an *automated system*, and the latter describing *autonomous systems* which can make "decisions independent of human control" [12]. As we are referring to swarm robots in this work, the terms are somewhat more interchangeable, as to automate a swarm on a particular task requires the granting of some degree of autonomy (for example performing decentralised allocation, where an automated algorithm runs across robots who then autonomously resolve their tasks [4]). We tend to prefer

the term autonomy as it is more relevant to classical swarm paradigms, but remind the reader that there can be some distinction dependent on use case.

Similar to calibrating trust, it is important that any assistive autonomous system is sufficiently performant to merit use. Human performance in the presence of such an agent is highly sensitive to the reliability and performance of that agent [25]. Some degree of imperfection is an expectation (otherwise there would be no need for a human), and humans are typically willing to tolerate occasional failure if it represents an adequate time- and effort-save. In fact, the most trusted systems are often those which sit in between 'too poor to trust' and 'too reliable to bother checking' [10].

Another important factor is the level of workload. Prior works have found that under high task load, more reliable automation increases operator performance, however under low task load, more reliable automation actually *decreases* operator performance. This is likely due to the over-trust issue discussed above, and presents a difficult challenge to designers, as users may defer to the agent even when not overloaded [22].

The level of chosen autonomy will necessarily represent a trade-off between different desirable qualities as alluded to above, and no single solution is likely to best suit every scenario [14]. This motivates a flexible approach where the level of autonomy is not fixed, rather it can be chosen by the operator. Such flexible approaches have been explored to the extent where agents can be granted autonomy where desirable [5], or else autonomous agents can be overridden at any level while they attempt to perform autonomously [30]. So-called mixed-initiative approaches to swarm control allow a human to switch the entire swarm between manual and autonomous modes [18]. To our knowledge, however, no prior work has examined the behaviour of an operator when autonomous assistance is structured into a two-tier hierarchy and the operator is free to use it selectively. Such a design allows us to ask not only whether the assistive system improves performance, but *when* operators choose to engage it and what advantage that engagement confers.

We term the two structural dimensions of selective autonomy the **Depth** and **Breadth** of autonomy. **Depth** refers to how autonomous some agent is, pertaining directly to Sheridan and Verplank's LOAs [26]. **Breadth** refers to the scale of autonomy, whether that be an individual asset, a mission's worth of assets, or the entire reserve. The present work extends this framework by stacking two agents vertically in a hierarchy—one operating at the within-mission (tactical) level and one at the cross-mission (strategic) level—and asking how the operator engages with, and elects to intervene in, autonomy at each level.

Metrics exist to assess the difficulty of tasks in the context of swarms, such as efficiency and effectiveness, and it has been suggested that autonomy should vary depending on these [8]. As well as the complexity of the task, a more direct assessment of cognitive complexity via operator state metrics such as EEG has been employed [1], and such signals could in principle inform a meta-level recommendation system.

[Drafted gap paragraph — my pass, for your check; rewrite freely. Intended as the closing paragraph of Related Work. Requires the supplement bib for the four keys.]

Most recently, the one-operator, many-robot paradigm has moved from concept to field demonstration. Adams et al. showed that a single trained commander can supervise a swarm of one hundred heterogeneous ground and aerial robots across real urban field exercises, albeit with frequent workload peaks [2], while the MOSAIC framework demonstrated a single operator coordinating a heterogeneous five-robot team in a lunar-analogue mission by shifting individual platforms between driver-, task-, and mission-level autonomy [19]. In parallel, variable autonomy has consolidated into a recognised subfield in its own right, with dedicated venues [28] and emerging design guidelines and research agendas [24]. Across this line of work, however, autonomy is treated as a single dial — a *level* to which a robot or team is set — and the evidence base consists of demonstrations and design agendas rather than controlled operator studies. What remains unexamined is the setting in which assistance is available simultaneously at *distinct decision tiers* — cross-mission resource allocation and within-mission planning — and the operator chooses, moment to moment, how far to engage each. The present study addresses exactly this gap: a controlled, within-subjects examination of how the locus of operational demand shifts operator engagement, workload, and trust across a two-tier assistive hierarchy.

---

**Step 4 — Reference integration notes (proposals, not drafted text).** New keys live in `refs_supplement_new.bib` (add it via `\bibliography{refs,refs_supplement_new}`). Each cluster gives *where* to place it and *what* it buys; the wording is yours. Key nuance for the whole trust literature below: our agent is now **perfect** — cite the trust-in-*imperfect*-autonomy work as calibration/per-tier-trust *backdrop and motivation*, not as error-detection, since we inject no agent error.

- **C1 — ORCHID lineage** (Intro near `ramchurn2016disaster`, and/or opening Related Work). Keys: `jennings2014human` (HAC vision; "flexible autonomy"/"agile teaming"), `ramchurn2021trustworthy` (successor framing — its UAV disaster-response pipeline mirrors our two-tier setup and its "errors enter at every pipeline stage" point motivates the per-tier question), `soorati2022intelligent` (group agenda). *Angle:* frame the study as building on the ORCHID human-agent-collectives programme; `ramchurn2016disaster` (already cited) is its flagship.
- **C2 — task-allocation substrate** (short new paragraph in Related Work, by the automation/autonomy discussion). Keys: `ramchurn2010cfstp` (CFSTP: heterogeneous agents, deadline/workload tasks — the formal root of our strategic/tactical allocation), `capezzuto2020anytime` & `capezzuto2021largescale/capezzuto2021ar` (modern scalable extensions), `macarthur2011distributed` (already in bib, uncited), `ramchurn2010robocup`, `wu2015agile`, `baker2016planning` (UAV SAR planning). *Angle:* one or two sentences establishing the agent's allocation task as a well-studied MAS problem; cite the lineage, don't dwell.
- **C3 — single-operator / heterogeneous-swarm HSI** (near the LOAs / one-to-many discussion). Keys: `soorati2021designi` (in-group HSI interface; HARIS precursor), `abioye2024adaptive` (your own group; workload-adaptive HSI → RQ2/RQ3), `adams2025supervise` (single operator, 100 heterogeneous robots; now also in the gap paragraph — best used as an Intro feasibility anchor, an RQ2 workload-at-scale *contrast* (their physiological multi-dimensional estimation vs. our NASA-TLX), and qualitative motivation for the tactical tier: operators were "red-lined" precisely when tactical allocation autonomy was unavailable. Not a design comparator — no tiers, no trust, n=2 field observation), `chandarana2018determining` (multi-job-type swarm sizing). *Angle:* situate the one-operator + heterogeneous-asset setup and the workload stakes.
- **C4 — variable/adjustable autonomy** (attach to the Depth/Breadth paragraph and the mixed-initiative cites). Keys: `methnani2024charge` (survey — anchor), `theodorou2024editorial`, `ramesh2023experimental`, `verhagen2024meaningful` (firefighting; SAR-flavoured), `reinmund2024variable`, `oberacker2026mosaic` (MOSAIC; now also in the gap paragraph — the depth axis operating in the field: one operator slides a heterogeneous five-robot team across driver/task/mission autonomy levels, but allocation is folded into mission-level autonomy with no separately-operable strategic tier; arXiv preprint, verify status at submission); `tecuci_seven_2007` (already in bib, mixed-initiative). *Angle:* the RQ3 grounding ("where should autonomy sit across levels"); links Depth/Breadth to the variable-autonomy subfield.
- **C5 — trust across tiers** (extend the trust-calibration paragraph; motivates RQ3 per-tier trust and RQ4). Keys: `wischnewski2023measuring` (subsystem-failure → whole-system trust spillover — the key precedent for measuring trust *per tier*), `wildman2024trust` (multilevel trust in 3+-member HATs), `rittenberg2024trust` (between-subjects reliability manipulation), `naiseh2023explanation`, `centeiojorge2023exploring`; `parasuraman2010complacency` (already in bib). *Angle:* frame as system-wide vs. component-specific trust; justify measuring trust separately for each tier. NB the closest analogue — a multilevel "agent reliability pairing" study (Am. J. Psychology) — I could not fully verify; see the supplement TODO before citing.
- **C6 — selective use / recommendation acceptance** (the "elect to deploy them selectively" sentence). Key: `fink_let_2024` (already in bib) — giving users control over whether to accept a recommendation raises acceptance; direct support for the selective-deployment design and RQ2.
- **Optional (thesis-level, not this paper):** the AtomicOrchid platform papers and Fischer's "in-the-loop vs on-the-loop" work sit better with your HARIS/platform chapter (RQ1) than here; omitted from the supplement to keep scope tight.

---

# 3 Scenario and Task Design

## 3.1 Operational Scenario

We built a simulated search-and-rescue operation in which a single operator directs a heterogeneous fleet of 33 drones from one hub. Missions keep arriving at positions on a map; each consists of two to six tasks, and each task needs a

fixed team of drone types. For every mission the operator makes two decisions in sequence:

- a **strategic** decision — *how many* drones of each type to commit to the mission, taken from the shared reserve that every other mission also draws on; and
- a **tactical** decision — *which* committed drone does *which* task, and in what order.

An assistant is available for each decision (Section 4); using either is always optional. Drones can fail mid-task, in which case the operator repairs the plan or abandons the mission. The score rewards completed tasks and penalises the time missions spend unfinished, so both getting it right and getting it done quickly matter.

## 3.2 Asset Types

The fleet comprises 11 drones of each of three types (Table 4). The types are named to the operator by role rather than colour — **Fast** (blue; 11 u/s), **Lifter** (red; 10 u/s) and **Camera** (green; 9 u/s) — and drones are labelled accordingly (e.g. `Fast-7`). Speeds are not shown to participants; the tutorial describes the types only as fastest, standard and slowest. Drones committed to a mission leave the reserve at once and return to it when their task chain is complete. The mission generator's archetype weights (below) were tuned in simulation so that aggregate demand, in drone-seconds, is approximately equal across the three types in both scenarios, so no single type is the systematic bottleneck.

## 3.3 Task Types

There are five task types of increasing complexity and value (Table 1). Each has a primary composition; three have a substitute composition with one fewer drone of the paired type, which takes about 2.5 times as long. The planner shows the substitute as an "OR" line on the task badge.

**Table 1:** Task types. Execution times are for the primary composition / the substitute.

| Type | Name | Primary | Substitute | Time (s) | Points |
|---|---|---|---|---|---|
| T1 | Recce | 1 Fast | — | 10 / — | 20 |
| T2 | Recon | 2 Fast | 1 Fast | 15 / 38 | 40 |
| T3 | Supply Drop | 2 Lifter + 1 Camera | 1 Lifter + 1 Camera | 25 / 62 | 60 |
| T4 | Precision Supply Drop | 1 Lifter + 2 Camera | 1 Lifter + 1 Camera | 30 / 75 | 80 |
| T5 | Search & Service | 1 Fast + 1 Lifter + 1 Camera | — | 45 / — | 100 |

## 3.4 Missions

Missions arrive by a Poisson process. Each has a **category** that sets its number of tasks and how quickly its penalty grows — A Routine (2 tasks), B Moderate (3), C Significant (4), D Critical (5), E Mass Casualty (6) — and an **archetype** drawn at generation (Fast-, Lifter-, Camera-leaning or Mixed, weights 38/26/21/15) that sets which task types make up that count (Table 2). A mission therefore needs one drone type (e.g. an all-Recce Fast mission), or all three; only 12 of 606 strategic decisions concerned a two-type mission. Mission zones are circles of radius 80 units at least 150 units from the hub and 200 units from any other active zone. Where the findings speak of **mission size** they mean the number of drones the mission needs: the sum of its tasks' primary requirements, which the task badges display and which is the same for every operator who met that mission.

## 3.5 Score

The score $S$ is completion reward minus an accrued time penalty, floored at zero:

$$S = \max\left(0,\ \sum_{m\in\mathcal{M}}\sum_{k\in\mathcal{T}_m} w_k\,\mathbf{1}[k \text{ completed}]\ -\ \sum_{m\in\mathcal{M}}\sum_{k\in\mathcal{T}_m} r_m\,\frac{w_k}{\sum_{k'\in\mathcal{T}_m} w_{k'}}\left(t_k^{\text{end}} - t_m^{a}\right)\right) \tag{1}$$

where $w_k$ is the points value of task $k$ (Table 1), $r_m$ the penalty rate of the mission's category (0.05–0.40 points/s; Table 4), $t_m^a$ the mission's arrival time and $t_k^{\text{end}}$ the task's completion time, or the current time while it is unfinished. The penalty is thus charged per task, weighted by the task's share of its mission's points, from the mission's arrival

**Table 2:** Task list for each mission (category × archetype), as generated. Task counts grow with category; the archetype fixes the mix.

| Cat. | # | Fast-lean | Lifter-lean | Camera-lean | Mixed |
|---|---|---|---|---|---|
| A | 2 | T1, T2 | T3, T5 | T4, T5 | T1, T3 |
| B | 3 | T1, T2, T2 | T3, T3, T5 | T1, T4, T4 | T1, T3, T5 |
| C | 4 | T1, T1, T2, T2 | T3, T3, T5, T5 | T1, T4, T4, T4 | T1, T2, T3, T5 |
| D | 5 | T1, T1, T2, T2, T5 | T3, T3, T3, T5, T5 | T1, T4, T4, T4, T5 | T1, T2, T3, T3, T5 |
| E | 6 | T1, T1, T2, T2, T2, T5 | T3, T3, T3, T5, T5, T5 | T4, T4, T4, T1, T5, T5 | T1, T2, T3, T3, T4, T5 |

until that task completes; it is displayed in 15 s steps. If a mission is abandoned, its parent's penalty clock stops and its unfinished tasks are re-queued as a new *residual* mission (id suffix `-R`) that accrues from that moment, so abandoning transfers work rather than losing it. Because the four mission streams (Section 6.1) differ in how many points they offer, raw scores are compared only within a game.

### 3.6 Drone Failures

Any deployed drone can fail. In the main build (`study-v1.4` onwards; Appendix B) failures are a live hazard of $1/900$ per drone per second, rolled on a fixed one-second simulated grid, identical for every drone type and mission. A drone that fails after its part of a task is done (e.g. the Lifters of a Supply Drop after the drop) exits gracefully. Otherwise the task reverts to pending and the mission enters **recovery**: its card turns red and the tactical planner opens in recovery mode (Figure 5). The operator may reassign the drones still on that mission (never the hub reserve, which is what makes abandoning a real option), press Suggest for a repair, or abandon the mission. A mission with an open recovery takes no further failures until 30 s after it is resolved (`v1.5` onwards). On average a session had about four drone failures (Table 3).

## 4 The Two Assistants

Both assistants are deliberately simple, transparent rules rather than learned models, and in every session analysed neither had errors injected (Section 4.3). In the interface they are called the *Strategic Assistant* and the *Tactical Assistant*; the words "AI" and "agent" are not used on screen.

### 4.1 Strategic Assistant: Strategy Cards

When the operator clicks *Allocate* on a queued mission, a panel opens beneath its card and two strategy cards load, each after an independent delay of about 4–5 s (Figure 3b). Each card shows the drones it would commit per type, the projected mission completion time (ETA), the reserve that would remain, and three bars: Speed (the best ETA of the two divided by this card's ETA), Reserve (share of the current reserve left behind) and Resilience (spare drones above the minimum needed).

- **Aggressive** — enough drones of each type for every task to run at once (the parallel sum of the primary compositions), with at least one spare of each type the mission uses; capped by the reserve.
- **Conservative** — the smallest team that completes every task in series (primary compositions, or substitutes where the reserve cannot cover a primary), plus one spare of each type the mission uses, plus 15% of the remaining reserve of *every* type. Tasks run mostly in series.

The 15% top-up is applied to every drone type, including types the mission does not use. On a one-type mission the Conservative card therefore often includes, say, a Camera drone on an all-Fast mission. We call this the **illogical card**: it was on screen for 132 of the 606 strategic decisions, almost all on one-type missions, and every participant met it. It was discovered during data collection and deliberately left unchanged so as not to split the cohort; Section 7.8 studies it.

The operator can deploy a card as offered (*Deploy Selected*), choose *Edit this allocation* (a counter per type, pre-filled

from the selected card), or choose *Set manually instead* (the same counters, empty), which is possible before the cards have loaded (Figure 3c). On deployment the committed drones leave the reserve at once and fly to the mission zone, where they wait at its edge for the tactical plan.

### 4.2 Tactical Assistant: Suggest

Once a mission has its drones, it appears as pending in the tactical planner on the second screen (Figure 4). The planner opens **empty**: no plan is shown until the operator asks for one. The operator can build the plan by hand — drag a drone onto a task; shift-drag chains a drone on to a further task after its first, and the sequence numbers update live — or press **Suggest**. Suggest builds a plan with a virtual-timeline scheduler: tasks are taken most-constrained first (T5 before T1) and each is given the earliest-free drones of the required types from the committed team, chaining drones across tasks where that finishes sooner. The plan appears one drone→task entry per second, and *Deploy* unlocks when it is complete. The operator can edit any part of it before deploying. *Deploy* is enabled only when every task has at least its substitute composition. The planner works in "plan-all" mode: the whole sequence is confirmed once.

Because Suggest routes every drone in one global task order, its plans cannot deadlock. An operator-built plan can (a cycle of chained drones each waiting on the other's task); in every session analysed, such deadlocks were detected and silently re-routed onto one order by the system, so a participant could never encounter one.

In recovery mode Suggest proposes a repair from the drones on the mission. It considers only drones that are not currently executing a task. When the only fix needs a busy drone — chaining it on after its current task, which the operator could do by hand — Suggest answers "No workable plan". We call this a **wrong refusal**. To the operator it is indistinguishable from a correct refusal. It was found after data collection and also left unchanged.

### 4.3 Assistant Reliability

The platform can inject errors into either assistant (a per-session error rate per tier). This was switched off ($\varepsilon_S = \varepsilon_T = 0$) for every session analysed; the one participant who ran with errors switched on (P09) is excluded. Neither assistant was therefore ever deliberately wrong. The assistants' only flaws are the two side effects of their rules described above: the illogical card on the strategic tier, which is visible on the card face, and the wrong refusal on the tactical tier, which is not. [The old draft's RQ4 ("value of intervention into a near-optimal agent") and the per-tier error manipulation do not apply to these data; see the RQ note in the Introduction.]

## 5 Platform and Interface

The platform is a browser application (React/TypeScript; no server) designed for two monitors side by side (Figure 1). All randomness is seeded, so a seed and a configuration reproduce a session exactly, and every operator action and assistant output is logged with a millisecond timestamp.

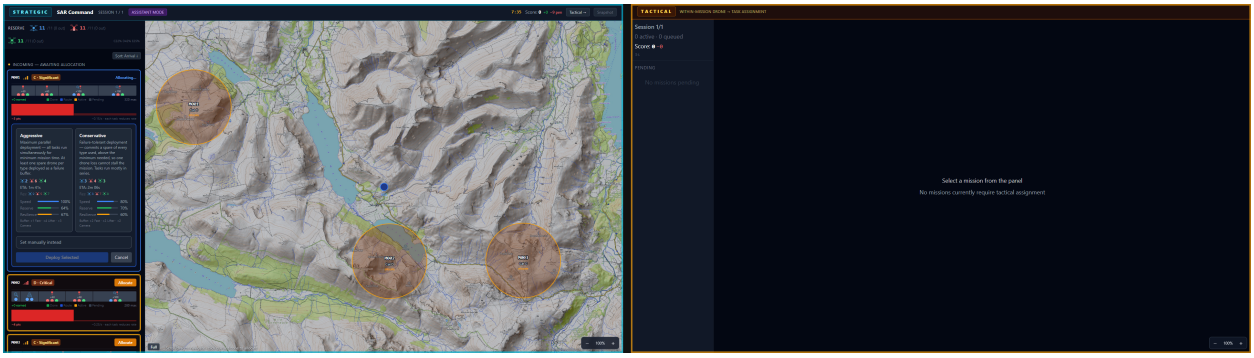



**Figure 1:** The two screens side by side. **Left, strategic screen:** reserve (top left), the queue of missions awaiting allocation (left column; the first is being allocated and shows its two strategy cards), and the strategic map with each mission's zone and the hub. **Right, tactical screen:** the tactical planner, here with no mission pending.

**Strategic screen.** The strategic screen (Figure 2) holds the reserve count per drone type, the mission queue and the strategic map. Each mission card (Figure 3a) shows the mission's id and category, a badge per task with its required drones and points, a progress legend (done, en route, active, pending) and a red penalty bar that grows while the mission is unfinished. The queue can be sorted by arrival or by criticality. About three and a half cards are visible at once; further waiting missions appear only on the map. The map can show zones only or every drone's path.

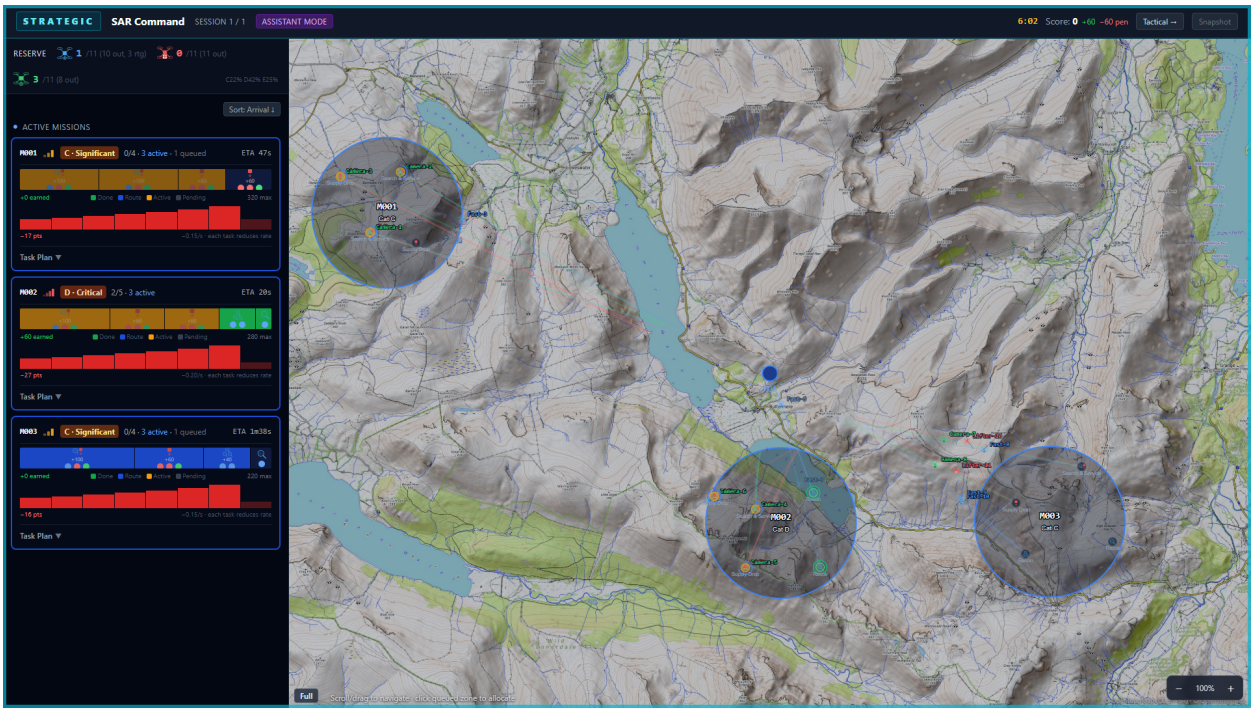



**Figure 2:** Strategic screen mid-session: missions in progress, drones in flight between the hub and the mission zones, and the penalty bars on the queued and active mission cards.

**Tactical screen.** The tactical screen lists missions awaiting a plan; selecting one opens the planner (Figure 4): the mission zone with each task as a waypoint labelled with its type and staffing (e.g. `1/2L 0/1C`, with its substitute below), the committed drones waiting at the zone edge, and a right-hand panel listing unassigned drones and the task schedule with each task's assigned drones and projected start and finish. Hovering a drone shows its full route.

**Failures.** When a drone fails, the mission's card turns red with a banner naming the lost drone, and the mission is listed as *reassign pending* on the tactical screen. The recovery planner (Figure 5) marks executing and travelling tasks, shows the failed drone crossed out, and offers *Suggest*, *Reassign* (enabled once every pending task is covered) and *Abandon Mission*.

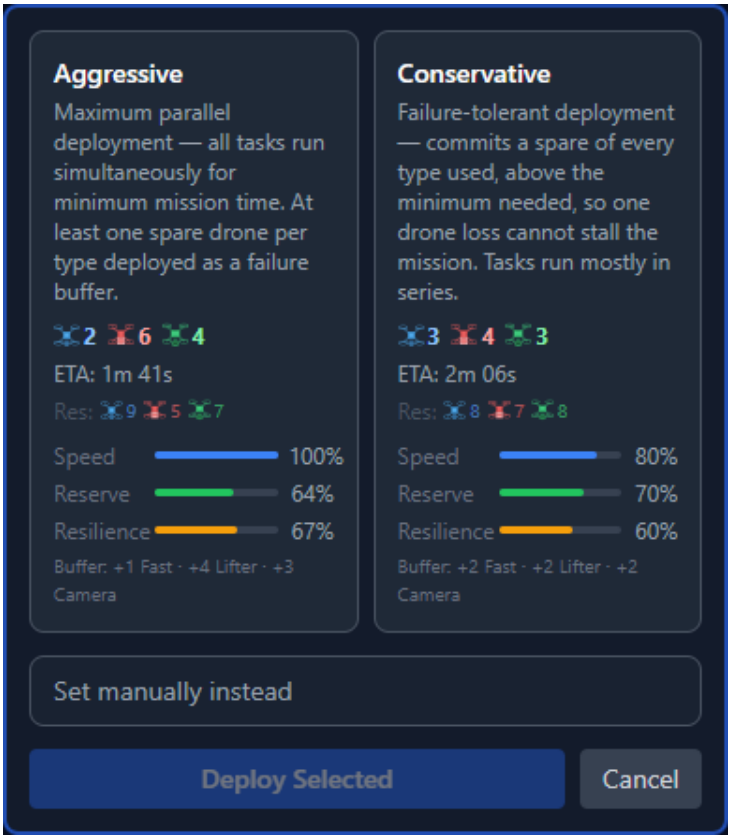
**Tutorial.** Before session 1 every participant completed a guided tutorial of 48 steps on a smaller fleet (6/6/6), with its own clock. It teaches the manual workflow first — allocate, plan, chain, deploy, recover from a failure by hand, and abandon a mission that cannot be recovered (two scripted failures) — and only then introduces the two assistants; the Suggest button is hidden until that lesson. The tutorial runs with error-free assistants.

# 6 Study Design

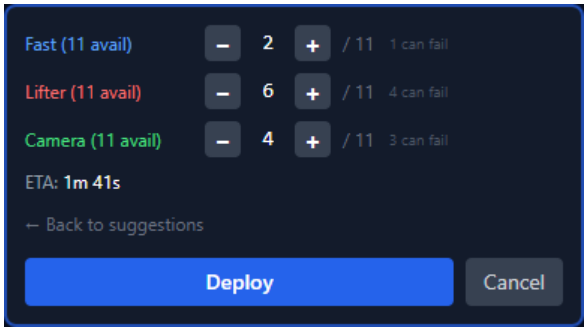
## 6.1 Design

The study used a within-subjects design with one factor, the **scenario**, i.e. where the operational demand sits. Each participant played one 8-minute (480 s) session of each scenario, in counterbalanced order (17 Strategic Heavy first,

**(a)** Mission queue: three missions awaiting allocation. Each badge shows a task's required drones and its points; the red bar is the accruing penalty.

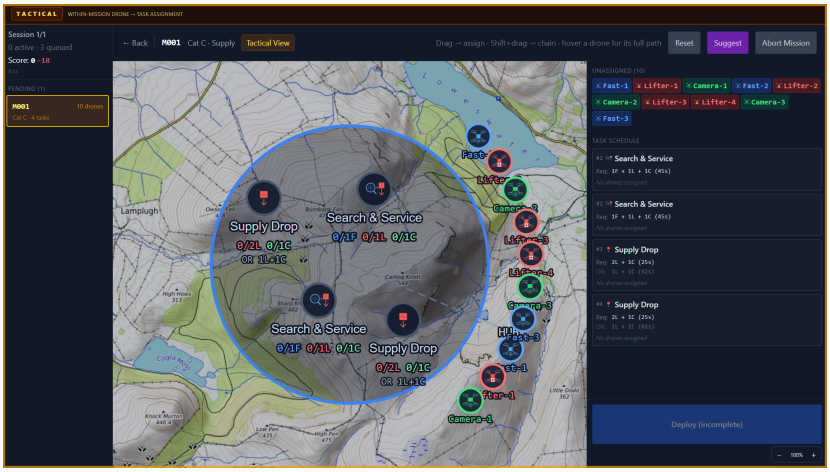



**(b)** The Strategic Assistant's two strategy cards for a Category C mission: drones per type, ETA, reserve left ("Res"), and the Speed, Reserve and Resilience bars.

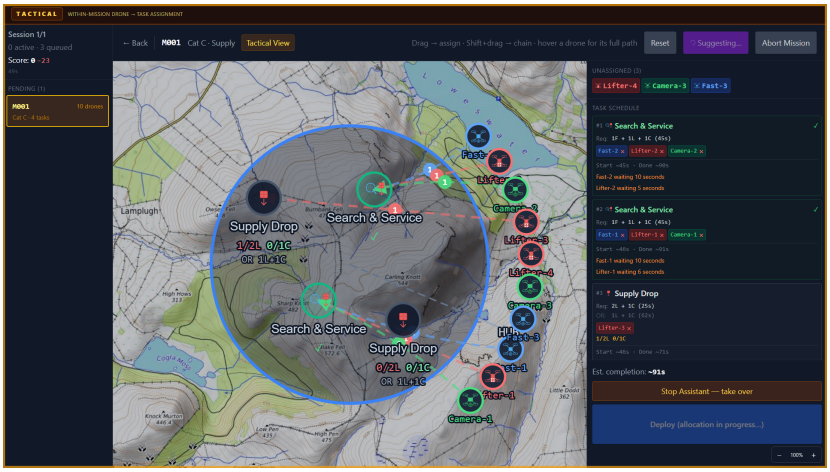



**(c)** Manual allocation (*Edit this allocation* or *Set manually instead*): a counter per type, how many of each could fail without stalling the mission, and the live ETA.

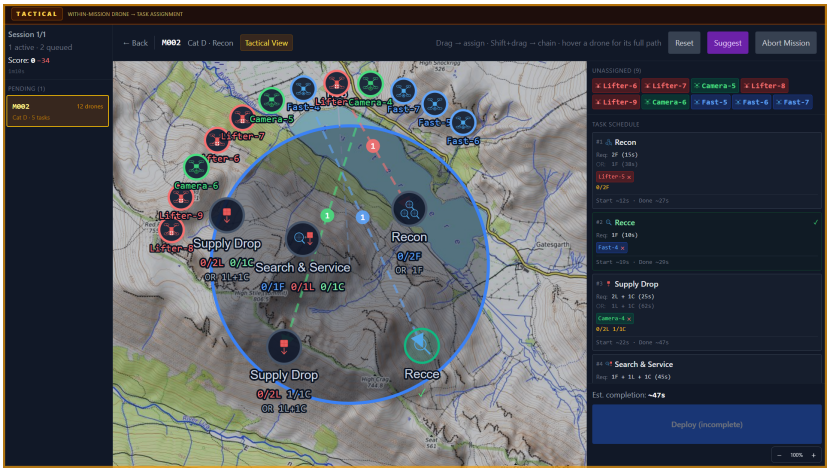
**Figure 3:** The strategic decision.




**(a)** The planner as it opens: every committed drone unassigned.



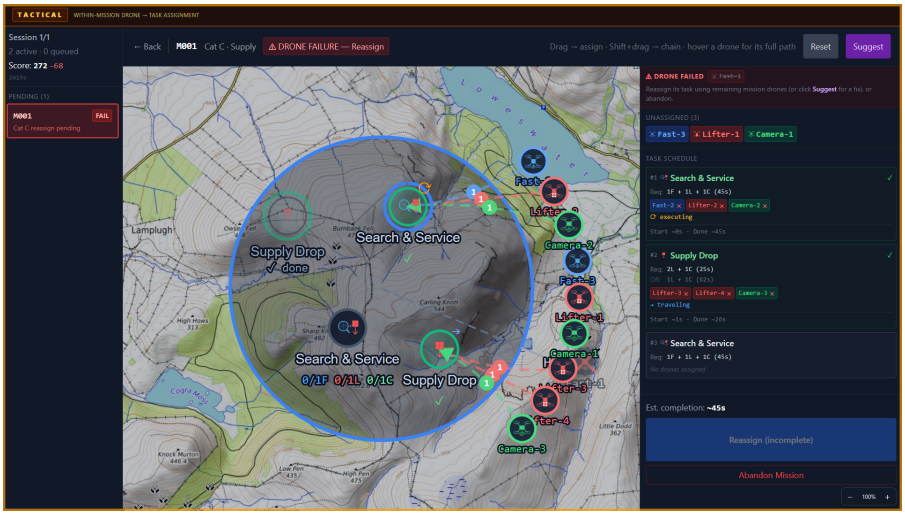

**(b)** Suggest filling in the plan, one drone→task entry per second; dashed lines and numbers show each drone's assigned task and order. *Deploy* stays locked until the fill completes.




**(c)** A plan being built by hand by dragging drones onto tasks.

**Figure 4:** The tactical decision in the tactical planner.

**(a)** Failure flagged in the queue.




**(b)** Recovery planner after Fast-1 failed: one Search & Service task is unstaffed; the remaining mission drones are listed as unassigned.

**Figure 5:** Drone failure and recovery.

16 Tactical Heavy first). The fleet, the assistants and every other parameter were the same in both.

- **Strategic Heavy** — missions arrive often (mean inter-arrival 37 s) and are mostly small (category weights A–E 40/38/16/5/1): about 13.5 missions per session, loading the cross-mission allocation decision.
- **Tactical Heavy** — missions arrive less often (75 s) and are mostly large (5/13/28/38/16): about 7.5 missions per session, loading the within-mission planning decision.

The two scenarios were tuned in simulation to comparable difficulty: a scripted near-optimal operator completes about 83% of tasks in both at the final failure rate.

**Four games, not two.** Every session used the same seed (42). A session's mission stream is seeded from the seed and a plan index, so there are exactly four fixed mission streams ("games"): Strategic Heavy 1 and 2 and Tactical Heavy 1 and 2. Up to P30 the plan index was the session number, so a participant's session-1 game was always the index-1 game of its scenario; from P31 (`study-v1.13`) the order was swapped, so later participants played the same games in the other position. The games differ in workload (tasks completed per session: SH1 25.1, SH2 28.9, TH1 21.5, TH2 32.8; Table 6), so every between-person comparison in the results holds the game fixed (z-scores within game, game means removed, or game as a fixed effect).

**Figure 6: Four games, not two scenarios.** Workload per session in each of the four fixed mission streams (Strategic Heavy 1 and 2, Tactical Heavy 1 and 2).

**Build versions.** The platform was updated during data collection (builds `study-v1.0` to `v1.13`); 24 of 33 participants played `v1.12` or `v1.13`. The changes that bear on the data are listed in Appendix B; the main one is that P01 and P02 played with half the later task points, which we rescale wherever a raw score is used.

## 6.2 Participants

Thirty-four people took part; 33 are analysed. P09 was run with assistant errors switched on (a pilot of the error manipulation) and did not follow the task, and is excluded throughout. The 33 were aged 21–44 (mean 28.7, SD 6.0); 25 men and 8 women. 28 held a postgraduate degree (23 master's, 5 doctorate) and 5 an undergraduate degree; 32 of 33 were from computing, AI, robotics or electronic engineering, and one from the armed forces. Self-reported experience was mostly moderate with simulations (20 moderate or extensive) and autonomous systems (24), and low with drones (21 none) and command or emergency management (24 none). [Recruitment route, ethics approval number, payment, and dates (sessions ran 2026-09).]

## 6.3 Procedure

1. **Consent and pre-study questionnaire.** Background and experience; a short comprehension check; and, from `study-v1.3`, three attitude scales: the AI Attitude Scale (AIAS-4, 10-point), a bespoke verification-propensity scale and a bespoke delegation-boundary scale (both 7-point).
2. **Tutorial** (Section 5).
3. **Session 1** (8 minutes; one scenario).
4. **Post-session questionnaires**: NASA-TLX and trust in each assistant (Section 6.4).
5. **A 30-second break screen.**
6. **Session 2** (8 minutes; the other scenario), then the same questionnaires.
7. **Debrief interview**, audio-recorded, with the researcher's written notes.

Participants were asked to think aloud while playing, and the researcher occasionally asked a short question during play; quotes that answer such a question are marked as following a leading question. Both screens, the cursor and the participant's voice were recorded throughout. [Total duration per participant; setting (lab, monitors); the debrief question list.]

## 6.4 Measures

**Event log.** Every session's first event dumps the full parameter set, and every operator action and assistant output is logged as it happens (606 strategic decisions and 630 tactical plans in all). The main behavioural measures are defined as follows.

- **Took a card**: a strategic decision deployed from a strategy card as offered. A card that was edited before deploying counts as a hand (manual) allocation.
- **Pre-emptive**: a manual allocation committed before the strategy cards had loaded.
- **Pressed Suggest**: Suggest was used at least once on that tactical plan (the planner starts empty, so this is always a deliberate request). **Kept unmodified**: a Suggest plan deployed with no change.
- **Handed over**: either of the two.
- **Decision time**: from opening the strategic panel (or the tactical planner) to the choice, minus the forced wait (the card reveal; one second per Suggest entry), unless stated otherwise.
- **Queue depth**: other missions that had arrived and had not yet received a strategic decision when the panel opened, rebuilt from the log.
- **Mission size**: drones needed (Section 3.4); also tasks, drone types, drones committed.

**Questionnaires.** After each session: **NASA-TLX** [6], six sliders from 0 to 20, reported as the unweighted mean (performance runs from 0 = perfect to 20 = failure; from `v1.12` the slider carried a note that it runs the other way, after two participants were confused by it). **Trust** in each assistant separately: six statements rated 1–7 ("The [Strategic/Tactical] Assistant is reliable", "I trust the … Assistant's recommendations", "… performs well", "I feel confident using …", "… provides useful guidance", "I would follow … suggestions without hesitation"), averaged.

[These six items are bespoke (adapted in spirit from [9], but not that scale) — describe and justify, or cite a source.] Technology-acceptance scales exist in the platform but are only given after a third session, so none were collected.

**Qualitative material.** Think-aloud and debrief audio was transcribed automatically (faster-whisper, word timestamps) and aligned to the event log by the on-screen session clock. Debriefs and the written interview notes were coded against a codebook of 28 codes written before coding (C01–C28; e.g. C03 "small myself, big to the assistant", C11 "hands over when overloaded", C12 "noticed over-allocation or unneeded types"). Each code is present or absent per participant, with every supporting excerpt checked to occur verbatim in the source. Quotes in the results were selected from these sources and checked verbatim. [Coding was done by a single coder, an AI model (Claude), with a second AI pass ($\kappa = .96$); the only independent check (stated preference, round 1) gave $\kappa = .27$. A human second coder is needed before the "X of 33 said" counts are reported as more than descriptive.]

### 6.5 Analysis

The analysis is exploratory, with a pre-specified core: ten hypotheses (H1–H10) were written and signed off before any test was run, each with one primary test, Holm-corrected together; everything else is reported as exploratory. The findings were then refined over five review passes in which every result was checked and re-specified. [Describe the analysis process honestly here: the analyses were scripted with the help of AI coding agents and every result re-run and reviewed by the researcher (register: `docs/analysis/register`). Decide how much of this belongs in the paper.] Common elements:

- **Decisions within people.** Decision-level outcomes are modelled with generalised estimating equations (GEE; logistic for yes/no, Gaussian for times on the log scale) clustered on participant, with small-sample bias-reduced standard errors. Predictors of interest are split into a *within-person* part (the decision's value minus the person's own mean) and a *between-person* part (the person's mean), so "within person" estimates compare each operator only with themselves.
- **Game held fixed**: game (one of the four) and session as fixed effects, or z-scores within game.
- **Expected and adjusted rates.** Where a raw rate could reflect who reaches a situation rather than the situation itself, a logistic model predicts each decision's chance of handing over from the mission (drones, tasks, types), the minute of the session, the game and the person (their rate on all their *other* decisions, so a decision never predicts itself). *Adjusted* = the overall rate plus the mean of (observed − expected) in a bin.
- **Intervals**: 95% intervals by resampling people (whole participants, with replacement; 400–2,000 draws), so one heavy user cannot make an interval look tight.
- **Nulls** are reported with the effect size the data could have detected.

## 7 Results

[The results below, Section 8 and Section 9 are generated from the analysis register by `scripts/paper/build_paper.py` (`results.tex`, `qual.tex`, `practitioner.tex`). Each subsection is one finding: its summary, then one page per point, in the order point (grey: the register's wording, accurate but written for the review page, to be rewritten), figure, key numbers, method, what participants said, and the quotes you highlighted, as you trimmed them. A † after a quote means it has not been ticked as checked against the recording. Blue text is your own pass-5 role for the section (for the paper / extra / numbers only) and your notes, so a page can be cut by its tag. Every section not archived is here, including those you marked extra or numbers only; where your notes left the choice of graph open, both views are given as panels. Sidelined analyses are in Appendix C. Findings are ordered: who the operator is; when operators hand over (mission size, manual builds, practice); workload and the queue; the assistants' flaws; trust; what operators wanted.]

### 7.1 Overview

Table 3 summarises each scenario, and Table 6 each game. Strategic Heavy sessions had nearly twice as many missions and decisions as Tactical Heavy ones, but about the same number of tasks completed; Tactical Heavy offered more points and accrued more penalty. Operators took strategy cards on 60% of decisions in Strategic Heavy and 73% in

Tactical Heavy, and pressed Suggest on 72% and 79% of plans. Workload (NASA-TLX) did not differ between the scenarios.

**Table 3:** Descriptive overview by scenario: mean (SD) over the 33 sessions of each scenario. Every participant played one session of each, so $p$ is a paired Wilcoxon signed-rank test (33 pairs; fewer where a measure is undefined, e.g. no manual decisions). Behavioural measures use the definitions of Section 6.4. [a]P01 and P02 ran at half the later task weights; their points are doubled here.

| Measure (per session) | Strategic Heavy | Tactical Heavy | $p$ |
|---|---|---|---|
| Missions arrived (non-residual) | 13.5 (0.5) | 7.5 (1.5) | $<.001$ |
| Strategic decisions made | 11.4 (2.3) | 6.9 (1.5) | $<.001$ |
| Tactical plans confirmed | 11.2 (2.5) | 6.8 (1.5) | $<.001$ |
| Tasks completed | 26.5 (6.0) | 27.2 (9.0) | .902 |
| Completion points[a] | 1276 (316) | 1627 (608) | .003 |
| Time penalty (points) | 276 (59) | 395 (182) | .001 |
| Score (points − penalty)[a] | 1000 (364) | 1233 (564) | .042 |
| Strategy card taken (% of decisions) | 60 (27) | 73 (28) | .004 |
| Manual before cards loaded (%) | 19 (23) | 11 (20) | .039 |
| Card edited, then deployed (%) | 6 (16) | 6 (14) | .508 |
| Suggest pressed (% of plans) | 72 (26) | 79 (29) | .029 |
| Suggest plan kept unmodified (% of plans) | 69 (28) | 75 (32) | .084 |
| Drone failures | 4.0 (2.6) | 3.8 (2.6) | .636 |
| Missions abandoned | 1.2 (1.6) | 0.8 (0.9) | .165 |
| NASA-TLX, raw mean (0–20) | 8.8 (3.1) | 8.6 (3.5) | .607 |
| Mental demand | 11.7 (4.8) | 11.0 (6.1) | .456 |
| Temporal demand | 13.6 (3.9) | 12.4 (5.0) | .226 |
| Performance (0 = perfect) | 7.5 (4.9) | 8.8 (4.9) | .152 |
| Effort | 9.6 (4.7) | 8.9 (4.7) | .352 |
| Frustration | 6.0 (5.0) | 6.2 (5.8) | .467 |
| Trust, Strategic Assistant (1–7) | 4.38 (1.52) | 4.84 (1.33) | .098 |
| Trust, Tactical Assistant (1–7) | 5.52 (0.88) | 5.36 (0.93) | .537 |

## 7.2 Each operator has a personal style — individual, not a type

**Summary.** Operators differ a great deal in how they use the assistants, and the differences are personal. We describe each session by ten habits (how often a strategy card is taken, how often Suggest is pressed, how long a decision takes, and so on). From those habits alone, a person's second session resembles their own first (median profile r = .48) and essentially nobody else's (r = −.01): their own session is in the closest three of 33 four times as often as chance (41% vs 9%), although the second session is a different scenario, and who the operator is explains about six times more of card-taking than the mission does. Six of the ten habits carry over clearly; one (topping up drones on single-type missions) does not at all. Between people, use of the two assistants is only loosely linked ($\rho \approx .35$), so leaning on one is not a fixed part of one overall "delegate" strategy (though within a person the two do rise and fall together from session to session). The styles do not fall into types, and nothing measured beforehand (attitude to AI, views on human authority, experience) predicts them. Strong evidence for a stable personal style; good evidence against types; "no evidence" (not "evidence of no effect") for disposition.

*Strength of evidence (register rating): strong. Your verdict: core.*

### 7.2.1 The habits we grade them on

[Pass 5: numbers only]

To compare styles we describe every 8-minute session by ten habits, each something the logs record at every decision. Seven concern the strategic tier (the strategy cards): how often a card is taken as offered; how often cards are previewed before choosing; how long a decision takes; whether, when building by hand, the operator does it before the cards have even appeared; whether a card is edited before committing; and how many drones are committed compared with the

cautious Conservative card, on single-type and on multi-type missions. Three concern the tactical tier (Suggest): how often Suggest is pressed on a plan, whether a suggested plan is deployed unchanged, and whether Suggest is pressed when recovering from a failure. The chart shows each habit's spread over all 66 sessions in its own units. Most habits vary enormously between sessions (card uptake runs from 0% to 100% of a session's decisions); two are nearly universal (most sessions keep every suggested plan unchanged, and most never edit a card). The right-hand column previews the next results: how strongly each habit carries over from a person's first session to their second. (Figure 7)

**Figure 7: The habits we grade them on**

**Key numbers.**

- Card taken as offered: median 67% of a session's strategic decisions (middle half 47–91%, range 0–100%)
- Suggest pressed: median 86% of tactical plans (55–100%); suggested plan kept unchanged: median 100% (middle half 100–100%)
- Decision time beyond the card wait: median 6.5 s (5.4–10.3 s, range 1.6–43 s)
- Card edited before committing: median 0% (most sessions never); going manual before the cards appear: median 50% of hand-made decisions (0–75%)
- Drones vs the Conservative card: single-type −0.5 (−1.25 to 0); multi-type +0.6 (−0.1 to +1.3)

**Method.** Rows: 66 sessions (33 people × 2; P09 excluded). Each habit is a share or an average over that session's decisions. A habit with nothing to count in a session is undefined there: going manual before the cards (no hand-made decisions) in 15 sessions,

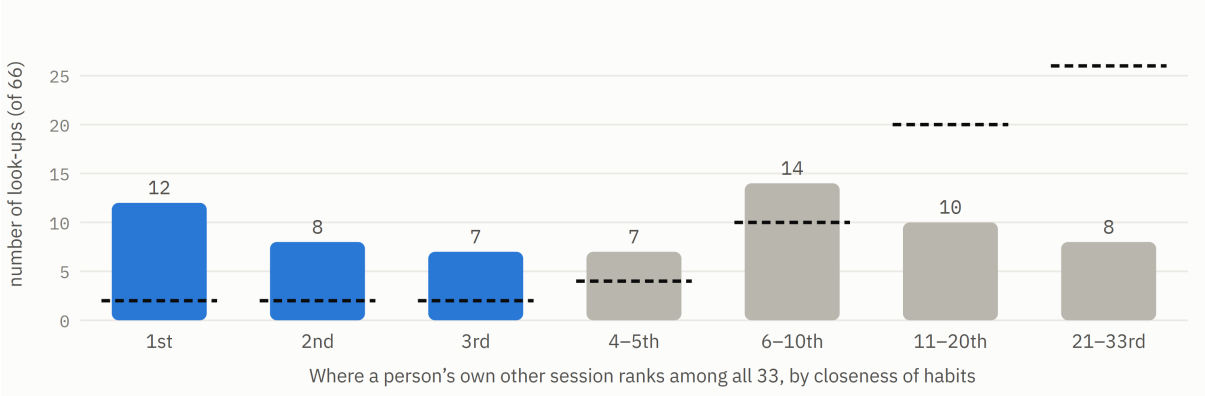
drones on single-type missions (no single-type mission opened) in 19, Suggest in recovery (no failures) in 4, keeping the Suggest plan (Suggest never pressed) in 2. For the profiles these are filled with the median of that game. Decision time is analysed as log(1 + seconds) and shown in seconds. For the comparisons that follow, each habit is turned into a z-score among the sessions of the same game (each of the four fixed games), so a profile says how a person compares with others who faced the same missions. The ten habits were fixed in round 1 (H1) before any grouping was tried; outcomes (score, trust, questionnaires) are never among them. $\rho$ = Spearman correlation between a person's session-1 and session-2 values, game-adjusted (see "How far each habit holds").

**What participants said.** Asked to describe their approach, 15 of 33 named a consistent strategy of their own (C01). In their own words it is mostly about the habits graded here: whether they wait for the cards or go manual before they load (P21, P32), whether they trim or edit a card, whether they start from Suggest and touch it up (P31, P21), and how they check (P14, P17). Few described a rule for when to press Suggest. Most said they pressed it on nearly everything.

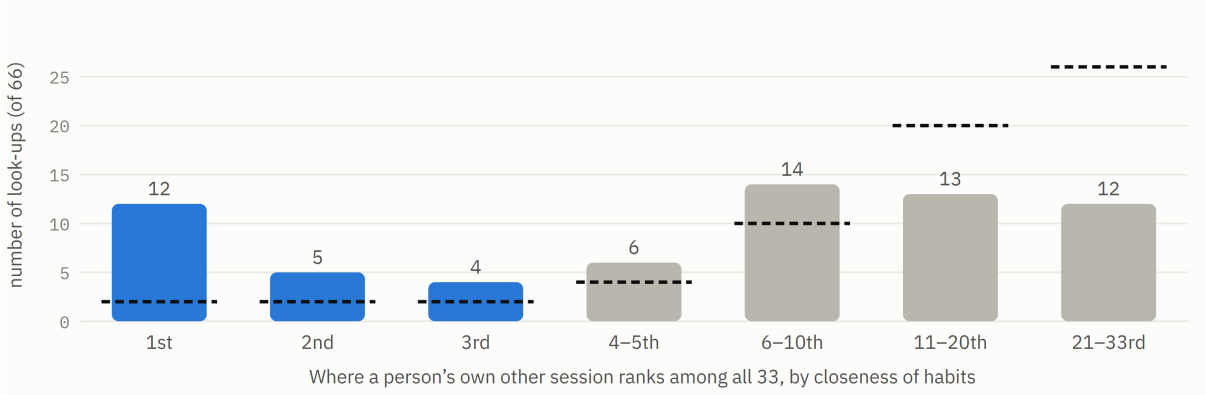
### 7.2.2 Where each person's own session ranks

[Pass 5: for the paper]

Each person's ten-habit profile from session 1 is compared with all 33 session-2 profiles, and the 33 are ranked from most to least alike (and the same the other way round, session 2 among the session 1s). If habits were not personal, a person's own other session would be equally likely to land anywhere in that ranking (dashed line). Instead it is the single closest match 12 times out of 66 (2 expected by chance) and in the closest three 27 times (6 expected), even though the second session is a different scenario. (Figure 8)




**(a)** All ten habits.




**(b)** With the reliance axis removed.

**Figure 8: Where each person's own session ranks**

**Key numbers.**

- Own session ranked 1st: 12 of 66 look-ups vs 2 expected; in the closest three: 27 vs 6 (41% vs 9%); permutation $p < .001$
- Average rank of own session 8.2 of 33 (chance 17.0); with the reliance axis removed 10.1 ($p < .001$)
- Profile correlation with own other session: median $r = .48$; with other people's: median $r = -.01$; own closer for 29 of 33

**Method.** Rows: 66 sessions (33 people × 2). Each session is the ten habits of the first section, each turned into a z-score among the sessions of the same game, so a profile says how this person compares with others who played that game. "Alike" = straight-line (Euclidean) distance between two ten-habit profiles. 66 look-ups (33 people, both directions); under chance every rank 1–33 is equally likely, so the dashed line is 66 × (ranks in the bin) / 33. The p-value shuffles who-is-who among the session-2 profiles 10

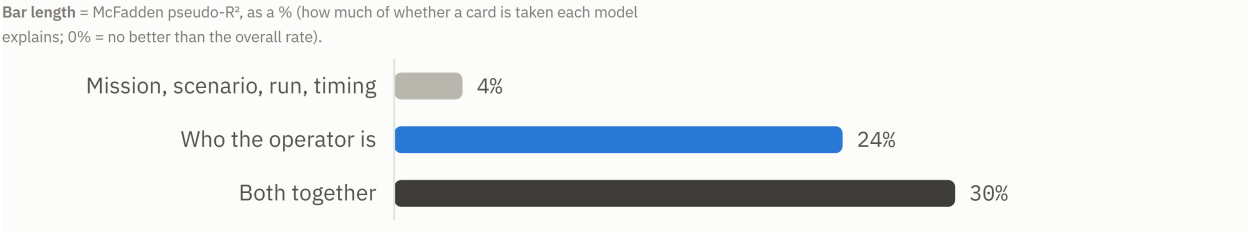
000 times. “Reliance axis removed” subtracts each profile’s score on the first principal component (see the axis numbers) before comparing.

**What participants said.** Nobody was asked to compare their two sessions directly. In the think-aloud, though, several people open session 2 with the same move and say the same thing as in session 1: P29 ’start conservative’ and does single-colour missions by hand, P11 builds all-blue missions themself, P21 hand-builds in parallel. That fits a recognisable personal style. The people who said they changed are in the retest section.

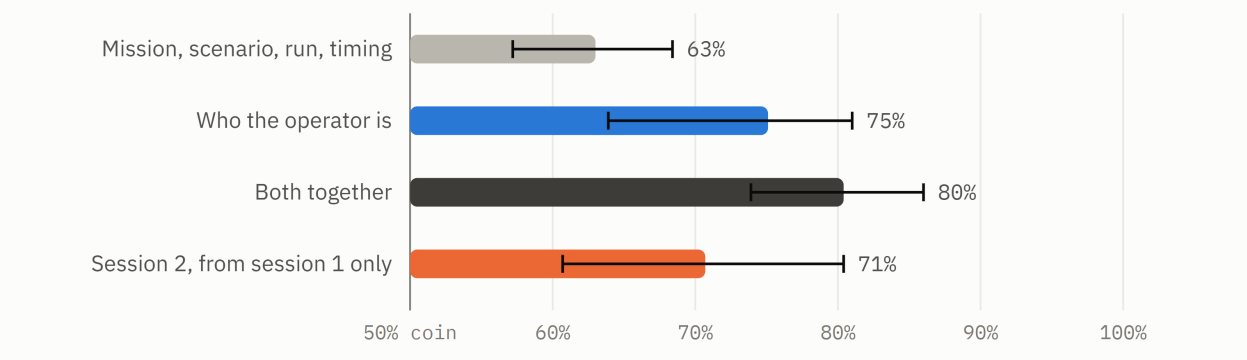
### 7.2.3 Who decides matters more than what is being decided

[Pass 5: for the paper]

Card uptake runs from 8% to 100% across people (median 61%). Knowing *who* is deciding predicts whether a card is taken about six times better than knowing everything about the mission and when it arrived. In plain terms: shown two decisions, one where a card was taken and one where it was not, and told only who made each, you would pick the right one 75% of the time (a coin: 50%; knowing only the mission: 63%). A person's session-1 rate on its own still does it 71% of the time for their session-2 decisions. (Figure 9)




**(a)** McFadden pseudo-$R^2$.




**(b)** The same comparison as a ranking test (AUC); whiskers are 95% intervals (people resampled).

**Figure 9: Who decides matters more than what is being decided**

**Key numbers.**

- Pseudo-$R^2$: mission + timing 4%; person 24%; both 30%
- Card uptake per person 8%–100%, median 61%; 6 people $\geq$ 90%, 3 people $\leq$ 25%
- Telling a taken from a not-taken decision (AUC): mission and timing 63% [57%, 68%]; the person, from their other decisions, 75% [64%, 81%]; both 80% [74%, 86%]; session-2 decisions from the session-1 rate alone 71% [61%, 80%] (321 decisions).
- Right-or-wrong guessing: always 'card' 66%; mission 65%; person 70%; both 77%.
- Person-only pseudo-$R^2$ with the person labels shuffled: 4% on average (95th percentile 6%), against 24% observed.

**Method.** 606 strategic decisions. Three logistic models of "card taken": (a) mission and timing only — drone types needed, drones needed, scenario, session 1 or 2, decision number; (b) the person only (one term each); (c) both. McFadden pseudo-$R^2$ = how much each improves on predicting the overall rate for everyone (0 = no better, 1 = perfect). Descriptive, not a test: with ~18 decisions each, the person-only model flatters itself a little; refitted 200 times with the person labels shuffled it scores 4% on average, so about 20 of its 24 points are real. Retest of raw card uptake session 1 vs 2: $\rho$ = .57 (.62 once game-adjusted, next section). **In plain terms** (the second view): AUC = the chance a model ranks a random card-taken decision above a random not-taken one. The person enters as their rate on all their OTHER decisions (on the log-odds scale), so no decision predicts itself; 'session 2 from session 1' uses only each person's session-1 rate. Intervals: people resampled 1,000 times, predictions held fixed.

**What participants said.** People often described their use of the assistants as part of who they are: 'naturally conservative' (P32), 'I'm more conservative' (P26), 'I always let the agent do it first' (P13). At the extremes, P14 used the assistants 'for almost every allocation' and P16 'manually overrode most of the time'. Against this, several gave the situation as the reason: workload (P32, 9 of 33 coded C11), being resource-constrained (P22), mission size (C03). P32 calls themself 'naturally conservative' and also says their use was 'very much about workload', so a person's style and the situation are both in play.

### 7.2.4 How far each habit holds from one session to the next

[Pass 5: for the paper]

One line per habit: how strongly a person's place on that habit in session 1 predicts their place in session 2 ($\rho$; 0 = no carry-over, 1 = same order of people). Six of the ten carry over clearly and survive correction for testing ten (coloured). Editing cards, going manual before the cards load and topping up drones on multi-type missions carry over weakly, and topping up drones on single-type missions does not carry over at all ($\rho$ −.09). "Keeps the Suggest plan" is high partly because 24 of 33 people kept every Suggest plan in both sessions. (Figure 10)

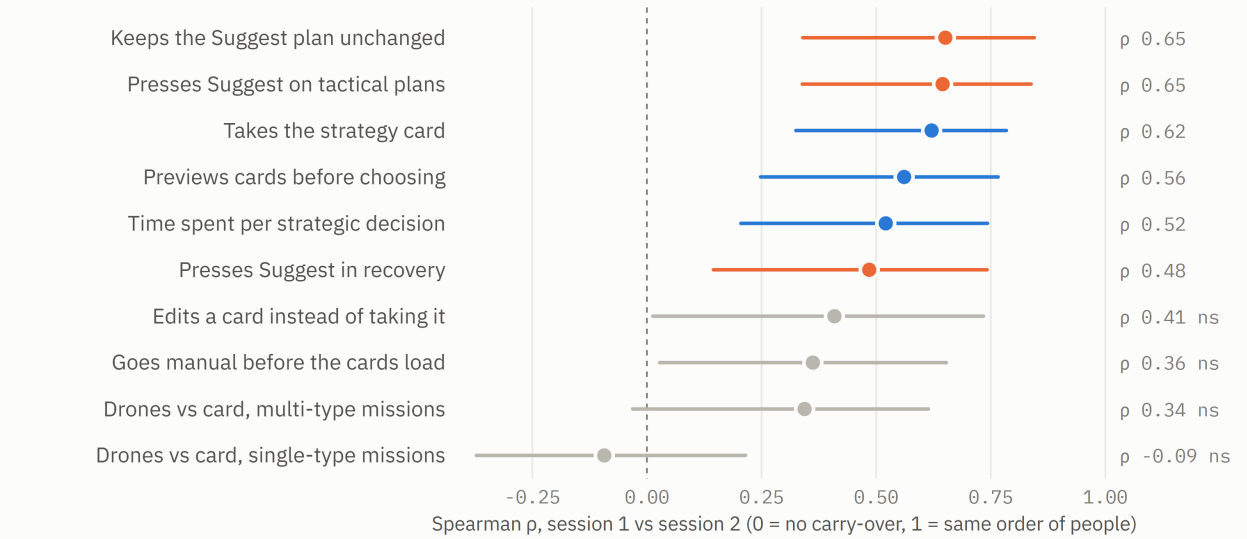



**Figure 10: How far each habit holds from one session to the next**

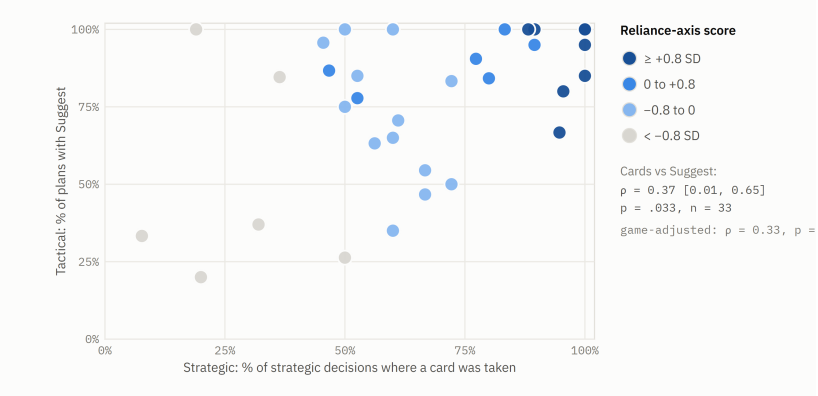
**Method.** 33 people. For each habit, Spearman $\rho$ between session-1 and session-2 values, each first z-scored within the game played; 95% intervals from resampling people 2000 times; Holm correction across the ten. The same $\rho$ values are the right-hand column of the habits chart in the first section.

**What participants said.** Some people say they kept their approach and some say they changed it. P18 'sort of changed from the first one' (led) and P30 'used more personalized' in session 2. Four people describe a real shift in the think-aloud: P06 and P31 came to rely on the assistants more and check less, while P35 and P15 went back to deciding for themselves after the strategy cards made errors. That fits habits that carry over clearly but not completely.

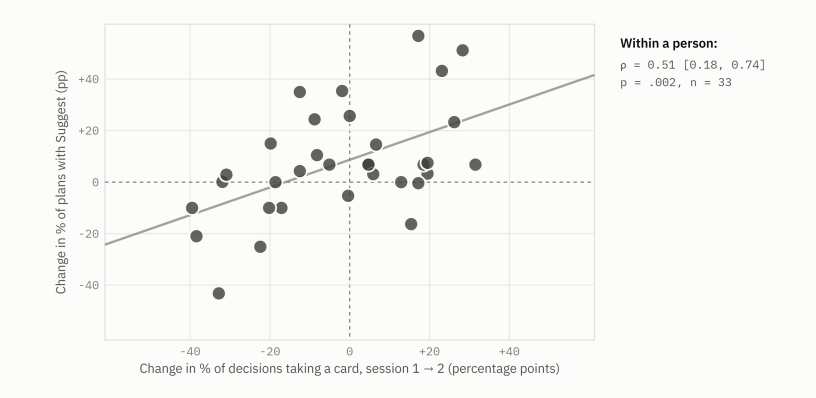
### 7.2.5 Strategic and tactical reliance go together only loosely between people

[Pass 5: for the paper]

Each dot is a person: x = share of strategy cards taken, y = share of tactical plans where Suggest was pressed. Between people the two are only weakly linked ($\rho$ = .37, and .33 once each is measured against others who played the same game): plenty of people press Suggest on almost every plan while taking only half the cards, so knowing how much someone relies on one assistant tells you little about the other. Part of the reason is a ceiling: Suggest is pressed on a median 86% of plans. The second view asks a different question: when the *same* person takes more cards in one session than the other, do they also press Suggest more? Yes ($\rho$ = .51). So use of the assistants is not one fixed trait spanning both tiers, but within a person there is a session-level "leaning on the assistants today" that moves both together. (Figure 11)




**(a)** Between people.




**(b)** Within a person: change from session 1 to session 2.

**Figure 11: Strategic and tactical reliance go together only loosely between people**

**Key numbers.**

- Between people, card uptake vs Suggest use: $\rho$ = .37 [.01, .65], p = .033 (raw); $\rho$ = .33 [−.03, .62], p = .058 (game-adjusted)
- Within a person, change in card uptake vs change in Suggest use, session 1 → 2: $\rho$ = .51 [.18, .74], p = .002
- Card uptake vs Suggest in recovery: $\rho$ = .42, p = .014; vs keeping the Suggest plan unchanged: $\rho$ = .20, p = .26
- Card taking and previewing: $\rho$ = .79 (largely mechanical: choosing a card means clicking it)

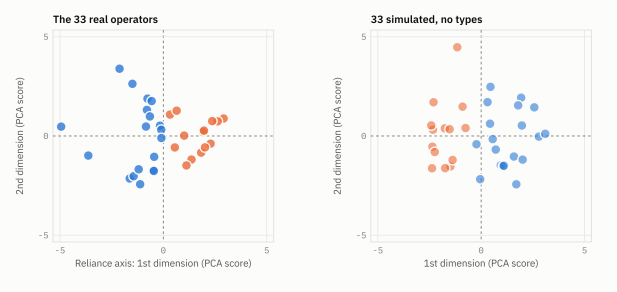
**Method.** Between people: one value per person, both sessions pooled (n = 33), Spearman $\rho$ with a Fisher interval; the game-adjusted version first subtracts the mean of everyone who played the same game from each session, then averages the person's two sessions. Within a person: the session-2 minus session-1 change in card uptake against the change in Suggest use, both game-adjusted (n = 33). Checked: the within-person link holds without game adjustment ($\rho$ = .56), and points the same way whichever scenario came first ($\rho$ = .47, p = .058 with Strategic Heavy first, n = 17; $\rho$ = .63, p = .010 with Tactical Heavy first, n = 16).

**What participants said.** Many describe relying on the two assistants differently, which fits the loose link between people. The most common split is to keep the strategic decision and hand over the tactical one: P22, P28, P30 and P32 say so, and P14 checked the tactical tier harder. A few did the reverse: P11 said they would use the strategic one more and built small plans themself, and P16 overrode the cards but 'more or less fully relied' on Suggest. P10 describes the two moving together: taking the Aggressive card makes Suggest the obvious next step.

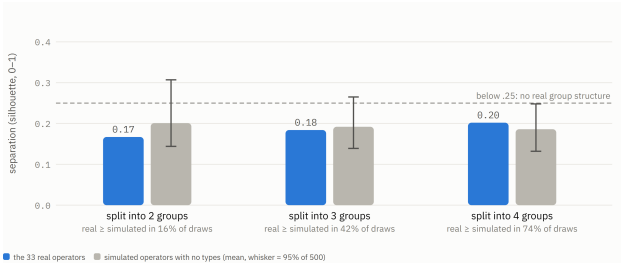
### 7.2.6 Individual, not types

[Pass 5: numbers only]

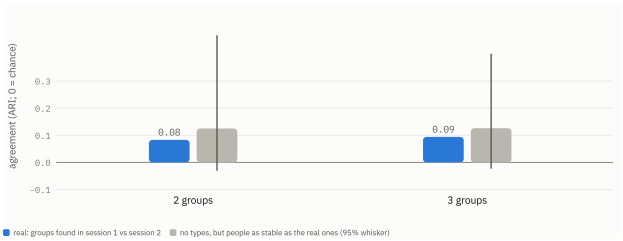
Distinct people, but not distinct kinds of people. If there were types (say "delegators" and "hand-builders"), people would sit in clumps with gaps between them, and splitting them into groups would find clean, well-separated groups. Three plain checks say they do not. **(1) The map:** placed by their two biggest habit dimensions, the 33 operators form one continuous cloud; it looks just like 33 simulated operators that have no types by construction, and the best two-group split simply cuts the cloud in half. **(2) Separation:** however many groups we ask for (2, 3 or 4), the real people separate no better than the no-types simulations do, and every value is below the usual threshold for real group structure. **(3) Recurrence:** grouping each session separately, the same people barely end up together twice, no more than in a smooth cloud where people keep their place. The archetype names (Delegator, Deliberate builder, Fast tinkerer) are best kept as labels for the ends of the main dimensions. (Figure 12)




**(a)** The people, mapped on the first two principal components.




**(b)** Cluster separation (silhouette) against no-types simulations.




**(c)** Recurrence of groupings across sessions (adjusted Rand index).

**Figure 12: Individual, not types**

**Key numbers.**

- Separation (silhouette), real vs no-types simulations: 2 groups .17 vs .20 [.14, .31]; 3 groups .18 vs .19 [.14, .27]; 4 groups .20 vs .19 [.13, .25]; a no-types sample separates at least as well 84%, 58% and 26% of the time
- Groups in session 1 vs session 2 (ARI): 2 groups .08, 3 groups .09; a stable no-types cloud gives .13 on average (95% of simulations −.03 to .47 for 2 groups, −.02 to .40 for 3), at least as high as the real value about 55% of the time; shuffled labels give up to .12–.13
- Round 1: two groups vs one, ΔBIC 1.9 (needed > 10; 42% of no-type simulations did at least as well)
- Archetypes (extras2): three explain 46% vs 43–59% in the no-types continuum (p = .83); same archetype both sessions 61% vs 30–79% (p = .32)

**Method.** Rows: the 33 people, each described by the ten habits pooled over both sessions and standardised. Grouping = k-means (the computer splits people into k groups so that each sits as close as possible to its group's centre; best of 20 starts). Separation = silhouette: for each person, how much closer they are to their own group than to the nearest other group, averaged (1 = tight clumps far apart; 0 = boundaries arbitrary; below .25 is conventionally "no substantial structure", Kaufman & Rousseeuw). Benchmark = 500 simulated samples of 33 operators drawn from one smooth cloud with the same distribution of each habit and the same correlations between habits as the real people, so they have no types by construction. (We use these rather than simply

shuffling each habit, because shuffling also destroys the links between habits, and a cloud of linked habits separates slightly better than shuffled data without having any types: shuffled data give about .14.) Recurrence: each session's profiles (z-scored within game) are grouped separately and compared with the adjusted Rand index (1 = the same groupings, 0 = chance), against (a) shuffled labels and (b) 500 simulated pairs of sessions with no types but the same session-to-session stability of every habit as the real people. The map uses the first two principal components (50% of the differences). Round 1's pre-registered test (mixture models compared by BIC) and the archetype simulations (extras2) reach the same conclusion; their numbers are kept below.

**What participants said.** Nobody described themselves or others as a 'type' of operator. People talked about their own rules, and several, such as P29, were still unsure of them partway through a session ('I haven't actually decided ... which ones are best to handle myself').

### 7.2.7 Nothing measured beforehand predicts the style

[Pass 5: for the paper]

Attitude to AI (AIAS-4), verification propensity, views on human authority over AI (delegation scale and its single items, including "a human should make the final decision"), and prior experience: 40 tests against five behaviours. Three reach p < .05 before correction, and <b>none survives Holm correction for the 40 tests. Three nominal hits is what chance alone produces: shuffling which person's questionnaire goes with which person's behaviour gives 2.0 such hits on average, and 3 or more in a third of shuffles. This is "no evidence", not proof of no link: the sample could only have detected fairly strong correlations ($|\rho| \geq .48$), and it is almost uniformly AI-positive (AIAS-4 averages 8.0 / 10) and from computing or engineering (32 of 33). Within this sample, the large differences in style are not explained by what people say they think about AI. (Figure 13)

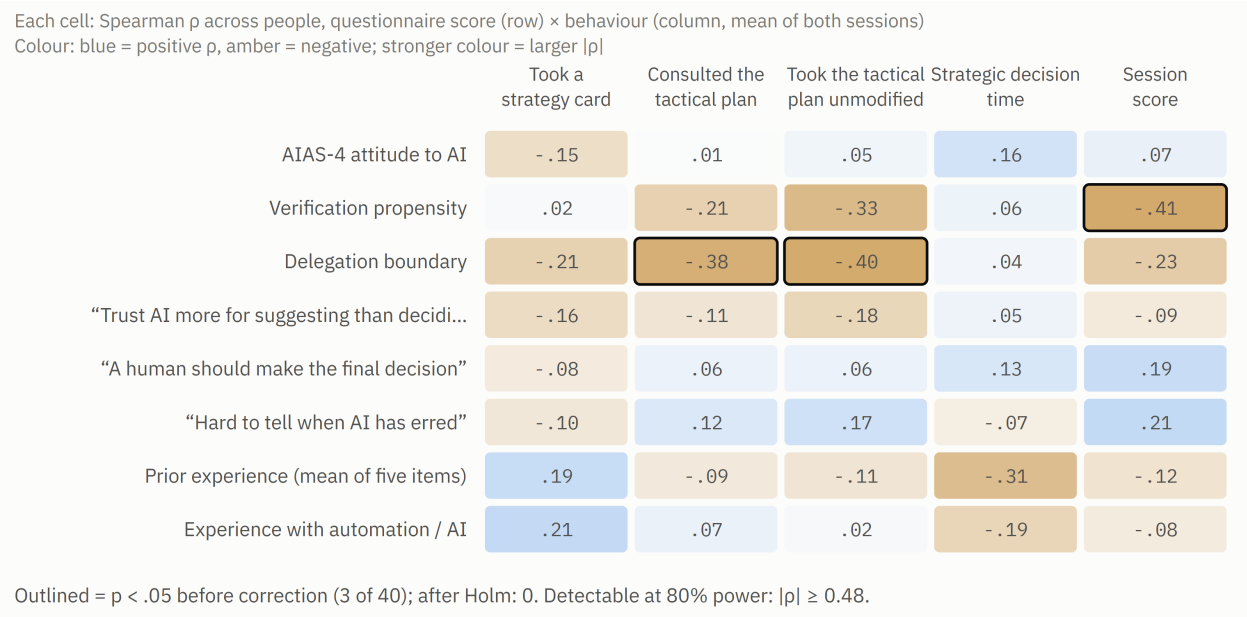



**Figure 13: Nothing measured beforehand predicts the style**

**Key numbers.**

- 0 of 40 tests survive Holm; 3 nominal (verification × score $\rho$ −.41; delegation × Suggest consulted −.38 and × plan kept −.40)
- Chance alone gives 2.0 nominal hits on average; 3 or more in 34% of 2000 shuffles of who-goes-with-whom; a p as small as the smallest one (.021) turns up somewhere in the grid in 51% of shuffles
- AIAS-4 × card uptake $\rho$ −.15; "A human should make the final decision" × card uptake $\rho$ −.08
- AIAS-4 odds ratio for card taking 0.80 per SD [0.50, 1.27]; detectable only at about ×2
- AIAS-4 mean 8.0 / 10 (SD 1.2); 32 of 33 from computing / engineering; detectable $|\rho| \geq .48$

**Method.** Per person: each questionnaire score (taken before session 1) against the average over both sessions of: card uptake, Suggest consulted, Suggest plan kept unmodified, strategic decision time, session score. Spearman $\rho$ per cell; Holm correction across all 40 together (build_findings_synthesis `analyse_disposition`). Detectable effect = the correlation the sample had an 80% chance to find at $\alpha$ = .05 (n = 32 with questionnaires). Round 1 H7's decision-level model gives the same answer for AIAS-4: odds of taking a card × 0.80 per SD [0.50, 1.27].

**What participants said.** When people talked about AI in general, it did not line up neatly with what they did. P04 does not trust the AI to be better than them but uses it because it is quicker. P24 says they really trust AI yet trims the cards to save drones. P15 calls the AI 'a little bit stupid' but hands it complex plans. P35 relies on it because they are 'not trained'. Most people explained their use by time pressure and job size rather than by a view about AI, which fits the null.

## 7.3 Small jobs by hand, big jobs to the assistant — on both tiers, and it is where the assistant saves time

**Summary.** The bigger the job, the more likely an operator hands it to an assistant, on both tiers. Measured the same way on both (drones the mission needs), the strategy card is taken on 56% of the smallest missions vs 75% of the largest, and Suggest is pressed on 54% vs 91%. Comparing each person only with themselves, every doubling of the mission multiplies the odds of handing over by about 1.6 (strategic) and 2.9 (tactical); 22 of 33 people hand over more on their biggest missions than their smallest on each tier. It survives every check we tried, including dropping the illogical Conservative cards and allowing for everyone having played the same missions. Three qualifications: on the strategic tier most of the effect is operators deciding small missions before the cards load; the strategic curve is lumpy (a dip at 5 drones, missions that mostly arrived in bursts in the Strategic Heavy games), so the dependable strategic contrast is one-type vs three-type missions; and handing over on one tier goes with handing over on the other (a mission allocated from a card gets Suggest at about twice the odds). The rule is sensible: the card is much quicker than building by hand on three-type missions and no quicker on one-type ones, and a Suggest plan is not slower: compared on missions of the same size it finishes about 15 s sooner, mostly because it is carried out faster.

*Strength of evidence (register rating): strong. Your verdict: core.*

### 7.3.1 Both tiers on one axis: the bigger the mission, the more it is handed over

[Pass 5: for the paper]

For every mission, the share of decisions where the operator handed over (took a strategy card as offered, blue, or pressed Suggest in the tactical planner, orange) by how many drones the mission needs. Both rise. The tactical tier rises more steeply and keeps rising, from 54% to 91%. The strategic tier climbs from about half on small missions to about three quarters, with a dip at 5 drones. That dip is seven particular all-Fast missions, all in the two Strategic Heavy games, and six of them arrived in bursts: in one game two of them arrived 11 s apart in a run of seven missions inside one minute (minutes 1.6 to 2.6), in the other four of them arrived within 80 s (minutes 3.7 to 5.0) alongside three other missions. So by the time operators reached them 3–4 other missions were waiting, and they decided fast and often before the cards loaded. Busy arrival is likely part of it, but not all: missions of other sizes in the same bursts got the card more often (a 3-drone and a 6-drone mission in those bursts got it 67% and 64% of the time), and holding the queue fixed barely changes the 5-vs-3 comparison (see "Why the 5-drone missions dip"). (Figure 14)

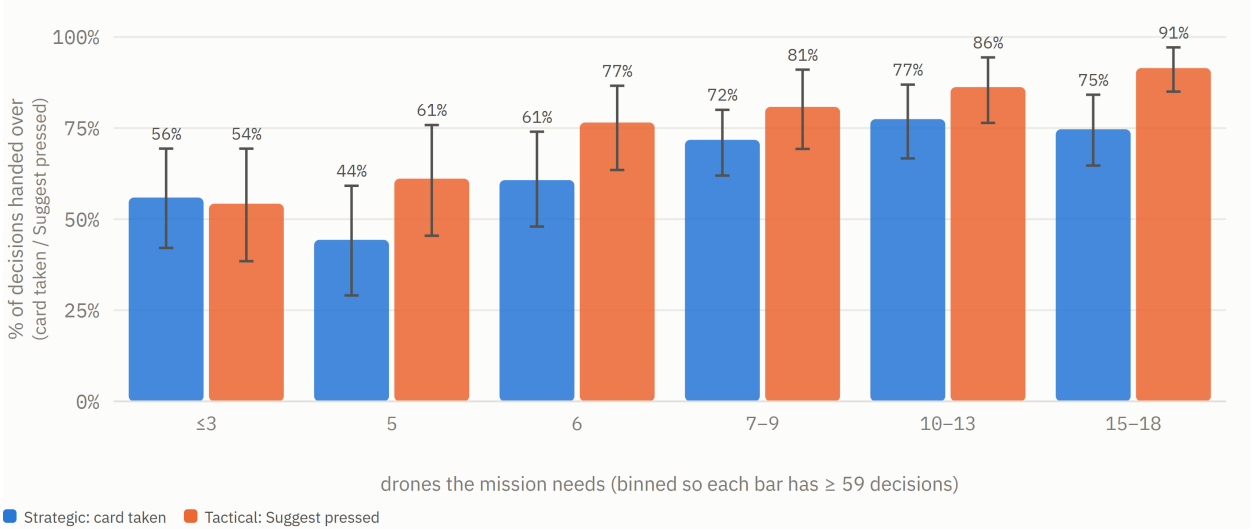



**Figure 14: Both tiers on one axis: the bigger the mission, the more it is handed over**

**Key numbers.**

- Strategic, per doubling of drones needed: odds ×1.62 [1.23, 2.15], p < .001 (606 decisions, 33 people)

- Tactical, per doubling of drones needed: odds $\times 2.91$ [2.03, 4.17], $p < .001$ (627 plans)
- Both tiers, one shared slope: $\times 2.07$ [1.64, 2.61], $p < .001$ (1,233 decisions); the tactical slope is steeper than the strategic one by $\times 1.38$ [0.99, 1.91], $p = .056$ — suggestive, not shown
- Raw rates, smallest to largest bin: strategic 56% $\rightarrow$ 75%, tactical 54% $\rightarrow$ 91%
- Arrival bursts (Strategic Heavy games, fixed per game): the seven 5-drone missions had on average 4.4 other missions arrive within 60 s of them, against 3.6 for the 3- and 6-drone one-type missions and 3.1 for every other mission; 6 of 7 arrived inside or straight after a burst; the seventh arrived 10 s into a game and got the card 69% of the time

**Method.** **Rows.** Strategic: all 606 strategy-card decisions (one per strategic choice). Tactical: all 627 non-recovery tactical plans whose mission size is known (630 minus 3 residual missions with no recorded need). **Size** = drones the mission needs: the sum of the primary requirements over its tasks — the number an operator reads off the task badges, the same for both tiers. Binned so each bin has $\geq 59$ decisions. **Intervals**: we resample whole people (2,000 times) and recompute the rate, so a person's decisions travel together and one heavy user cannot make an interval look tight. **Model**: a logistic GEE (a regression for yes/no outcomes that allows each person's decisions to be correlated) with size split into two parts — this decision's size minus the person's own average size (the *within-person* part) and the person's average (the between part); game and session are held fixed. The within-person odds ratio answers: "when the same person meets a mission twice as big, how much do the odds that they hand it over change?" Odds $\times 2$ means, for example, a 60% chance becomes 75%. Both tiers in one model: the two tables stacked, one shared within-person slope plus a tier term; the tier $\times$ size term tests whether the slopes differ.

**What participants said.** 10 of 33 state the size rule in debrief or interview (C03). The think-aloud adds at least six more who give the same reason while playing: P04, P06, P12, P20, P32 and P36. They do single-colour or few-drone missions themselves and hand over large, multi-colour, many-chain ones, usually because it is quicker. There is dissent. P30 thinks simple tasks are the ones the AI 'should be able to do even easier'. P29 switched direction mid-session ('if it's easy, then it's hard for it to do it wrong'). P20 kept complex orderings for themself, and P10 hands trivial one-option plans to Suggest.

**Quotes selected for this section.**

> *"I think if it's just one colour, I'm gonna do it, like, assign it myself, but if it's multiple colours, I think I'm gonna let, um, the AI choose for me, for the, the strategic side." — P29 (think-aloud s2 0:17:33)*

> *"either it's a really big mission, and it's like, oh, I can trust it with a large mission. [. . . ] Or it's a simple mission, and I'll be like, oh, I can just bang it up myself." — P11 (debrief 0:28:41)*

> *"I did feel like some of them I should do myself and I trusted it more with the complicated tasks" — P29 (debrief 0:24:54)*

> *"When they only needed a single colour of drone I tended to do it manually" — P34 (written interview)*

> *"I'll use the suggest, because there's a lot of drones here, so because there's a lot of drones to allocate" — P05 (think-aloud s1 0:09:36)*

### 7.3.2 Person-adjusted: every size measure, both tiers, one picture

[Pass 5: for the paper]

Each row is one within-person estimate: how the odds of handing over change when the same person meets a bigger job. Rows right of the line mean "bigger → hand over more". All size measures point the same way on both tiers; the two that do not clear the line are strategic tasks and the strategic slope once the cards are on screen. The bottom block holds the artefact checks. Dropping every illogical Conservative card leaves the effect in place. So does allowing for the fact that every operator played the same fixed missions, so one odd mission could otherwise make the interval look tight. With each mission as one point, the share handed over still rises with its size on both tiers. (Figure 15)

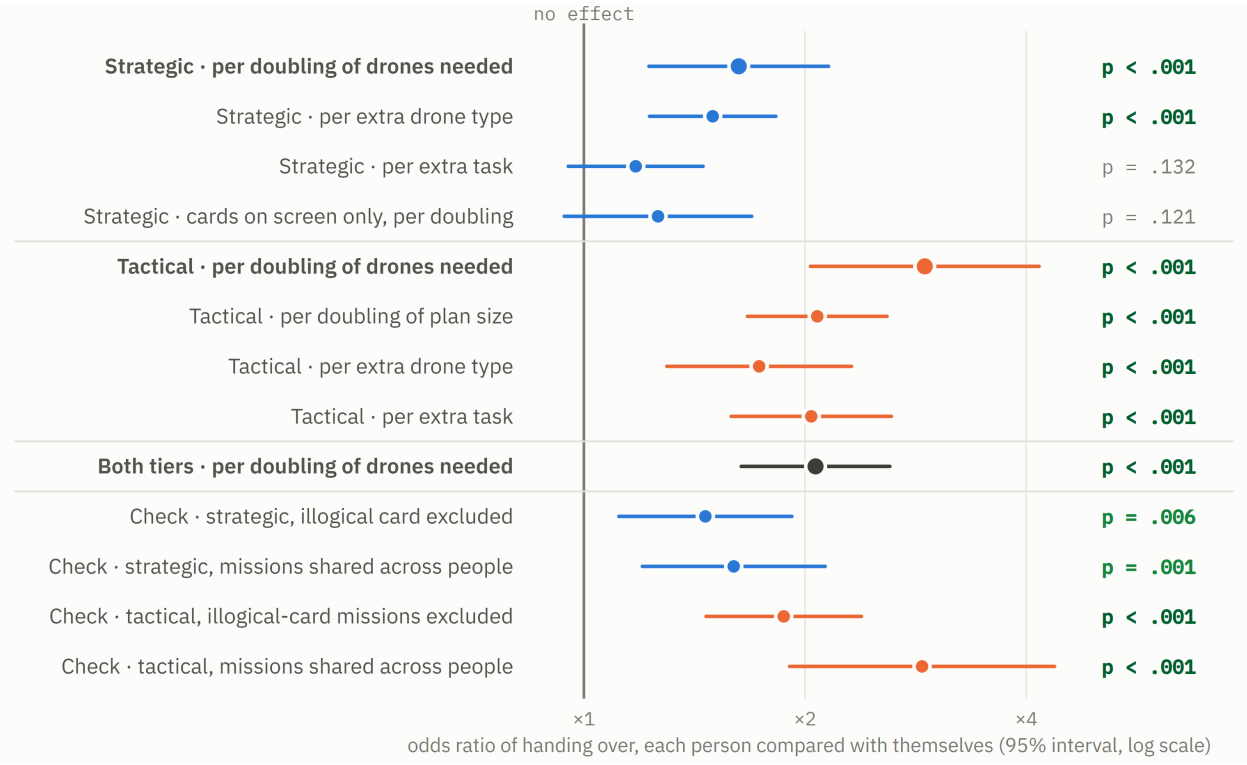



**Figure 15: Person-adjusted: every size measure, both tiers, one picture**

**Key numbers.**

- Strategic, illogical Conservative card excluded: ×1.46 [1.12, 1.92], p = .006 (474 decisions)
- Missions shared across people (interval clustered on person and mission): strategic ×1.60 [1.20, 2.13], p = .001; tactical ×2.89 [1.91, 4.37], p < .001 (65 missions)
- Each mission as one point (missions with ≥ 5 decisions): strategic $\rho$ = .63 over 41 missions, tactical $\rho$ = .69 over 41 (both p < .001)

**Method.** GEE logistic models clustered on person (exchangeable correlation, small-sample "bias-reduced" standard errors), size split into within- and between-person parts, game and session held fixed. Odds ratio = how many times the odds of handing over multiply per unit of size. Read ×1.6 per doubling as: a person who takes the card on 60% of 6-drone missions would be expected to take it on about 71% of 12-drone ones.

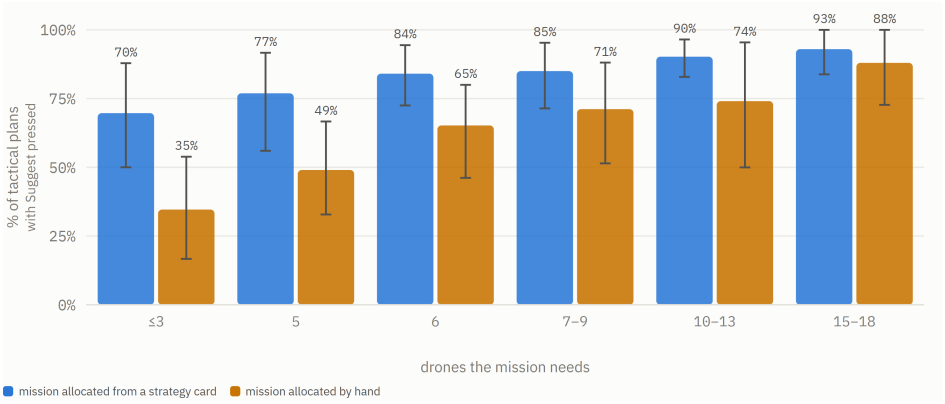
**What participants said.** No one spoke to the model checks themselves. One check bears on what people said, though. Several said part of their hand-building on one-type missions was to avoid the Conservative card's spare

drones of unneeded types (P19, P29, P21, P13). The finding's check that drops those illogical cards is the right control for that stated reason.

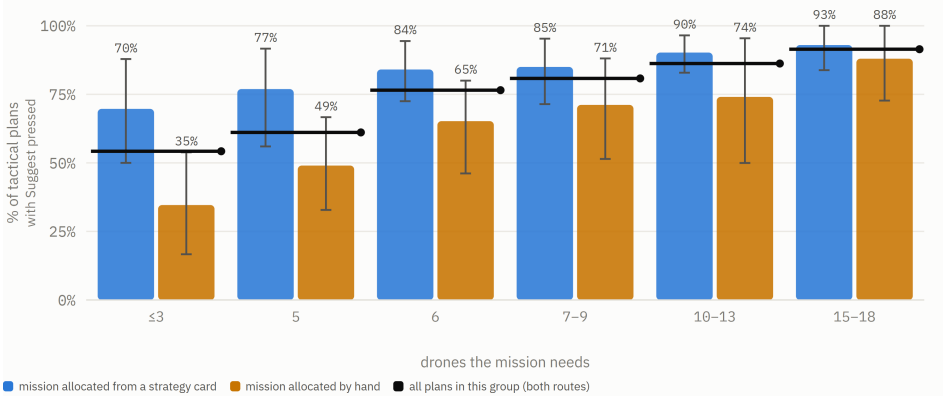
### 7.3.3 Tactical tier: Suggest rises with mission size, and goes with the strategic choice

[Pass 5: for the paper]

In the tactical planner the dose-response is smooth whichever way size is measured: drones the mission needs, drones committed, tasks, drone types or plan size. The two bar colours split plans by how the mission was allocated strategically. A mission allocated from a strategy card gets Suggest more often than one allocated by hand, at every size and on every measure: **handing over at one tier goes with handing over at the other**. Comparing the same person's missions of the same size, a card-allocated mission gets Suggest at about twice the odds (×2.10). The gap is wide on small missions (70% vs 35% at ≤3 drones) and small on the biggest (93% vs 88%), but that narrowing is both rates approaching 100%; tested properly, the gap does not reliably shrink with size. Operators say this too: one participant (P10) described, while playing and again afterwards, that once they had taken the Aggressive card "anything but Suggest" made no sense, because a generous allocation leaves only one sensible way to assign the drones; two others (P05, P30) talked through taking a card and then pressing Suggest on the same mission. P10's reason, that it is the generous card that makes Suggest the obvious next step, does not show up across everyone: Suggest follows the Aggressive and the Conservative card at the same rate. The link is card vs hand, not which card. We use drones needed as the main measure: it is fixed by the mission, the same for every operator and what the task badges show. Plan size and team size also contain the operator's own strategic choice: for the same mission, a card commits about 11% more drones than a hand allocation, so part of the strategic route is built into those x axes. (Figure 16)




**(a)** Size measured as drones needed.




**(b)** With a marker on each pair at the rate for all plans in that size group.

**Figure 16: Tactical tier: Suggest rises with mission size, and goes with the strategic choice**

**Key numbers.**

- Within person, per doubling of drones needed: odds ×2.91 [2.03, 4.17]; per extra task ×2.04 [1.59, 2.62]; per extra drone type ×1.73 [1.30, 2.31]; per doubling of drones committed ×2.51 [1.83, 3.45]; per doubling of plan size

$\times$2.08 [1.67, 2.59] (all $p < .001$)

- Allocated from a card vs by hand, same person, size held: $\times$2.10 [1.30, 3.40], $p = .003$ (drones needed); $\times$1.78 to $\times$2.12 on the other four measures, all $p < .05$
- Aggressive card vs Conservative card, same person, drones needed held: $\times$0.88 [0.45, 1.75], $p = .72$ (400 plans); raw Suggest rate after Aggressive 81% (200), after Conservative 89% (203), after a hand allocation 63% (227)
- Does the gap shrink with size? Route $\times$ size: $\times$0.76 [0.30, 1.92], $p = .56$ (drones needed); $p = .28$ to .98 on the other measures: no reliable narrowing
- Suggest pressed, card-allocated vs hand-allocated: $\leq$3 drones 70% vs 35%; 7–9 drones 85% vs 71%; 15–18 drones 93% vs 88%
- Same mission, card vs hand: $\times$1.11 more drones committed and $\times$1.12 larger plan (both $p < .001$); drones needed, tasks and types identical by construction
- Person level: people who take more cards also press Suggest more (Spearman $\rho = .37$, $p = .033$, 33 people)
- P10 in the log: Suggest on 8 of 9 Aggressive-allocated missions (never took the Conservative card) vs 6 of 9 hand-allocated

**Method.** **Rows**: the 630 non-recovery tactical plans that were confirmed (the planner starts empty, so pressing Suggest is a deliberate choice); 627 have a known mission size. **Measures**: drones needed (sum of the task badges' requirements), drones committed (the team the strategic choice sent), tasks (residual missions left out), drone types, plan size (team $\times$ tasks). **Size model**: logistic GEE clustered on person, size split into this plan's size minus the person's own average (within) and that average (between), game and session fixed. **Route**: the same model plus "allocated from a card" (also split within/between person), so its odds ratio compares the same person's card-allocated and hand-allocated missions of the same size. **Narrowing test**: a route $\times$ size term (ratio of odds ratios; below 1 = the gap narrows as size grows). **Same-mission check**: regression of each measure on route with one dummy per mission, so only operators who met the same mission are compared.

**What participants said.** Three people tie the two tiers together the way the logs do (card-allocated missions get Suggest more). P10 says repeatedly that once they have taken the Aggressive card there is 'only really one way' to assign, so Suggest is the obvious next step. P05 and P30 take a card and then press Suggest on the same mission. The same continuity runs the other way for P21, who lays out a plan in their head while building the allocation by hand and says it would 'disrupt my thought process' to switch to Suggest for the tactical step; P30 likewise finds it easier to give the assistant resources and let it optimise. The opposite pairing is said too. P18 takes a card and then plans a routine mission by hand, and P26 and P28 build strategic allocations by hand but press Suggest. On size, the reasons given are drag-and-drop effort and the number of combinations or chains.

**Quotes selected for this section.**

> *"Now I'm using the suggestion because I took the aggressive approach which means I know that I have enough drones." — P10 (think-aloud s2 0:13:08)*

> *"Better to use tac agent when using the aggressive strat agent" — P10 (written interview) (paraphrase)*

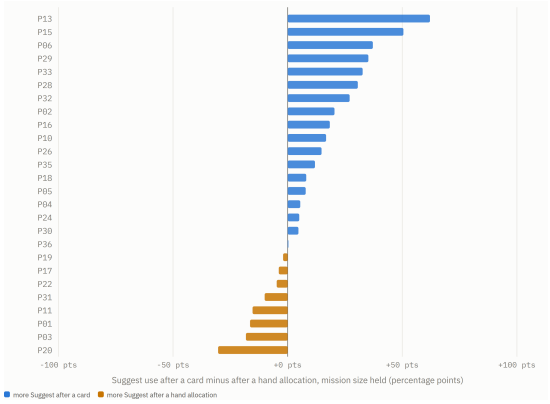
> *"If I look at a mission with 15 drones and I need to move two or three chains, that's just going to take me about a minute, and it's just no point." — P11 (debrief 0:29:41)*

> *"Dragging 14 drones is going to take way more time than hitting the assign and waiting for eight seconds." — P10 (think-aloud s2 0:13:43)*

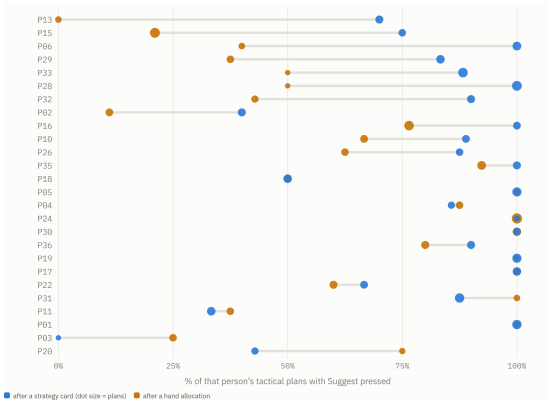
### 7.3.4 Card then Suggest, person by person: a broad tilt, but a few people carry much of it

[Pass 5: numbers only]

Is the card → Suggest link down to a few people? Partly. Each bar is one person: how much more often they pressed Suggest on missions they had allocated from a card than on ones they allocated by hand, with mission size held (the second view shows the two raw rates). Of the 26 people who did both often enough to compare, **17 lean towards Suggest after a card, 8 the other way, 1 neither**, so it is not one or two people. But it is uneven: the three biggest gaps (P13, P06, P15) make up about half of the total, and the person-adjusted odds ratio (×2.10 for everyone) falls to ×1.63 without those three (still p = .03) and to ×1.46 without the top five (p = .08). Leaving out any single person never moves it outside ×1.90–×2.40. So: a general tendency, strongest in a handful of people. Seven people cannot show a gap at all, six because they nearly always took a card before planning. (Figure 17)




**(a)** Each person's gap, mission size held.




**(b)** Each person's two raw rates.

**Figure 17: Card then Suggest, person by person: a broad tilt, but a few people carry much of it**

**Key numbers.**

- People with ≥ 2 plans each way: 26. Gap (size held) positive 17, negative 8, none 1; sign test p = .11; mean gap +11 points.
- Card vs hand odds ratio, everyone ×2.10; leaving out any one person ×1.90 (without P15) to ×2.40 (without P11), always p < .05.
- Without the largest gaps: top 1 (P13) ×1.97, p = .006; top 3 (P06, P13, P15) ×1.63 [1.05, 2.52], p = .03; top 5 ×1.46 [0.96, 2.22], p = .08; top 8 ×1.23, p = .34.
- The three largest gaps are 52% of the summed gap. Not comparable: 6 people almost only planned after a card, 1

almost only after a hand allocation.

- Also holding fixed whether the mission needs one drone type: $\times$2.13 [1.30, 3.49]; and the number of tasks too: $\times$2.06 [1.24, 3.43], p = .005.

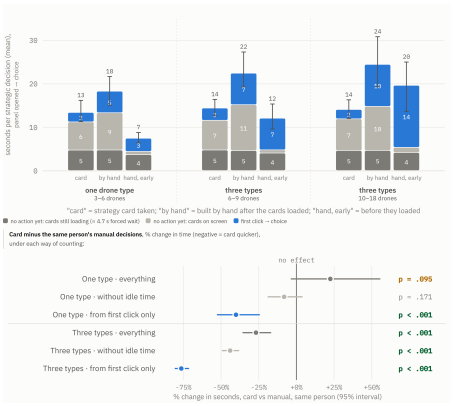
**Method.** Rows: the 627 tactical plans with a known mission size, split by how the mission was allocated (strategy card or by hand). For each person with at least two plans each way, the gap is the mean of (pressed Suggest − expected) after a card minus the same after a hand allocation, where 'expected' comes from a person-clustered model of Suggest on mission size (drones needed, within and between person), game and session, without the route; so the gap compares missions of the same size. The odds ratio is the section above's route model (card vs hand, same person, size held), refitted leaving out one person at a time, and then leaving out the people with the 1, 3, 5 and 8 largest gaps. The sign test asks whether more people lean one way than chance.

**What participants said.** The people at the ends of the chart, in their own words. **Strongest tilt** (P13, P15, P06, then P29): none of them talks about linking the two steps. They describe a rule for the whole mission: P15 hands over when there are 'too many decisions', P29 does one-colour missions themself and lets the assistant take multi-colour ones, and P06 drags and drops less as the work piles up. P13's hand allocations are disagreements with over-stocked cards ('why would I spend more resources'), and they say 'I always let the agent do it first' on the tactical step, yet none of their five plans after a hand allocation used Suggest. Their words do not explain their gap. **Leaning the other way** (P20, P03, P11, P31): their hand allocations are of a different kind. P20 builds by hand when the card cannot be met ('I've run out of blues') and then presses Suggest because the mission has many tasks. P03 works every allocation out first and checks the card against it. P11 rarely presses Suggest at all and does not want 'to be reduced to pressing allocate, suggest, deploy'. P31 lets Suggest get 'the majority of the way' whatever happened at the strategic step. **Only one route**: P21, who almost always allocated by hand, gives the clearest account of carrying a plan from one step to the next ('it's going to disrupt my thought process'). The card-only people (P07, P12, P14, P25, P27, P34) mostly give speed and urgency as the reason; P14 used the assistants 'for almost every allocation'.

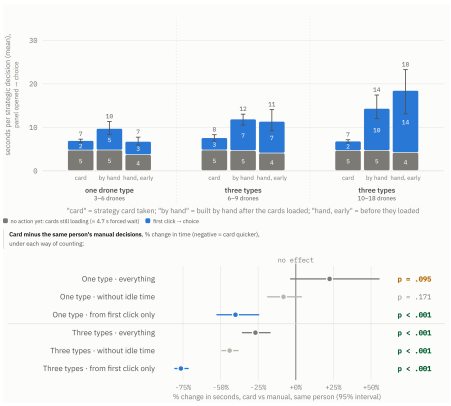
### 7.3.5 Why the rule makes sense: the card pays only on big strategic jobs; Suggest is not slower

[Pass 5: extra: may make it in]

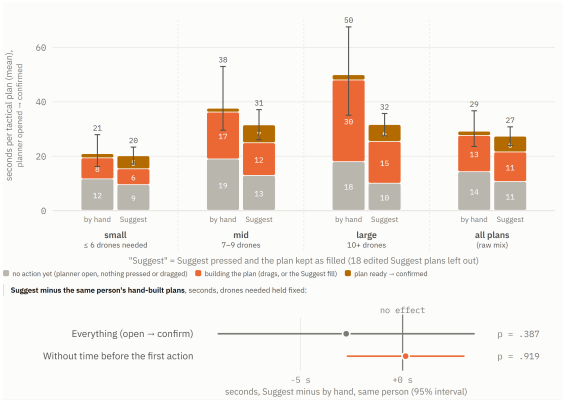
Each decision's time split into the time before the operator's first action and the time after it, by mission size, as stacked bars. **First action** = the first click on that decision: highlighting a card or changing a manual count (strategic; taking a card always needs a highlight first), pressing Suggest or dragging a drone (tactical). Before it, the strategic panel has two parts: the cards still loading (a forced wait of about 4.7 s; deciding by hand early can cut it short) and **idle time** with the cards on screen. The "Time counted" toggle drops the idle time, or everything before the first click. Strategic: counting everything, a card is much quicker than a manual build on three-type missions (−27%) and no quicker on one-type ones (+22%, not reliable). Card-takers sit idle about as long as manual builders (7 s vs 9–11 s on average), so dropping idle time makes the card look better everywhere; on one-type missions it then ties with a manual build (−8%), and on three-type missions the card is 44% quicker. Deciding before the cards load (7 s) beats the card when everything is counted and is level with it once idle time is dropped, on one-type missions. Tactical, on all 627 plans: by hand and with Suggest take about the same planner time (29 vs 27 s on average), and the same with the idle time removed (15 vs 17 s); held to the same person and size the difference is under 1 s. On large missions hand-building takes longer (32 vs 22 s without idle time), but the size × Suggest term is not reliable. So the speed story is clear on the strategic tier, and on the tactical tier Suggest saves no planner time. Its time saving is in carrying out (next section). (Figure 18)



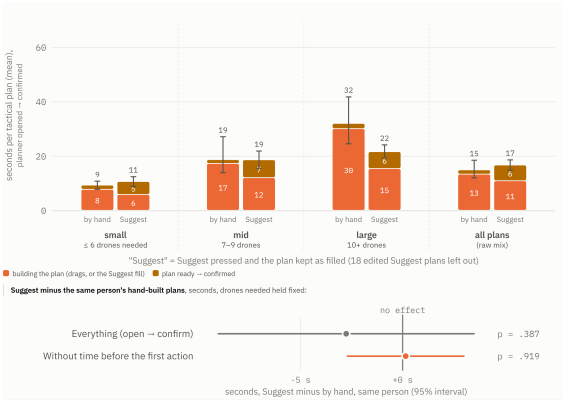

**(a)** Strategic tier, all time counted.




**(b)** Strategic tier, idle time removed.




**(c)** Tactical tier, all time counted.




**(d)** Tactical tier, idle time removed.

**Figure 18: Why the rule makes sense: the card pays only on big strategic jobs; Suggest is not slower**

**Key numbers.**

- Strategic, card vs manual, same person, % change in time: everything: one-type +22% [−3, +55], p = .09; three-type −27% [−35, −17], p < .001

- Without idle time: one-type −8% [−19, +4], p = .17; three-type −44% [−49, −38], p < .001. From the first click only: one-type −40%, three-type −76% (both p < .001)
- Strategic means (s), loading + idle + acting: one-type card 4.7 + 6.5 + 2.2 = 13.3; by hand after cards 4.7 + 8.6 + 5.0 = 18.2; before cards 3.7 + 0.7 + 3.0 = 7.4. Three-type 10–18 drones: card 4.6 + 7.3 + 2.1 = 14.0; by hand 4.6 + 10.1 + 9.7 = 24.4
- Tactical means (s), idle + building + ready → confirmed: all plans by hand 14.2 + 13.3 + 1.6 = 29.1 (144) vs Suggest 10.6 + 10.9 + 5.8 = 27.2 (465); large missions 17.9 + 30.0 + 1.9 = 49.8 (22) vs 10.0 + 15.3 + 6.2 = 31.5 (159)
- Tactical, Suggest minus by hand, same person, drones needed held: everything −2.8 s [−9.0, +3.5], p = .39; without idle time +0.2 s [−2.7, +3.0], p = .92; Suggest × size (without idle) −1.3 s per doubling, p = .47
- Tactical plans idle for more than 10 s: 54% had another mission's strategic panel opened before the first action (176 plans); all plans 22%

**Method.** **Rows**: strategic, the 606 card decisions (the last panel opening before the choice, as everywhere in this finding); tactical, the 627 non-recovery plans with a known mission size (kept Suggest plans vs hand-built; the 18 edited Suggest plans are in the table only). **Segments**, from the event log: strategic, panel opened → first click (split at the moment the cards appeared: before = "cards still loading", after = "idle, cards on screen") and first click → choice; tactical, planner opened → first Suggest press or drag ("idle"), → plan ready (last drag, or the end of the one-entry-per-second Suggest fill), → confirmed. These are the same phases as the next section, which adds carrying out. Bars are means (means add up, medians do not), whiskers 95% intervals of the bar's total from resampling people. Sizes: strategic, one-type missions (3–6 drones) and three-type missions of 6–9 and 10–18 drones; tactical, drones needed ≤ 6, 7–9, 10+. **Models**: strategic, Gaussian GEE on log seconds, card × three-type, all manual decisions pooled as the comparison, the person's own card rate, drones, game, session and minute held fixed, run once per way of counting; tactical, GEE on seconds, Suggest split within/between person, drones needed, game and session fixed. **Caveat**: idle time mixes reading the cards or plan with being busy elsewhere, and the log cannot tell them apart. For tactical plans idle for more than 10 s, 54% had another mission's strategic panel opened in that time, so long idle time is often attention elsewhere.

**What participants said.** Speed is the most common reason given for using the assistants: 15 of 33 coded C04. Many add that the plan is not better, just faster (P04, P17, P11). The card's loading wait is the main complaint. Six were coded C06, and in the think-aloud P18, P21, P29 and P32 go manual because they 'can't wait' for the cards, mostly on small missions. That matches the finding that the card pays off only on big strategic jobs. P20 first said the strategic assistant saved them more time, then corrected themself on remembering its loading stage.

**Quotes selected for this section.**

*"I think the key strength when I [am] using the agent is it can help me to save the time and energy to those specific deployments" — P28 (debrief 0:32:10)*

*"the point is, it takes me so much time to untangle that, that maybe it probably will not be faster if I do it myself manually. So as long as the AI makes sure that everything is covered once." — P17 (think-aloud s1 0:07:07)*

*"Manually, [I] can't wait for the processing, to be honest." — P21 (think-aloud s1 0:09:32)*

*"I think even if the tactical agent gave me a dodgy suggestion and the mission failed and I lost points, I probably would have gained more points from the time saved..." — P11 (debrief 0:30:11)*

*"The top advantage in this situation was the speed of response." — P31 (debrief 0:29:10)*

*"I think I was definitely a bit slower than the, than the algorithm there." — P22 (think-aloud s1 0:07:11)*

*"You know, it's quicker to get a suggestion, but it is less clicking, and it seems to do what I would do." — P32 (think-aloud s1 0:10:07)*

*"So, it just seems quicker to just send some myself than wait for the recommendation. I might just be impatient." — P32 (think-aloud s1 0:03:09)*

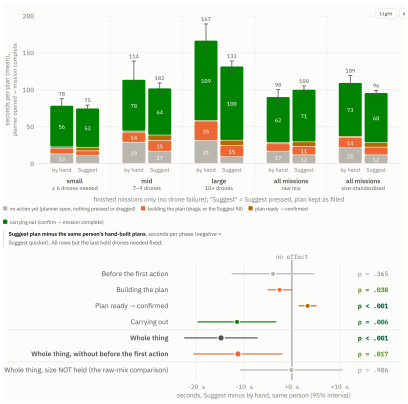
*"So, when we are trying to do with the tactical screen, I find the agent is quite slow." — P27 (debrief 0:29:10)*

*"Yeah, when I located, I think sometimes the AI will take a little longer. And I just mess up my strategy" — P26 (debrief 0:28:37)*

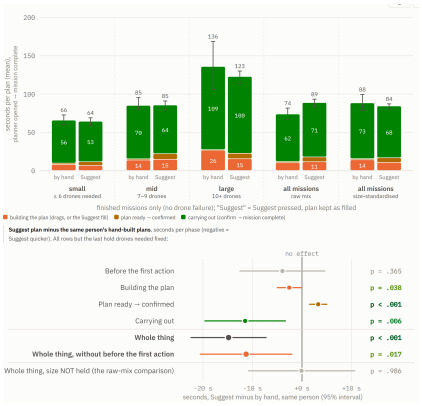
### 7.3.6 Where the tactical time goes: hand-built and Suggest plans are close, with a small edge to Suggest in carrying the plan out

[Pass 5: for the paper]

Each tactical plan split into four phases: before the first action (idle, or busy elsewhere), building the plan (drags, or the one-entry-per-second Suggest fill), plan ready → confirmed, and carrying out (confirm → mission complete), for every size group side by side; the toggle leaves out the time before the first action. **The headline is how close the bars are.** The operator's own time on a plan is the same either way: Suggest saves about 3 s in building and gives it back waiting between ready and confirmed (net about +1 s). The only reliable difference is in carrying the plan out, about 11 s sooner for a Suggest plan (roughly a tenth of a mission's time), so the assistant's plans are very slightly better rather than much faster to make. Read the large group with care: only 6 hand-built large plans finished cleanly. The raw "all missions" bars look the other way (hand-built 90 s, Suggest 100 s); that is a mix effect, not a contradiction: 78% of finished hand-built plans are small missions, against 42% of Suggest plans. Given the same mix of sizes, hand-built plans take 109 s and Suggest 96 s. So the common feeling that the assistant is "faster" is only weakly supported: it does not save the operator time, and the plans it makes finish a little sooner. Each tier's use still looks sensible: Suggest is used where plans are big, which is where its small edge in execution adds up. And relying on the assistants more does not make an operator do better: within a person, card missions are completed in full no more often than their own hand-built ones (odds ×0.81, p = .27; raw 60% vs 62%), and session score is flat in sessions where they rely more (−0.10 SD per 25 points of card uptake, p = .61). The ready → confirmed gap is when operators turn to other missions: 45% of Suggest plans had another mission's strategic panel opened while the planner was open, against 19% of hand-built ones. (Figure 19)




**(a)** All four phases.




**(b)** Without the time before the first action.

**Figure 19: Where the tactical time goes: hand-built and Suggest plans are close, with a small edge to Suggest in carrying the plan out**

**Key numbers.**

- Suggest minus hand-built, same person, drones needed held (315 finished plans): before the first action −4.0 s [−12.5, +4.6], p = .37; building −2.6 s [−5.0, −0.1], p = .038; ready → confirmed +3.3 s [+1.5, +5.0], p < .001; carrying out −11.5 s [−19.6, −3.4], p = .006; whole −14.8 s [−22.4, −7.3], p < .001
- Whole thing without the time before the first action: −11.3 s [−20.5, −2.1], p = .017; whole thing with size not held: −0.1 s [−10.8, +10.6], p = .99
- Means (s), hand vs kept Suggest: small 78.5 vs 74.8 (63 vs 97 plans); mid 113.6 vs 102.0 (12 vs 59); large 166.6 vs 131.6 (6 vs 75); all, raw 90.2 vs 100.2; all, size-standardised 109.3 [102.7, 119.4] vs 95.7 [91.9, 99.1]
- Size mix of finished plans: hand-built 78% small, 15% mid, 7% large (mean 5.7 drones needed); Suggest 42%, 26%, 32% (mean 8.8)
- All 627 plans, first three phases: before −2.9 s, p = .29; building −3.2 s [−6.2, −0.1], p = .04; ready → confirmed +3.3 s [+2.0, +4.6], p < .001
- Carrying out, including missions with a drone failure (378): −8.1 s [−17.1, +0.9], p = .08: same direction, less certain
- Projected completion vs the assistant's own plan at open, median: hand-built +10.3 s [3.7, 28.3] (144 plans), kept Suggest +1.6 s [0.8, 4.0] (468), edited Suggest +38.7 s (18)
- Interleaving, round-1 specification re-run: odds ×4.83 [2.24, 10.40], p < .001 (630 plans, 33 people); raw 45% vs 19%; 28% vs 11% actually allocated another mission
- Operator time, building + ready → confirmed, Suggest minus hand-built, same person, size held: −2.6 s + 3.3 s ≈ +0.7 s (negligible)
- From the retired "who does better" finding: within a person, card missions vs own manual missions, completed in full odds ×0.81 [0.55, 1.18], p = .27 (raw 60% vs 62%); session score per +25 points of card uptake −0.10 SD [−0.47, 0.27], p = .61. Between people, heavier card users complete more (×1.32 per 25 points, p = .004): the difference is in the people, not the advice

**Method.** **Rows**: confirmed non-recovery tactical plans. **Phases** from the event log, the same as in section 4: planner opened (*tactical_opened*); first action = first Suggest press or first drag; plan ready = last drag, or for a Suggest plan the end of its fill (last press + 1 s per drone-task entry), whichever is later; confirmed (*tactical_confirmed*); mission complete (*mission_completed*). **Carrying out** can only be measured when the mission finished in full with no drone failure: 315 of 630 plans; the chart shows those (81 hand-built, 231 kept Suggest; edited Suggest plans left out, 3 survive). Size groups by drones needed: small ≤ 6, mid 7–9, large 10+. Bars are means, whiskers 95% intervals of the bar's total from resampling people. **Size-standardised**: each route's mean in each size group, weighted by the share of all finished plans in that group (small 51%, mid 23%, large 26%), so both routes get the same mix; this answers "how long would each route take on the same missions?". Only 6 large hand-built plans finished cleanly (5 people), so that bar and the standardised hand-built bar are the least certain. **Model**: Gaussian GEE of seconds in each phase on "Suggest pressed" (split within/between person), log drones needed, game and session. Seconds add up, so the phase differences sum to roughly the whole. The bottom row drops drones needed from the model: still within person, but with size not held the difference vanishes (−0.1 s), which shows the raw reversal comes from size, not from who uses which route. **Projected completion** and **interleaving** as before.

### 7.3.7 What operators say

[Pass 5: for the paper]

Ten of 33 state the rule unprompted (C03: small/simple myself, big/complex to the assistant); 15 value the assistant for speed rather than accuracy (C04, the most common reason given); 8 describe working on something else while it runs (C05). The behaviour matches the words. People who state the rule press Suggest less overall (59% vs 86% of plans) and their size slope is, if anything, steeper on the tactical tier ($\times$1.43 steeper, p = .10) — not reliably different from everyone else's, so stating the rule is not what makes someone follow it.

**Key numbers.**

- C03 10 of 33; C04 15 of 33; C05 8 of 33
- C03 vs others, size slope ratio: strategic $\times$1.28 [0.69, 2.38], p = .43; tactical $\times$1.43 [0.93, 2.20], p = .10

**Method.** Codes from the qualitative coding (docs/analysis/qual/codes_matrix.csv); excerpts are pulled by code on the page. Slope comparison: the within-person size model with a size $\times$ C03 term (ratio of the two groups' odds ratios).

**What participants said.** 10 of 33 state the size rule unprompted (C03), 15 value the assistant for speed rather than accuracy (C04) and 8 work in parallel (C05). The clearest statements tie all three together: small jobs are quicker by hand, big ones are a 'computation problem', and the wait frees attention for the next mission. Dissent from P30, P20 and P29 is in the first section. P32 said their use followed workload, not how hard a mission looked.

### 7.3.8 In their words: working in parallel while the assistant plans

[Pass 5: extra: may make it in]

To close the finding: what people said about pressing Suggest and turning to other work while it fills in. It sits here rather than under the time-phase chart because it is about how people used the assistant, not about the chart's numbers.

**What participants said.** 8 of 33 describe starting Suggest and working elsewhere while it fills in (C05). The think-aloud shows more of it (P19, P32, P36). Nobody said a Suggest plan finishes the mission sooner. Two people said the opposite about plan quality: P31 adjusted a Suggest plan they thought visited tasks in a poor order, and P21 found it lacked the cross-task redundancy they built themself.

## 7.4 Manual allocations are anchored on the cards — and the 'trimming small missions' was mostly dropping the illogical spare

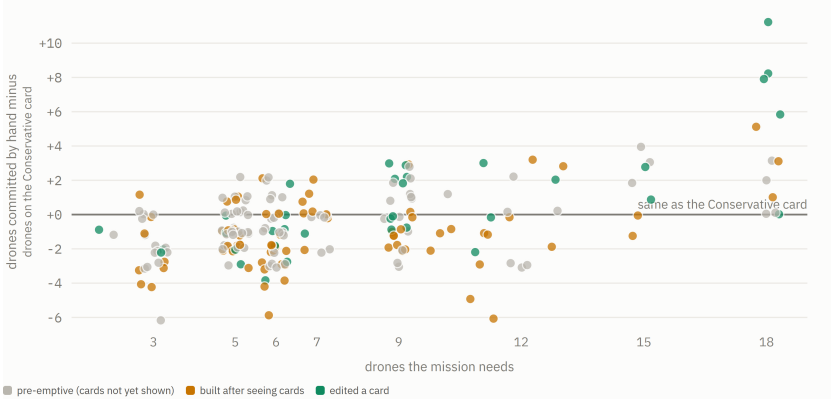
**Summary.** When operators allocate by hand after seeing the cards, their numbers track the cards: one more drone on the Conservative card means about half a drone more in the manual build, beyond what the mission needs. Operators who decided before the cards appeared show no such link — a clean control, so this is anchoring, not both tracking the mission. Manual builds sit at or below the cards (44% below both, 33% equal to one, 14% between, 8% above). The earlier "trim small missions, pad big ones" mostly dissolves: on one-type missions the 'trim' is almost entirely dropping the Conservative card's spare drones of types the mission does not need.

*Strength of evidence (register rating): moderate. Your verdict: keep.*

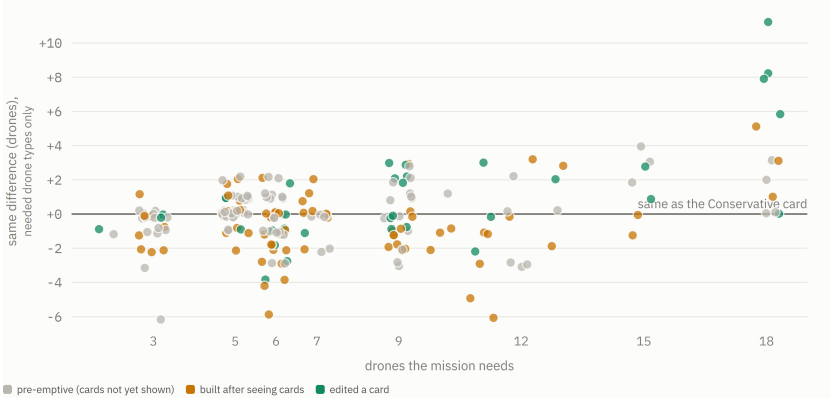
### 7.4.1 Each manual decision against the Conservative card

[Pass 5: for the paper]

Every manual allocation (209) as one dot: drones committed minus drones on the Conservative card, against the drones the mission needs. Switch to "needed types only" and the band of one-type missions moves up to zero: on the types the mission actually needs, manual builds on small missions match the Conservative card (+0.05 drones on average); the −1.2 drones on all types is the card's spare of each unneeded type being left off. On big three-type missions manual builds sit about one drone above the Conservative card (and still far below the drones the mission lists, since drones are reused across tasks in series). (Figure 20)




**(a)** All drone types.




**(b)** Needed drone types only.

**Figure 20: Each manual decision against the Conservative card**

**Key numbers.**

- Mean manual − Conservative: −0.55 drones overall = −0.51 on unneeded types + −0.04 on needed types
- One-type missions: −1.17 overall, +0.05 on needed types, −1.22 on unneeded; three-type with 10+ drones: +0.95
- Per doubling of mission size the difference grows by +1.51 drones [0.53, 2.49] (after seeing the cards) and +1.09 [0.47, 1.72] (pre-emptive): the size pattern is just as strong in people who never saw the cards, so it is not a reaction to the card
- Needed types only, after seeing the cards: +0.93 per doubling [−0.18, 2.04], p = .10

**Method.** Rows: strategic choices of type manual. The cards' true asset counts per colour come from the last *strategic_modal_opened* before the choice; the operator's counts from *assetsChosen*. "Needed types" = colours with a primary requirement in the mission. Kinds: pre-emptive (decided before the cards loaded — the cards existed in the log but were never on screen), edited a card (started from a card, then changed it), built after seeing the cards. Size slope: Gaussian within-person GEE of the difference on log2 drones needed.

**What participants said.** The think-aloud shows what was happening on the cards. People trimming small missions mostly said they were removing the card's drones of types the mission did not need (P10, P19, P21, P29, P13). Others cut one or two drones to protect the reserve (P24, P28). P20 shows the anchoring directly: they take the Aggressive card's numbers and does 'one less than each of those'. On big missions short of drones, P10 'send[s] everything I have', above the cards. One dissent: P29 found their own leaner builds sometimes had to be aborted.

**Quotes selected for this section.**

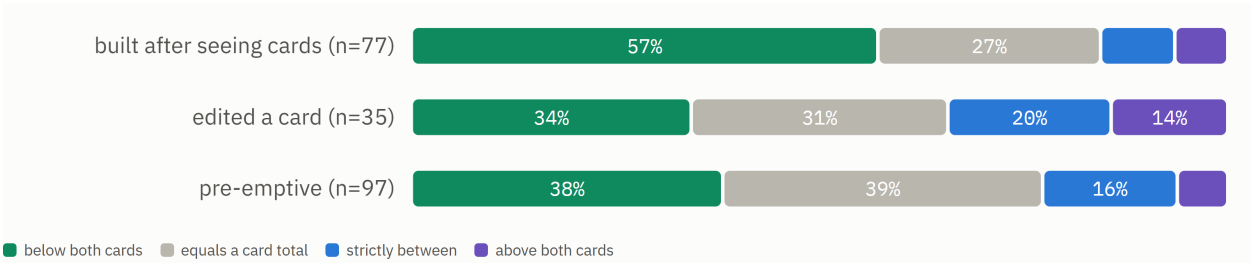
> *"All right. Let's just send everything I have. Edit, give me five red ones. Five green ones. Hit it." — P10 (think-aloud s2 0:17:46)*

> *"I'm trying to reserve more and more drones in the base, rather than send them out to finish the task quickly, because I cannot predict when the next task will appear." — P28 (debrief 0:28:05)*

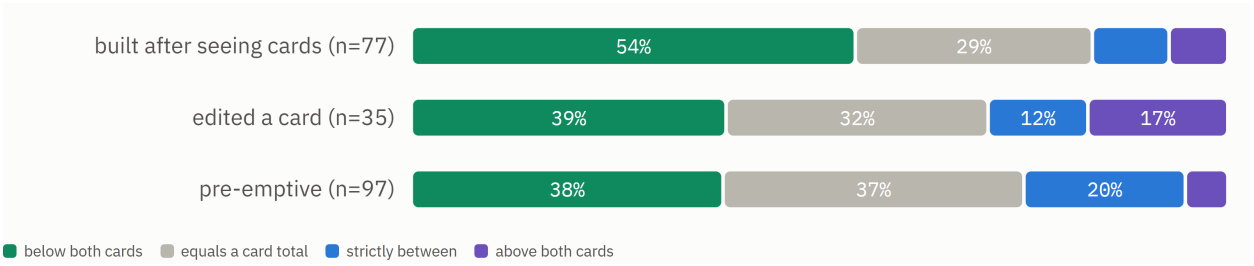
### 7.4.2 Where manual builds sit relative to the two cards

[Pass 5: for the paper]

Mostly below both cards or equal to one of them, rarely above. Pre-emptive builds — made without seeing the cards — land equal to a card total more often than builds made after seeing them (39% vs 27%), so 'equal to a card' is partly just the obvious allocation, not copying. Part of that gap is the missions: pre-emptive builds were on smaller, more often one-type missions. Balanced for the mission (second view), it is 37% vs 29%, too close to tell apart. (Figure 21)




**(a)** As built.




**(b)** Balanced for mission size, drone types, the gap between the cards, game, session and minute.

**Figure 21: Where manual builds sit relative to the two cards**

**Key numbers.**

- All 209: below both 44%, equal to a card 33%, strictly between 14%, above both 8%
- Built after seeing (77): 57% below, 27% equal, 9% between, 6% above; pre-emptive (97): 38%, 39%, 16%, 6%
- Mission mix: pre-emptive builds needed 6.9 drones on average (after seeing 7.6, edited 9.3) and 52% were one drone type (38%, 23%).
- Balanced: after seeing 54% below, 29% equal, 9% between, 7% above; pre-emptive 38%, 37%, 20%, 5%; edited a card 39%, 32%, 12%, 17%.
- Pre-emptive minus after seeing, balanced: equal to a card +8 points [−10, +23] (raw +12); below both −16 [−33, +4] (raw −19).

**Method.** Each manual total compared with the two card totals (all types): below both, equal to either, strictly between, above both. Counts per kind of manual decision. **Balanced view**: one logistic model per position (below both / equal / between / above) on the kind of build + drones needed + number of drone types + the gap between the two card totals + game + session + minute. Every one of the 209 builds is predicted as each kind in turn and averaged, so the three kinds are compared on the same mix of missions; the four shares are rescaled to 100%. Intervals: people resampled 1,000 times, models refitted. The person is not held fixed (too few builds each).

**What participants said.** 6 of 33 wanted something between the two cards (C08), and P17 says in play that they aim 'between' them. Most of what people say describes trimming a card or matching it, which fits 'below or equal'.

'Equal to a card' was often their own count arrived at first: P30 and P03 work out a number while the cards load and see whether a card matches, and P32 stopped waiting once the cards matched their count.

**Quotes selected for this section.**

> *"But whenever I clicked manual, it's either been for the cause of 'they are sending too much', or I want it to be somewhere between those two levels." — P17 (debrief 0:23:38)*

> *"I felt like sometimes I could find like a middle ground that had like maybe a little bit of gambling in there, essentially, that you wouldn't have the necessary coverage." — P22 (debrief 0:28:57)*

### 7.4.3 What operators say about building by hand

Four accounts of manual allocation sit alongside the numbers. Holding drones in reserve (C07, 6 people) fits the 'below both cards' mass — though in the logs these six are not lower than others on needed types (+0.36 vs the Conservative card). Wanting a middle ground or more options (C08, 6 people) fits the wish, not the behaviour: only 10% of their manual builds land strictly between the cards. Treating a card as a first draft (C09, 4 people) shows in the logs — 16 of their 33 manual decisions are edited cards. Testing the assistant against themselves (C10, 5 people) fits slower, deliberate manual builds (round 1: 20 s vs 12.6 s). All four describe manual as working *with* the advice, not rejecting it.

**Key numbers.**

- C07: 55 manual decisions from 6 people, mean +0.36 vs Conservative on needed types, 21 edited cards
- C08: 40 from 5 people, 10% strictly between the cards
- C09: 33 from 4 people, 16 edited cards
- C10: 40 from 5 people

**Method.** Codes from docs/analysis/qual/codes_matrix.csv; per-code manual numbers from the same rows as the dot chart (small groups, descriptive only).

**What participants said.** The four accounts are all present in people's own words. They hold drones back (C07, 6), want a middle ground (C08, 6), treat a card as a draft (C09, 4: 'you can fix bad, you can't fix nothing') and test themselves against the assistant (C10, 5). The think-aloud shows the draft habit more widely than coded: P10, P14 and P31 edit cards, and P21 and P31 use Suggest as a first draft on the tactical tier. The one person who tested themself said it ended in favour of the assistant (P29).

## 7.5 Practice makes operators faster, not better at choosing when to hand over

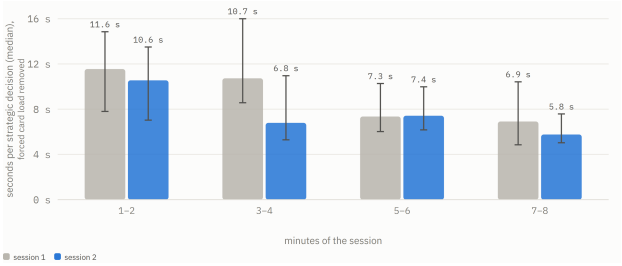
**Summary.** Decisions speed up within each session: about 6% faster per minute on both tiers, and not because the queue gets longer. On the strategic tier most of that gain is gone by the start of the next session (median 11.6 s at the start of session 1, 6.9 s at its end, 10.6 s at the start of session 2), so it reads as a within-session warm-up more than durable learning. On the tactical tier session 2 starts about where session 1 ended. The size rule does not sharpen with practice: the big-minus-small hand-over gap is the same or smaller in session 2. The six people who say they got better did improve their scores (all six, against 37% of others), but they did not come to use the assistants more, and across everyone a bigger score gain does not go with more use.

*Strength of evidence (register rating): moderate. Your verdict: keep.*

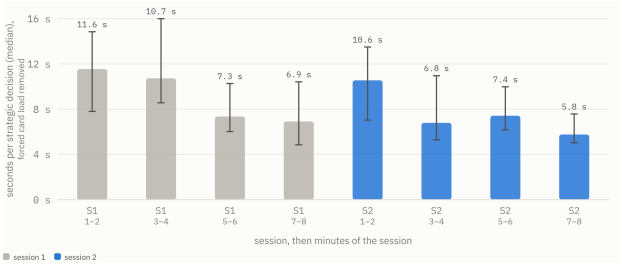
### 7.5.1 Faster within a session; on the strategic tier the gain mostly resets by the next

[Pass 5: numbers only]

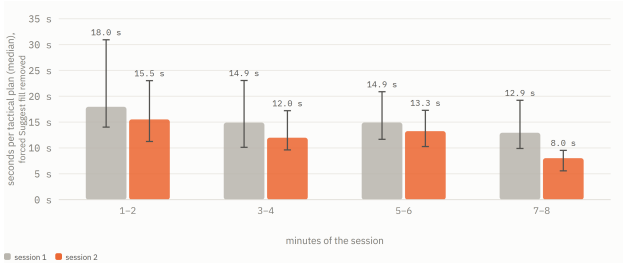
Median net decision time through each session, by pair of minutes. On the strategic tier decisions get much faster through a session (11.6 → 6.9 s in session 1) and then start nearly as slow again (10.6 s), before speeding up once more (5.8 s). The model keeps about half of the gain across the break, but uncertainly. On the tactical tier the first session's speed-up is weaker and session 2 starts about where session 1 ended, then speeds up again. Holding the number of waiting missions fixed does not remove the within-session speed-up, so it is not just time pressure from a longer queue. (Figure 22)




**(a)** Strategic tier, sessions side by side.




**(b)** Strategic tier, all of session 1 then all of session 2.




**(c)** Tactical tier, sessions side by side.

**Figure 22: Faster within a session; on the strategic tier the gain mostly resets by the next**

**Key numbers.**

- Strategic, medians: session 1 minutes 1–2 11.6 s [7.8, 14.5] → minutes 7–8 6.9 s [4.9, 10.2]; session 2 10.6 s [7.0, 13.5] → 5.8 s [5.0, 7.6]
- Strategic, model: end of session 1 −39% [−51, −24]; start of session 2 −20% [−37, +2], p = .07; start of session 2 vs end of session 1 +31% [0, +72], p = .053 (slower again)
- Tactical, medians: 18.0 s [14.0, 30.9] → 12.9 s [10.1, 19.2]; session 2 15.5 s [11.3, 21.9] → 8.0 s [5.6, 9.6]
- Tactical, model: end of session 1 −24% [−45, +4], p = .09; start of session 2 vs end of session 1 +2% [−33, +56], p = .92; session 2 slope −7% per minute vs −4% in session 1 (difference p = .26)

- Within-session slope with the queue held fixed: strategic −6% per minute, tactical −8% per minute (both $p < .001$)
- Overall (unchanged): strategic −5.7% per minute [−7.8, −3.6], session 2 vs 1 −19% [−30, −6]; tactical −5.7% per minute [−9.1, −2.2], session 2 vs 1 −24% [−36, −9]

**Method.** **Net time** = seconds to decide minus the forced wait (the card reveal for card choices; 1 s per entry for Suggest plans). Bars: medians of all decisions made in minutes 1–2 and 7–8 of each session, with 95% intervals from resampling people. **Model**: Gaussian GEE on log net time, clustered on person, with minute of session, session 2 and their interaction, plus route and mission size; game held fixed. From it: the end of session 1, and the start and end of session 2, each relative to the start of session 1, and the start of session 2 relative to the end of session 1 (a positive value = slower again after the break). A log-scale coefficient converts to "% faster". The start of a session is also the start of a game (first missions, short queue), so part of a within-session speed-up may be the session's own rhythm rather than learning.

**What participants said.** People describe getting faster and finding their rhythm: P10 'find your stride', P30 'got into the rhythm', P35 'getting familiar', P11 'get the hang of it quite quickly'. Several state a lesson from session 1 at the start of session 2: P22 'don't send everything straight away', P06 'the task will get a lot pretty quickly'. P18 is a partial dissent: the second session felt better prepared but more stressful.

**Quotes selected for this section.**

> *"there was definitely an improvement further throughout session one, and in session two, you really find your stride" — P10 (debrief 0:22:57)*

> *"i think the second scenario i used more personalised, i think it was easier because i had to i sort of got into the rhythm of having to anticipate other problems" — P30 (debrief 0:23:52)*

> *"Right. So now we also know from the last one, don't send everything straight away. Cause otherwise we end up, we're waiting for drones." — P22 (think-aloud s2 0:17:11)*

> *"And I could definitely feel that there is absolutely a learning effect." — P10 (debrief 0:22:39)*

> *"I think with having some experience, yeah, I was able to quickly understand this" — P05 (think-aloud s1 0:11:14)*

### 7.5.2 Does 'I got better with practice' match the logs? For score yes; not for speed or assistant use

[Pass 5: numbers only]

Six people say they got better, faster or more confident (C22: P10, P11, P15, P21, P30, P32). All six raised their within-game score from session 1 to 2, against 10 of 27 others. Their tactical planning sped up more than others' on average (5 of 6 faster), but not reliably; their strategic decisions did not speed up more. They also **did not come to use the assistants more**: strategy cards taken changed by −1 point (others −4) and Suggest by +11 points (others +7), with 3 of the 6 using each assistant more in session 2. Across all 33 people, a bigger score gain does not go with a bigger rise in either kind of use. So "got better" is right about outcomes, but it is not "relied on the assistants more". (Figure 23)

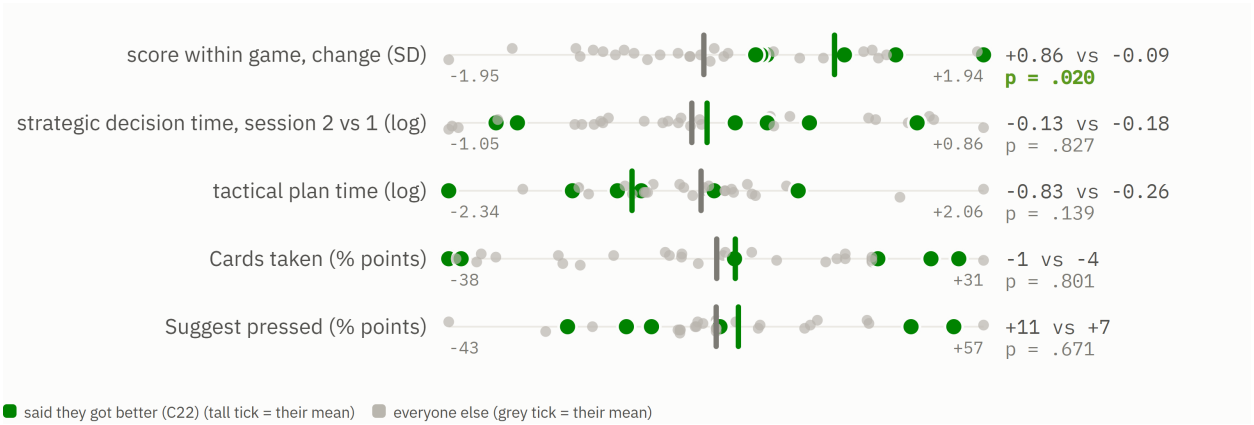



**Figure 23: Does 'I got better with practice' match the logs? For score yes; not for speed or assistant use**

**Key numbers.**

- Score change: C22 +0.86 SD vs others −0.09, p = .020
- Tactical time change (log): −0.83 vs −0.26, p = .14; strategic: −0.13 vs −0.18, p = .83
- Change in cards taken (game-adjusted): C22 −1 point vs others −4, p = .80; Suggest pressed: +11 vs +7 points, p = .67
- Used more in session 2: cards 3 of 6 C22 vs 12 of 27 others; Suggest 3 of 6 vs 19 of 27
- Everyone: score change vs change in card-taking $\rho = -.07$ (p = .71); vs change in Suggest $\rho = -.18$ (p = .32)

**Method.** Per person: session 2 minus session 1 in (a) score standardised within game, (b) log median net strategic time, (c) log median net tactical time, (d) share of strategy cards taken and (e) share of plans with Suggest pressed. For (d) and (e) each session's rate is first taken relative to that game's average rate across everyone, so playing a different game in session 2 does not count as a change of habit (raw changes agree). Group mean vs the other 27, two-sided permutation p (20,000 reshuffles). The across-everyone check is a Spearman rank correlation between score change and change in use (33 people). With P01/P02 left out of the game means, P21's score change is slightly negative (five of six, not six).

**What participants said.** 6 of 33 say they got better (C22). Their reasons are understanding the task (P32), experience (P11, P15) and optimising how they worked with the assistants (P21), not leaning on the assistants more. Others outside C22 describe practice changing how much they use the assistants, in both directions. P06 and P31 relied more and checked less, P28 expected to rely less once familiar, and P35 used the cards less after spotting errors. That mixed picture fits the logs: no overall rise in use with practice.

## 7.6 A busier moment does not push people onto the assistants — even though they say it does

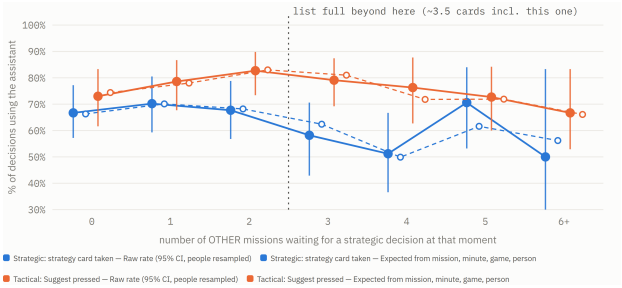
**Summary.** Comparing each operator with themselves, having more missions waiting does not change how often they take a strategy card or press Suggest: the line is flat once we account for what each decision was (mission size and drone types), when it came in the session, which game it was and who made it. This is "no evidence of an effect", with enough data to rule out large effects (strategic: at most about +15 points of card-taking going from 0 to 5 waiting). Nine of 33 said in the debrief that they hand over when overloaded; their own behaviour moved with the queue no more than anyone else's. In play they said overload-type things at slightly busier moments, but their reliance around those moments did not change.

*Strength of evidence (register rating): a null result. Your verdict: core.*

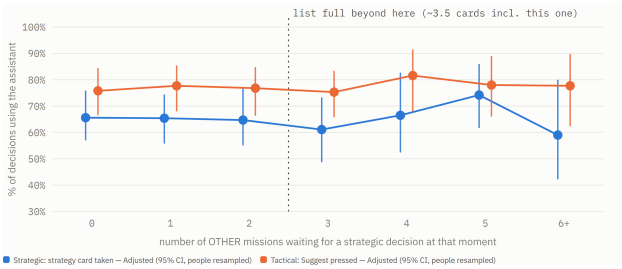
### 7.6.1 Queue depth against reliance — raw, and adjusted

[Pass 5: for the paper]

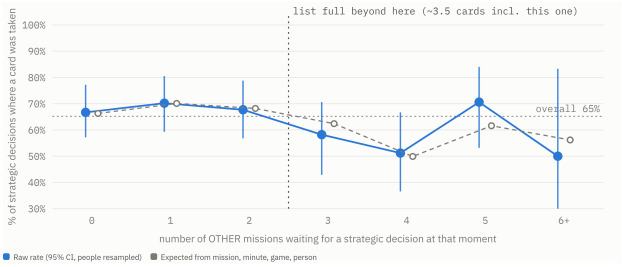
Each point is the share of decisions where the operator took a strategy card (or pressed Suggest), grouped by how many OTHER missions were waiting for a strategic decision at that moment. Whiskers are 95% intervals. The dotted vertical line marks where the queue list runs out of room (about 3.5 cards visible at rest, counting the one being opened): beyond it, waiting missions are only on the strategic map. Nothing changes at that line either. (Figure 24)



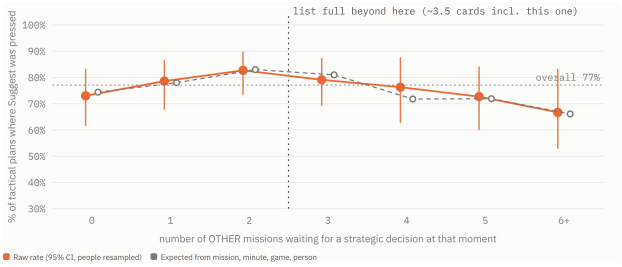

**(a)** Both tiers together, raw with their expected rates (dashed).




**(b)** Both tiers together, adjusted.




**(c)** Strategic tier, raw with the expected rate (dashed).




**(d)** Tactical tier, raw with the expected rate (dashed).

**Figure 24: Queue depth against reliance — raw, and adjusted**

**Key numbers.**

- Strategic raw: 67% (0 waiting), 70%, 68%, 58%, 51%, 71%, 50% (6+); adjusted: 66%, 65%, 65%, 61%, 67%, 74%, 59% — all intervals overlap.
- Tactical raw: 73%, 79%, 83%, 79%, 76%, 73%, 67%; adjusted: 76%, 78%, 77%, 75%, 82%, 78%, 78%.
- The queue rises with the minute (Spearman $\rho = .49$) but hardly with mission size ($\rho = -.08$): the adjustment mostly removes WHO reaches deep queues and WHICH GAME they are in.
- Deep queues are a Strategic Heavy phenomenon: 90–97% of decisions at 4+ waiting are in those two games.

**Method.** **Rows:** 564 strategic decisions on original (non-residual) missions and 630 tactical plans, 33 people (P09 excluded). **Queue** = other missions that had arrived and not yet been given a strategic choice when the decision panel opened, rebuilt from the event log (agrees with the logged count for 100% of decisions). **Expected**: a logistic model predicts each decision's chance of a card (or Suggest) from four things only — the mission (drones needed, number of tasks, number of drone types), the minute of the session, the game (one of four fixed scenarios), and the person (their own rate on all their OTHER decisions, so a decision never predicts itself). The queue is deliberately left OUT. **Adjusted** for a bin = the overall rate + the average of (observed − expected) in that bin. In plain words: 'given who made these decisions, on which missions, at what point in which game, how much more or less did they rely on the assistant than we would expect?' If load mattered, the deep-queue bins would sit above what their missions and people predict; they do not. **Worked example (strategic, 4 waiting, 41 decisions):** raw 51% looks like a drop from 67% at 0 waiting. But 90% of those decisions were in the two Strategic Heavy games, on missions averaging 6.9 drones (vs 8.7) with fewer three-type missions (54% vs 66%), made by people whose usual rate is 56% (vs 66%). The model expects 50%; observed is 51%; adjusted ≈ 67%. **Intervals**: people resampled with replacement 400 times, the expected-model re-fitted each time, so the interval includes the uncertainty of the adjustment itself.

**What participants said.** 9 of 33 say they hand over more when busy (C11). Almost all of them mean the harder session or 'when workload gets higher', not the moment a queue builds. One clear voice goes against it in play: P18, asked during a stressful session, said 'I think less… I'm too focused to look at it'. P21 felt pressure pushed them past the strategy card rather than onto it. P29 and P05 describe pressure as the thing that tips them towards the assistant. That is the claim the adjusted data do not bear out.

### Quotes selected for this section.

*"You can have a lot of tasks to be overloaded and try and offload some of that work to AI and then for the very complex stuff, maybe you don't." — P20 (debrief 0:28:46)*

*"So I think if, at first glance, I felt really overwhelmed by the amount of drones to be assigned. I would use the assistant" — P18 (debrief 0:26:17)*

*"there was no more active missions coming in, so therefore I could take more time and ensure that it was correct. But perhaps if I was more time pressured, then I'd ask it to suggest." — P05 (think-aloud s1 7:31)*

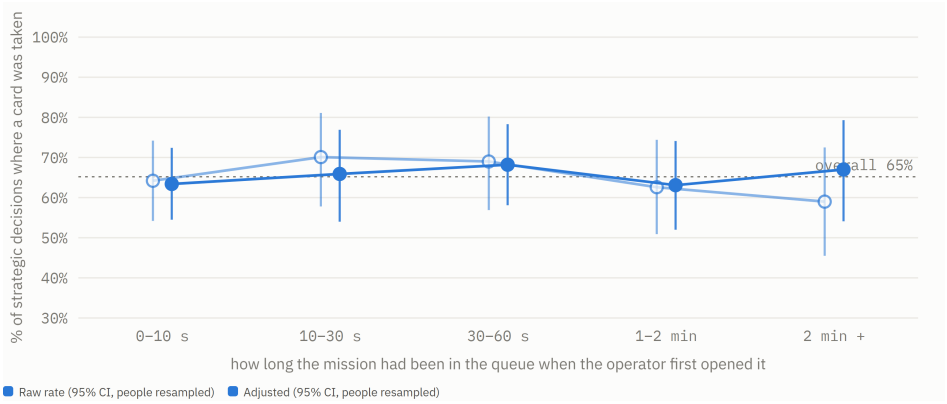
*"But in an urgent situation and the number of the swarm is not enough, so I need to consider about the strategies and I think I need the help of AI assistant." — P35 (debrief 0:28:19)*

*"I think the first one, I feel the pace is very, very fast. So, I can't do it very well in a very short time. So, I used a lot of AI suggestions." — P33 (debrief 0:27:33)*

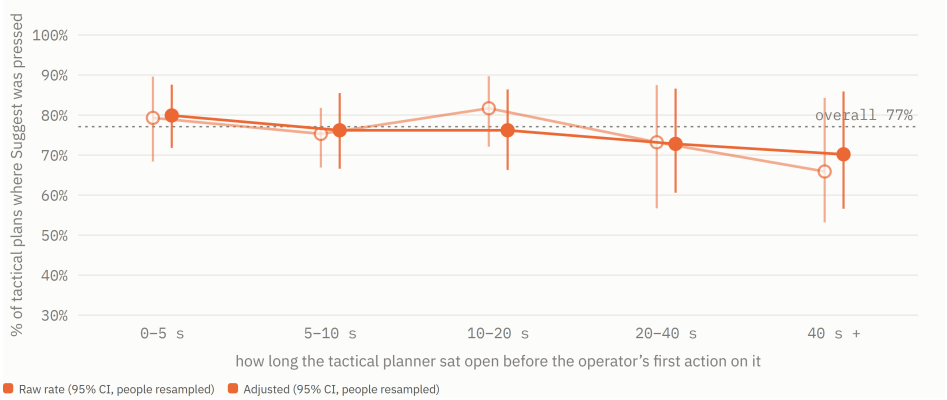
### 7.6.2 A mission left waiting is not handed over more often

[Pass 5: numbers only]

Do people hand a decision over when they realise they have left it waiting? Here each decision is grouped by how long its mission had been waiting: on the strategic tier, from the mission's arrival to the first time the operator opened it; on the tactical tier, from the planner opening to the operator's first action on it (the plan the operator left sitting while they did something else). **The lines are flat.** Strategic card use is 63–68% (adjusted) whether the mission waited 10 seconds or more than two minutes. Tactical Suggest use, if anything, drifts down when a plan was left open longer (80% → 70%), the opposite of the idea, but not reliably. (Figure 25)




**(a)** Strategic tier: how long the mission had waited in the queue.




**(b)** Tactical tier: how long the plan had been left open.

**Figure 25: A mission left waiting is not handed over more often**

**Key numbers.**

- Strategic (mission left in the queue): adjusted 63% at 0–10 s, 66% 10–30 s, 68% 30–60 s, 63% 1–2 min, 67% at 2 min+ (n 173, 117, 100, 91, 83). Within person, odds ×1.10 per doubling of the wait [0.96, 1.28], p = .17; holding the queue fixed ×1.10, p = .19.
- Tactical (plan left open): adjusted 80% at 0–5 s, 76% 5–10 s, 76% 10–20 s, 73% 20–40 s, 70% at 40 s+ (n 270, 182, 93, 41, 44). Within person, odds ×0.80 per doubling [0.60, 1.06], p = .12; holding the queue fixed ×0.76, p = .10.

**Method.** Rows: every strategic decision on a real (not re-queued) mission, 564 from 33 people, and every tactical plan, 630. Wait: strategic, mission arrival to the first time its strategy panel was opened; tactical, planner open to the first drag or Suggest. 'Adjusted' is the overall rate plus the average of (observed − expected) in that bin, where the expected value comes from mission

size and drone types, minute of the session, the game and the person's own rate on their other decisions (the same model as section 1). Intervals resample people. The odds ratio compares each person with themselves: log2 of the wait (+5 s), centred within person, in a model with the same controls, people clustered; a second version also holds the queue depth fixed, since long waits come at busy moments.

## 7.7 Busy operators speed up a little but do not change what they pick

**Summary.** The earlier story — that with a long queue operators abandon order and pick smaller missions — does not survive a fair comparison. Long queues happen almost only in the two Strategic Heavy games, where every mission is small, so the opened mission is smaller simply because everything waiting is. Compared with what was actually waiting, operators keep the same priorities at any queue length: a strong pull towards the oldest mission, a mild preference for larger ones, and only falling 'oldest-first' rates because chance falls as more missions wait. What does change is pace: a bit more decisions per minute when more is waiting (suggestive).

*Strength of evidence (register rating): mixed. Your verdict: keep.*

### 7.7.1 Order: still oldest-first well above chance

[Pass 5: for the paper]

The share of decisions that open the oldest waiting mission falls as the queue grows (59% with one other waiting, about 30% with five) — but so does the chance of hitting the oldest by picking at random (50% → 17%). Measured against chance, the pull towards the oldest stays roughly the same. 'One of the top 3 in the list' is near chance at every depth, so the list position itself does little beyond 'oldest first'. (Figure 26)

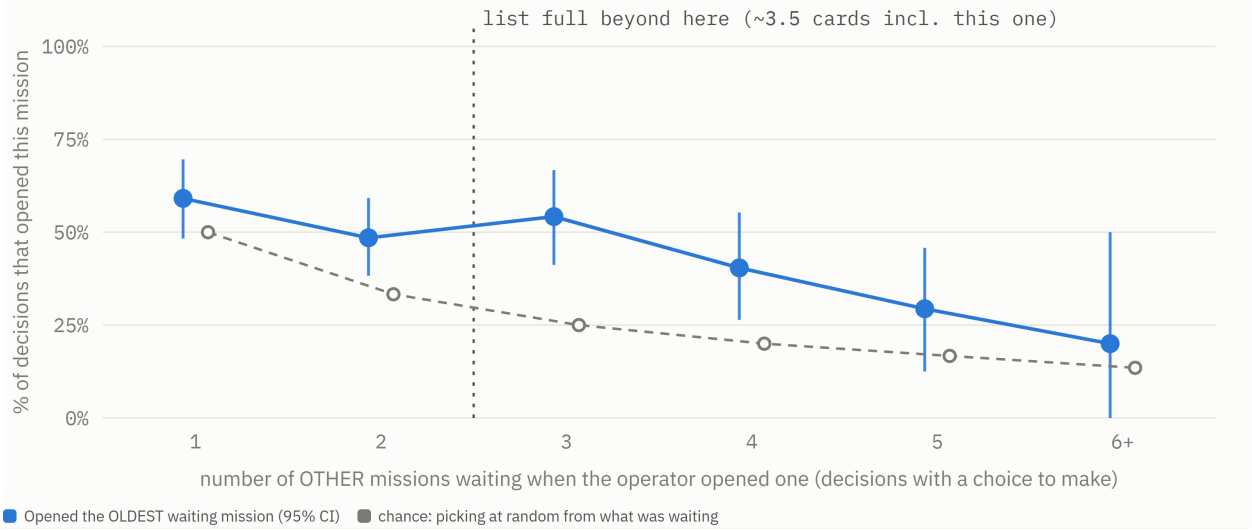



**Figure 26: Order: still oldest-first well above chance**

**Key numbers.**

- Opened the oldest: 59% (1 other waiting; chance 50%), 48% (33%), 54% (25%), 40% (20%), 29% (17%), 20% (14%).
- Within person, the excess over chance does not shrink with depth: +2.4 points per extra mission [−1.0, +5.7], p = .16.

**Method.** 414 strategic decisions where at least one other mission was waiting (33 people). For each, the set of missions waiting when the panel opened (arrival order from the log). Chance = 1 / (number waiting) for 'oldest'; 3 / (number waiting) for 'top 3'. Intervals: people resampled 400 times. Within-person test: (picked oldest − chance) against queue depth, person-centred, with minute and game. Caveat: the list could be re-sorted by score and that toggle was never logged.

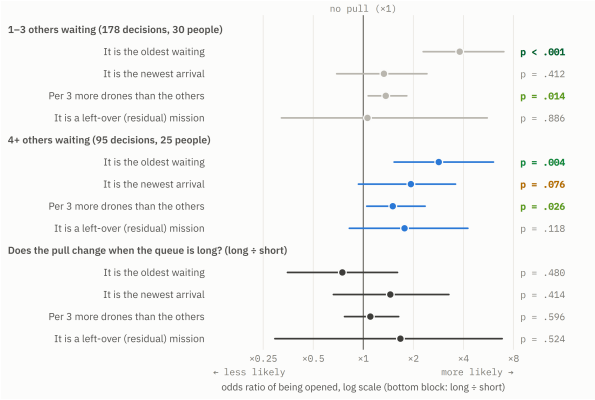
**What participants said.** Nobody says they took missions oldest-first. Five say, in play or after, that criticality comes first ('I'll go with the most critical first'): P20, P22, P31, P19 and P27. P27 adds 'can it be completed' as the first

filter. The order model has no criticality term, so this stated rule is untested. P19 wished for a sort control, and P27 for smart ordering, unaware that an order-by-criticality button existed.

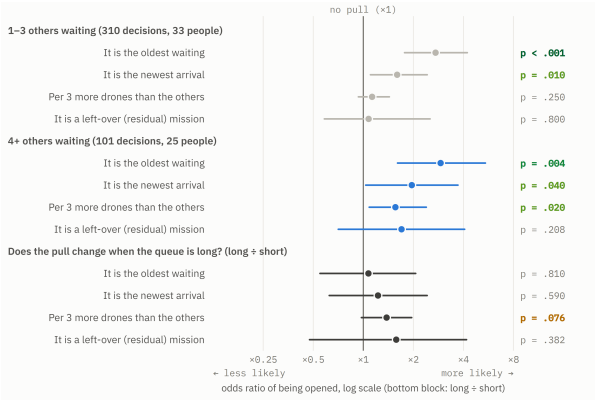
### 7.7.2 What makes a waiting mission get opened, short vs long queues

[Pass 5: numbers only]

Putting it together in one model of the choice among what was waiting: being the oldest roughly triples a mission's chance of being opened, larger missions are mildly preferred, and being the newest arrival helps a little. These pulls are statistically clear where marked, and none of them changes significantly between short (1–3 others waiting) and long (4+) queues: the "long ÷ short" rows all have intervals spanning 1. Under pressure, priorities do not visibly change, though with 95 long-queue decisions only a large change could have been detected. (Figure 27)




**(a)** Strategic Heavy games only.




**(b)** All four games.

**Figure 27: What makes a waiting mission get opened, short vs long queues**

**Key numbers.**

- Strategic Heavy games, short queues (178 decisions): oldest ×3.80 [2.29, 6.99], p < .001; newest ×1.33 [0.69, 2.41], p = .41; +3 drones ×1.36 [1.07, 1.82], p = .014; residual ×1.06, p = .89
- Strategic Heavy games, long queues (95 decisions): oldest ×2.84 [1.53, 6.06], p = .004; newest ×1.93 [0.94, 3.58], p = .076; +3 drones ×1.50 [1.05, 2.35], p = .026; residual ×1.77, p = .12
- Long ÷ short, Strategic Heavy games: oldest ×0.75 [0.35, 1.60], p = .48; newest ×1.45, p = .41; size ×1.10, p = .60; residual ×1.67, p = .52 (none differs)
- All four games: oldest ×2.72 (short) and ×2.92 (long), both p ≤ .004; newest ×1.60, p = .010 and ×1.96, p = .040; size ×1.13, p = .25 and ×1.56, p = .020; long ÷ short for size ×1.38 [0.97, 1.95], p = .076, all others p ≥ .38

**Method.** Conditional logit: each decision is a choice among the missions waiting at that moment; the model estimates how much each feature (oldest, newest, size relative to the set in units of 3 drones, residual) multiplies a mission's odds of being the one opened. Fitted separately for short (1–3 others) and long (4+) queues, in the Strategic Heavy games only (like with like) and in all four. Intervals and p-values: people resampled 1000 times (the short- and long-queue fits on the same resampled people), so they allow for each person making many decisions; p = twice the share of resamples on the far side of "no pull". The "long ÷ short" rows are the ratio of the two odds ratios with their own interval and p, i.e. a direct test of whether the pull changes with the queue.

**What participants said.** Besides criticality, people describe choosing by feasibility: they skip what the reserve cannot cover yet and come back (P17, P27, P30). P19 jumped to a newly arrived higher-priority mission, which matches the small 'newest' pull. Nobody mentions preferring larger missions, and nobody says their way of choosing changed when the queue was long.

## 7.8 The dip in strategy-card use belongs to the people who later said they noticed the illogical card, early in the session; it shows up whether or not they had met that card yet

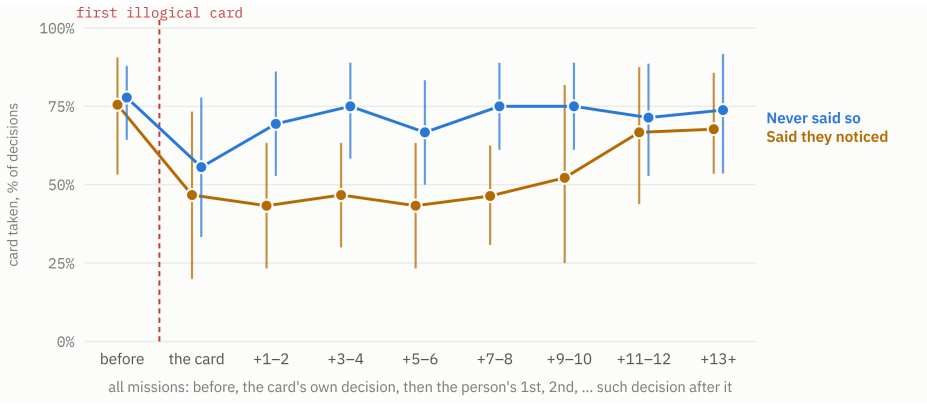
**Summary.** The Conservative card sometimes adds a drone type the mission does not use (22% of decisions, nearly all on one-type missions); all 33 people met it, usually on their 1st or 2nd decision. The 15 who said they noticed it take the cards less afterwards, and on the flawed one-type missions they learn to skip them (0 takes in 13 by their 4th–6th such mission). But the **dip** in their use is not tied to the card. Counted simply by decision number, noticers take the cards on 80% of their 1st session-1 decision and 46–50% of decisions 3–5, then 83% at decision 6, while never-said people rise to 89–94% in the same stretch. People who played the tactical-heavy game first had *not yet met* an illogical card in decisions 3–5 (95% of those decisions came before it), and their noticers dip just as much ($-21$ points per person, against $-32$ for those who had met it; game order makes no difference, p = .79). In that stretch noticers look at the cards and turn them down, on three-type missions too. Pooled over everyone the dip disappears, because the two groups move in opposite directions. Noticers climb back later in session 1 (+24 points per person) and start session 2 lower again (49%), but never-said people also start session 2 a little lower, so a weaker carry-over for noticers is not something these data can show (difference at the break $-6$ points [$-32$, +20]). Our best reading: "noticed" marks people who scrutinise the cards, and the dip is that scrutiny at the first busy stretch of the study, not a reaction to the illogical card. What triggers it at that moment we cannot say: mission mix, load, reserve shortage and game do not account for it. Suggestive (15 vs 18 people, window read off the curve). This finding also holds, sidelined, the flaw-visibility comparison and the recovery material merged in from two other findings.

*Strength of evidence (register rating): suggestive. Your verdict: core.*

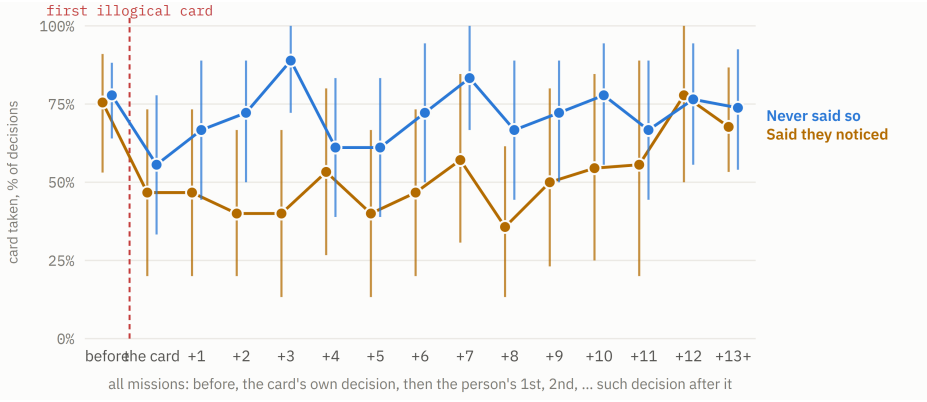
### 7.8.1 The event study, aligned on the first illogical card (every step, pairs or threes)

[Pass 5: for the paper]

Each person's decisions are numbered *within* the kind of mission plotted: their 1st, 2nd, … decision after the first illogical card. With every step shown, each point is then about 15 single decisions and its interval spans roughly 30 points either way. Pairs keep the shape with intervals about a third narrower, so pairs is the default. On **all missions**, noticers go from 76% before the card to 43–52% for their next ten decisions, recovering to 67–68% only from the 11th on. Never-said people stay at 67–75% throughout. Part of the noticers' low on all missions is the flawed one-type missions themselves, where they increasingly refuse the card. On **three-type missions**, noticers go 74% $\rightarrow$ 43% (1st–2nd) $\rightarrow$ 59–63% $\rightarrow$ 75%: a short dip and recovery. On one- and two-type missions almost nobody has a decision before the card, because the card only ever appears there, so only the after-card trend can be read. **The steps before the card** add little: most people have only one decision before it, so $-2$ and $-3$ are 8 decisions each ($\pm 35$ points). The last step before the card ($-1$) is lower than $-3$/$-2$ in *both* groups (64% and 56%), so it is not a noticer warning sign. **Read with the next section:** the aligned dip is not where it seems. (Figure 28)




**(a)** All missions, every pair of decisions, observed.




**(b)** All missions, every step.

**Figure 28: The event study, aligned on the first illogical card (every step, pairs or threes)**

**Key numbers.**

- All missions, noticers (15): 76% before (n 53) → card 47% (n 15) → pairs +1–2 to +9–10: 43%, 47%, 43%, 46%, 52% → 67% (+11–12) → 68% (13th+, n 31)
- All missions, never said (18): 78% before (n 54) → 56% at the card → 67–75% for every later pair
- All missions, before → after, adjusted: noticers −14 points, p = .08; never said +9, p = .14; difference −14 [−30, +1], p = .06
- Three-type, noticers by pair: 74% before (n 51), 43% (n 30), 59% (n 29), 62% (n 24), 63% (n 19), 75% (n 12)
- Why expected is lower for noticers: their own-rate term (take rate on all their other decisions) is 56% vs 70%; 79% of a noticer's decisions come at or after the card, where they took 50% per person vs 72% (p = .007); before the card the groups were alike (68% vs 63%, p = .71)
- Mission-only expected (own-rate term set to the average): after the card 68% noticers vs 69% never-said; before 80% vs 81%
- Steps before the card, all missions: noticers ≤−4 87% (n 23), −3 88% (n 8), −2 50% (n 8), −1 64% (n 14); never-said 82%, 100%, 88%, 56% (n 22, 8, 8, 16)
- −1 against −3/−2, the 16 people with three steps (all tactical-heavy-first): adjusted −13 points [−40, +13], p = .29
- Matched view (8 noticers with 3+ steps, 6 of whom met the card at the start of session 2): 87%, 88%, 50%, 63% before → 38% at the card → 31–44% for the next ten decisions

**Method.** Rows: strategic decisions of the 33 people, split into those who said they noticed (debrief or interview code C12, or a strict in-play remark; 15 people) and those who never said so (18). 'Before' and 'after' are relative to the first illogical card. That decision itself is excluded from both windows and shown as its own point ('the card') in the all-missions view. Change = the 'after' coefficient in a linear model of card uptake, with practice (decision number) held fixed and people clustered for the uncertainty. Raw version: uptake ~ after + practice + game. Adjusted version: (observed − expected) ~ after + practice. The interaction asks whether the after-step differs between the groups. Label shuffle: the said/never-said labels were shuffled across people 5,000 times, to see how often a random split of the same people shows as large a difference in their own after-minus-before. Each person counts once in that test, so it is exact. Steps before the card: $-1$ is each person's last decision before it, $-2$ the one before that, and '$\leq -4$' pools everything earlier; the matched view keeps only the 16 people (8 per group) with at least three such decisions. Mission-only expected: the same model's prediction with each person's own-rate term set to everyone's average.

**What participants said.** 12 of 33 said in the debrief or interview that the strategy cards over-allocated or sent unneeded types (C12), and 12 commented on it during play (7 strictly about a card sending an unneeded type). The first sight is often puzzlement rather than a verdict ('unless I'm missing something', 'do fast drones have a higher chance of failing?'), and some took the card anyway (P04) or noticed only after taking it (P20, P29). Dissent: P10 decided the redundancy was sensible; P27 was not sure it was a fault.

**Quotes selected for this section.**

*"Oh, this just sends a bunch of extra of these even though it doesn't need it?" — P19 (think-aloud s2 0:36)*

*"Why is it that the green ones, there's no green or red, unless I'm missing something." — P17 (think-aloud s1 5:25)*

*"And then when it came to assigning them, I noticed, oh, okay, it's made a mistake." — P20 (debrief 0:26:11)*

*"where we can go conservative, but we don't need any, in this one we don't need any lifter or cameras." — P30 (think-aloud s1 1:01)*

*"But I find that the first task it allocated too much Lifter and Camera. So I think I'm less confident for it." — P35 (think-aloud s1 1:22)*

*"there was one example in the first round where it was required all fast drones. And I think the AI assistant recommended I add some red and green drones." — P20 (debrief 0:25:51)*

*"I mean, it was giving me the wrong type of drones. It required only blue drones and kept giving me red and green ones." — P21 (debrief 0:35:03)*

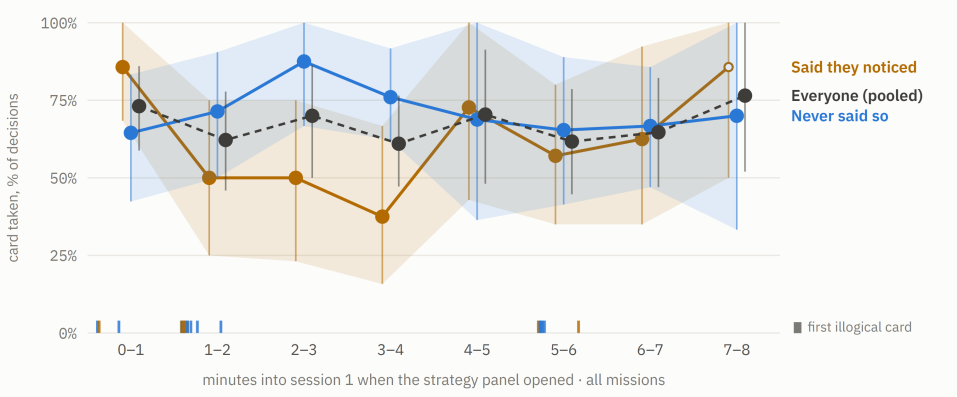
*"But in conservative, you don't need camera or lifters, yet it adds it as reserve. Even for tasks it doesn't do. That doesn't make sense." — P13 (debrief 0:29:55)*

*"deploy a conservative strategy for that one doesn't make sense, I don't think. Because I don't need any red or blue drones." — P20 (think-aloud s1 3:41)*

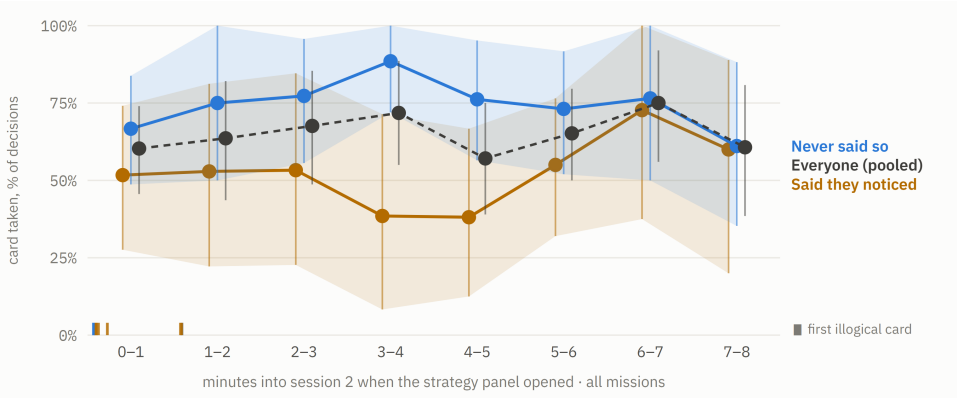
### 7.8.2 Card use through the session, noticers and never-said

[Pass 5: for the paper]

**What is plotted.** Every strategic decision is placed in the minute of the session when its strategy panel opened (0–1 is the first minute). Each point is the share of the decisions in that minute where the person took one of the two strategy cards instead of allocating by hand, with a 95% band. Nothing is lined up on the illogical card; the small ticks on the axis show when each person first met it. By default it shows **each group and everyone pooled** (dashed line). **Session 1**: noticers start at 86% in the first minute, fall to 50%, 50% and 38% over minutes 1–4, and come back to 73% in minute 4–5. Never-said people rise over the same minutes, 65% → 88%. Everyone pooled sits between the two and stays at 61–77%: the two groups move in opposite directions, so lumping them together hides the dip. That is the point of this chart. **Session 2**: noticers are lower throughout (52–53% in the first three minutes, 38% in minutes 3–5, then 55–73%); never-said people sit between 61% and 89%. Each minute holds 10–30 decisions per group, so single minutes have wide bands: the shape, not any one point, is the result. (Figure 29)




**(a)** Session 1: each group and everyone.




**(b)** Session 2: each group and everyone.

**Figure 29: Card use through the session, noticers and never-said**

**Key numbers.**

- Session 1, noticers by minute 0–1 to 4–5: 86% (n 21), 50% (n 16), 50% (n 14), 38% (n 16), 73% (n 11)
- Session 1, never said: 65% (n 31), 71%, 88%, 76%, 69%
- Session 2, noticers: 52% (n 29), 53%, 53%, 38% (n 13), 38% (n 21), 55%, 73%, 60%; never said 61–89%

**Method.** Rows: every strategic decision, binned by the minute of the session at which its panel opened (the last bin, 7–8 min, is thin). Rate = share of decisions where a card was taken; 95% bands from resampling people. Adjusted = the group's overall

rate plus the mean of (observed − expected) from the adjustment model, i.e. relative to each person's own habit given the mission in front of them. Game order: strategic-heavy-first players met the illogical card at about 67 s of session 1; 7 tactical-heavy-first players at about 315 s, the other 9 at the very start of session 2.

**What participants said.** Nobody describes a dip-and-recover in their own words, but the pieces are there: several noticed early in session 1 (P35 in the first two minutes, P13 at minute 3, P20 at minute 4) and went manual for a while, and those who say why they kept using the cards give speed — it is less reliable 'but at least it saves time' (P28). P20, asked if their trust was destroyed: 'No. Definitely not… most of the time it seemed accurate'.

**Quotes selected for this section.**

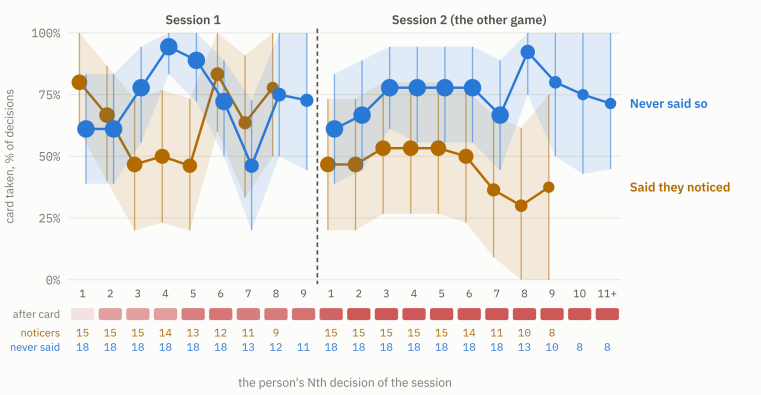
> *"I think it's less reliable than before, but I still use its suggestion, and that's the case. Yeah, because at least it saves time." — P28 (debrief 0:29:56)*

> *"I know there is a better choice, but I have no time to tune everything." — P28 (debrief 0:30:33)*

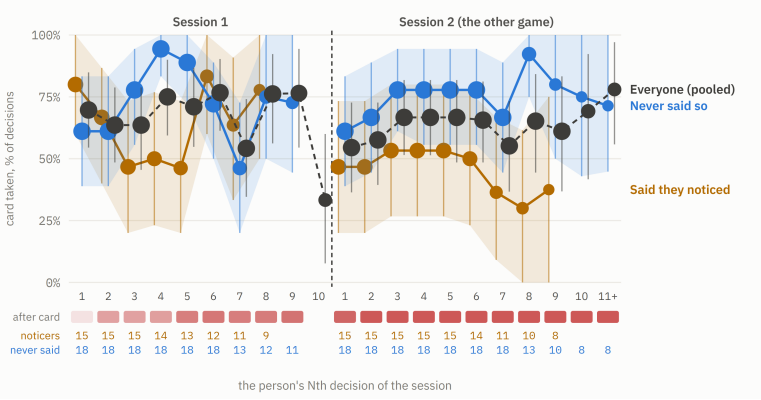
### 7.8.3 Card use by decision number, session 1 then session 2, not aligned to the card

[Pass 5: for the paper]

Each person's decisions are numbered in the order they made them within each session (or counting only three-type, one/two-type, or missions with no illogical card). **Session 1 and session 2 sit on one axis**, separated by a dashed line, with the two sessions' lines deliberately not joined (session 2 is a different game, after a break and a survey). **Later numbers are different people**: only those who made that many decisions reach them, and they are mostly players of the strategic-heavy game, which has more missions. So a point is drawn only if at least 8 people of its group reach it; dot size and the numbers under the axis show how many people each point rests on. **Session 1**: noticers 80%, 67%, 47%, 50%, 46%, then 83%, 64%, 78% for decisions 6–8; never-said 61%, 61%, 78%, 94%, 89%, 72%. Noticers dip at 3–5 and climb back (+24 points per person). The same shape holds in a fixed panel of the 9 noticers and 12 never-said people who made at least 8 decisions (noticers 44% at decisions 4–5, 78–89% at 6 and 8), so it is not a change of who is counted. **Session 2**: noticers 47–53% for six decisions, then 30–38% at 7–9; never-said people 61–92%. The late low is the mission mix: around decisions 7–8 of the strategic-heavy game half the missions show the illogical card, and noticers took it on 17% of those against 62% of the rest. Leaving those missions out, noticers and never-said people meet at decisions 7–8 (71% against 70–73%). Noticers start session 2 below where they ended session 1 (64% $\rightarrow$ 49%), but never-said people also start a little lower (76% $\rightarrow$ 69%); the difference at the break is small and uncertain ($-6$ points [$-32$, +20]), and $-9$ [$-30$, +12] for noticers after allowing for the mission mix. The very last numbers of each session (session 1 decisions 9–11+, session 2 decision 10 and 11+ for noticers) rest on 2–6 noticers and are hidden; the *Thin points* button shows them. (Figure 30)




**(a)** Noticers against never-said, session 1 then session 2.




**(b)** Each group and everyone.

**Figure 30: Card use by decision number, session 1 then session 2, not aligned to the card**

**Key numbers.**

- Everyone, session 1, decisions 1–6: 70%, 64%, 64%, 75%, 71%, 77% (n 33 → 30)
- Noticers, session 1: 80%, 67%, 47%, 50%, 46%, 83%, 64%, 78% (decisions 1–8, 15 → 9 people); never said: 61%, 61%, 78%, 94%, 89%, 72%, 46%, 75%, 73% (decisions 1–9, 18 → 11 people)
- Session 1 fixed panel (9 noticers, 12 never-said, decisions 1–8): noticers 78, 67, 56, 44, 44, 89, 56, 78%; never said 50, 42, 67, 92, 83, 67, 50, 75%
- Session 1 tail (hidden): decision 9, 6 noticers 83%; decision 10, 5 noticers 1 of 5 took (20% [4, 62]), 7 never-said 43% [16, 75]; 11+, 2 noticers 3 of 7 (one took 3 of 4, the other 0 of 3), 5 never-said 7 of 12. All strategic-heavy-game players; 57–67% of the 11+ decisions in the last 90 s
- Session 2 tail (hidden for noticers): decision 10, 5 noticers 60%; 11+, 4 noticers (all tactical-heavy first, so playing the strategic-heavy game) 12 of 13 = 92% [67, 99], against 8 never-said 20 of 28 = 71%. These 4 took cards on 42% of their decisions 1–6, below their group's 51%: not keen takers surviving
- Strategic-heavy game: share of decisions with an illogical card at decisions 7, 8, 9, 10, 11+: 48%, 55%, 30%, 4%, 5%. At decisions 7+, noticers took 17% of illogical-card missions (n 18, 9 people) and 62% of the rest (n 53); never-said 68% and 69%
- Session 2, no illogical card: noticers 71% at decisions 7 and 8 (7 people) against never-said 73% and 70%; earlier the gap is there (decisions 2–3: 33–47% against 67–89%)
- Session 2 fixed panel (8 noticers, 10 never-said, decisions 1–9): gap over the last 3 decisions −47 points [−81, −12] against −19 [−50, +12] before; on missions with no illogical card (panel to decision 6) −17 [−45, +12] against −30 [−61, +1]
- At the session break (last 3 of session 1 → first 3 of session 2): noticers 64% → 49%, −16 points [−38, +4], p = .17; adjusted −9 [−30, +12]; never-said 76% → 69%, −7 [−24, +7]; difference −6 [−32, +20], p = .66
- Recovery within session 1 (decisions 3–5 → 6+): noticers 48% (n 42) → 66% (n 50), +24 points per person [+5, +43], p = .04 (window read off the curve, so descriptive)

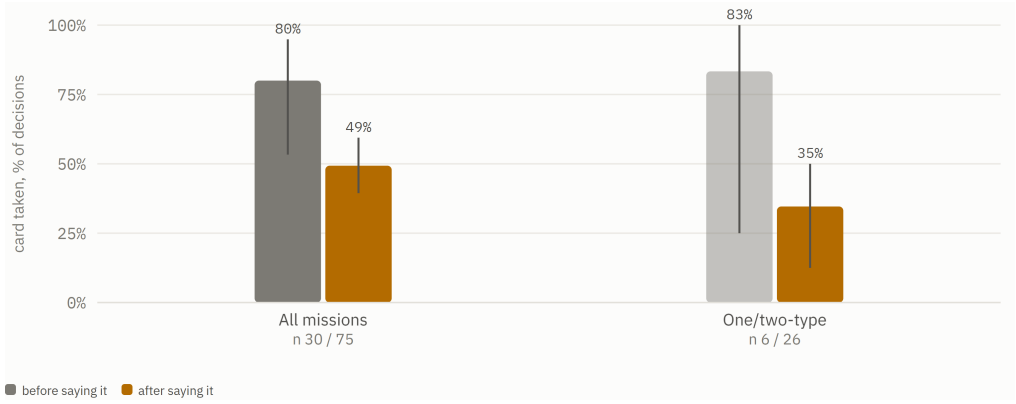
**Method.** Rows: every strategic decision; position = its rank among that person's decisions of the chosen kind in that session (positions past 10 pooled as 11+). Rate = share where a card was taken, 95% bands from resampling people, n decisions and people in each tooltip, with the point's share of three-type missions, illogical-card missions, strategic-heavy-game decisions and last-90-s decisions. **Minimum**: a point is drawn by default only if at least 8 people of its group (both game orders together) reach that number, about half of the smaller group. **Fixed panel**: per session and mission kind, K = the largest number that at least 8 people in each group reach; only people with at least K decisions, decisions 1–K (all missions: session 1 K = 8, 9 noticers and 12 never-said; session 2 K = 9, 8 and 10). 'No illogical card' drops the decisions on missions whose Conservative card showed an unused drone type, then renumbers. 'After card' strip = share of the position's decisions made at or after the person's first illogical card. Carry-over checks, per person then averaged (intervals from resampling people, p from a t-test over people): (a) recovery = the rest of session 1 (decisions 6+) minus decisions 3–5; (b) break = the first three decisions of session 2 minus the last three of session 1; (c) start-to-start. Group differences = the interaction in a linear model with people clustered; adjusted versions use observed − expected.

**What participants said.** 4 of 33 said they caught the assistant's mistakes more, or were more careful, in the second session (C15: P15, P17, P19, P33), and P15 and P33 add that in the first session there was no time to check. One said the opposite: P20 saw no strategic issues in the second run. No one connects it to which game they played.

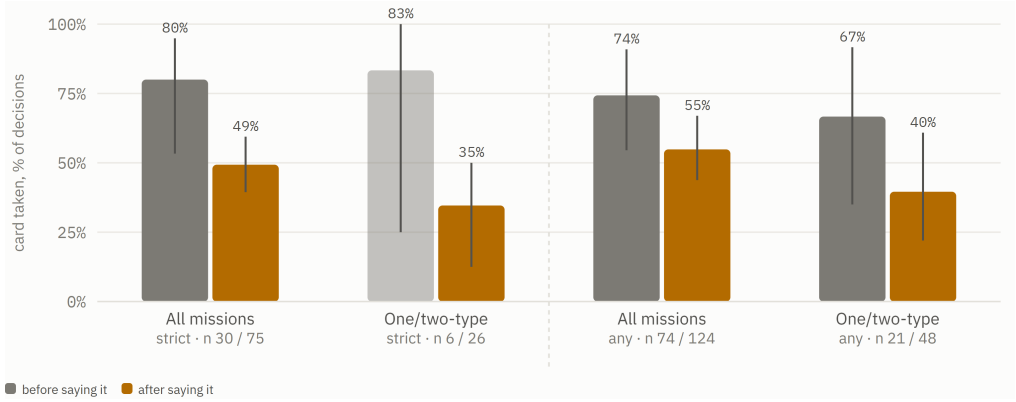
### 7.8.4 Lined up on the moment they said a strategy card was wrong: card use falls (7 people, a direction only)

[Pass 5: for the paper]

*This section is about noticing the strategy card's flaw; the next section is a different event (the Tactical Assistant failing to offer a recovery plan), so the two are not in conflict.* Here each person is anchored on the moment they said, during play, that a card was wrong. Seven people made a strict remark (a card sends types the mission does not need, or a specific card is wrong); five more made broader 'too many drones' remarks. For four of the seven, an illogical card was on screen as they spoke. P30's remark was about an illogical card that opened 5 s later, so it was moved onto that card. P29 spoke 28 s after taking the illogical card. P04's remark was about too many drones on a three-type card. The chart pools each set of people, before vs after the remark, with **all missions and one- and two-type missions side by side**. On all missions the seven went from 80% to 49% (adjusted −21 points [−49, +7], p = .11, timing-matched placebo p = .51; five fell and two rose). On one- and two-type missions, the ones that carry the illogical card, the drop is larger, 83% → 35%, but that 'before' bar is only 6 decisions from 4 people, so it is a direction, not a measured size. Seven people cannot settle it; the direction matches the main result. (Figure 31)




**(a)** Strict remarks: all missions against one- and two-type missions.




**(b)** Strict and any remark, side by side.

**Figure 31: Lined up on the moment they said a strategy card was wrong: card use falls (7 people, a direction only)**

**Key numbers.**

- Strict remarks (7 people), all missions: 80% (n 30) → 49% (n 75); adjusted −21 points [−49, +7], p = .11; 5 fell, 2 rose; placebo p = .51
- Strict, one/two-type: 83% (n 6, 4 people) → 35% (n 26, 7 people); three-type: 79% (n 24) → 57% (n 49)
- Any remark (12 people), all missions: 74% (n 74) → 55% (n 124), adjusted −4 points, p = .60; one/two-type 67% (n 21) → 40% (n 48)

**Method.** The spoken moments were hand-coded from the in-play narration before any behaviour around them was computed (extras.SPOKEN_STRAT, session time). A decision whose card was open at that moment is the anchor decision. Otherwise, 'before' and 'after' are split at the remark. The same models and placebo as above are used.

**What participants said.** The in-play remarks are short and pointed ('It's just wrong', 'You don't need all of that') and are usually followed by going manual (P19, P29, P35, P13), sometimes by taking the card anyway (P04). Four people make further in-play remarks that the anchor list leaves out, late in session 2: P19, P21, P35, and P32. P32 has no C12 code, so they are counted as never having said it, yet they said 'It's just wrong' about an unneeded Camera on a card (s2 5:45).

### 7.8.5 A different event: after the Tactical Assistant's first 'No workable plan', nothing changes on either tier

[Pass 5: numbers only]

*Not the same moment as the previous section.* Here each person is lined up on the first time the Tactical Assistant's Suggest, asked for a plan to recover from a drone failure, answered 'No workable plan'. That is a failure of the other assistant, and nothing to do with noticing the illogical strategy card. Does *strategic* use go down after the first refusal, with the problem cascading up the chain? **It does not.** After their first 'No workable plan' (18 people), card uptake actually went *up* in raw terms (68% → 76%). The expected value went up just as much, because these refusals tend to come later, when people have settled in. Adjusted, the change is +2 points [−6, +10], with 10 people falling and 8 rising. On three-type missions it is 0 points. Tactical Suggest use also rose raw (78% → 87%) and is flat adjusted (−1 point). After the first *wrong* refusal (14 people) the picture is the same: strategic −2.5 points [−16, +11], tactical 0. The first drone failure (33 people) changes nothing either. These are absences of evidence, with intervals around ±10 points. They are not proof of no effect, but no cascade is visible. (Figure 32)

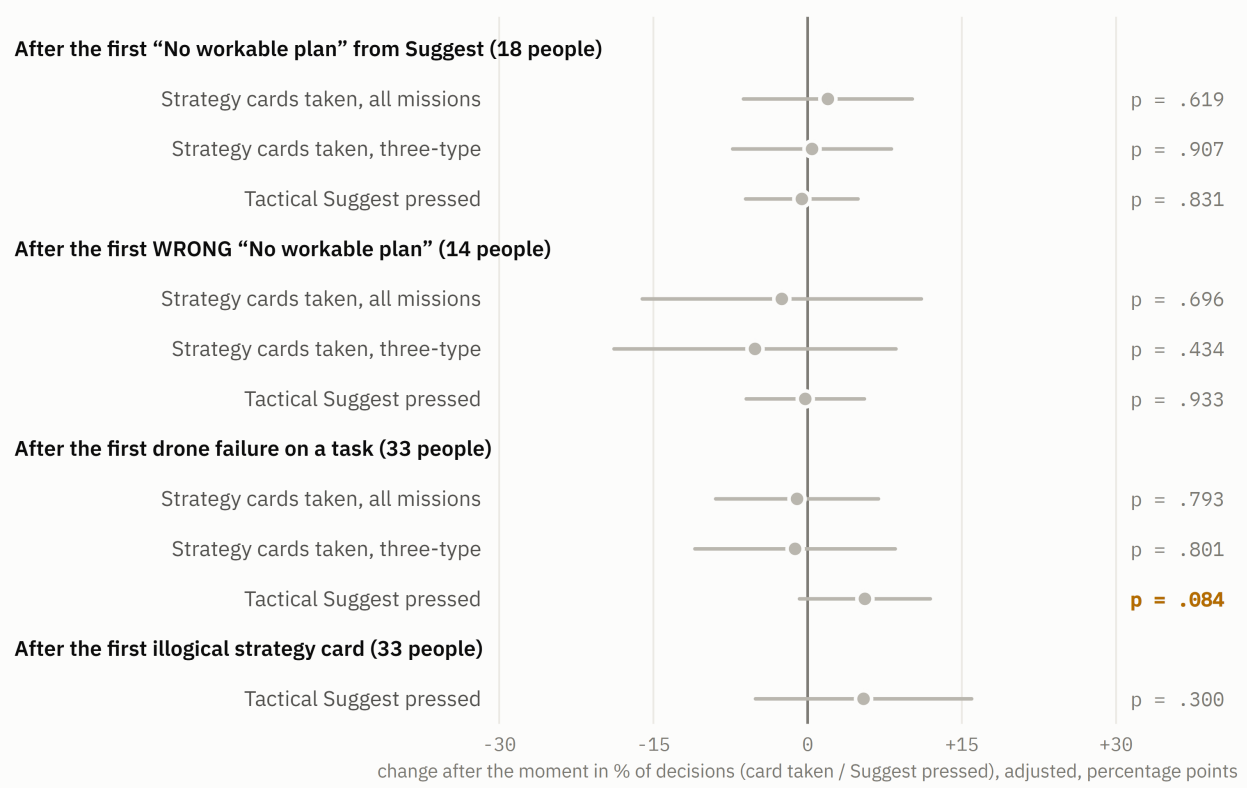



**Figure 32: A different event: after the Tactical Assistant's first 'No workable plan', nothing changes on either tier**

**Key numbers.**

- First refusal (18): strategy cards, all missions, adjusted +2.0 points [−6, +10], p = .62; three-type +0.4, p = .91; tactical Suggest −0.6 [−6, +5], p = .83
- First WRONG refusal (14): strategy cards −2.5 [−16, +11], p = .70; three-type −5.1, p = .43; tactical −0.2, p = .93
- First drone failure (33): strategy cards −1.0, p = .79; tactical +5.6, p = .08
- After the first illogical card, tactical Suggest +5.4 points, p = .30 (no spill-down either)

**Method.** Same rebuilt method as Section 7.8. Each person's strategic decisions (or tactical plans) are numbered within the

sample, after versus before the moment of their first refusal, with practice held fixed and people clustered. Adjusted = observed minus the expected value. The placebo moves each anchor to a random point drawn from the real anchor positions. All p-values are for the adjusted change.

**What participants said.** Five people reacted aloud to Suggest's 'No workable plan', and every one took it at face value and moved on to abandoning ('Oh well. We'll have to abandon it then') or doing it by hand. Nobody said a refusal made them doubt either assistant. That fits the null. The only counter-voice is P05's note that they would want to check a refusal themself, and P27 asked for the abort button to be highlighted when no plan exists.

### 7.8.6 Why the pre-registered test (H8) was dropped

H8 asked whether uptake falls after an operator 'realises' the card can be illogical. Its marker for that realisation was the first time a person went manual on an illogical-card mission and left out the unneeded type. The trouble is that, by that definition, every illogical-card decision *before* the marker was not such a manual choice. So the 'before' window was mostly forced to be card takes, near 100%. Any later manual choice then shows up as a drop, even if the person never changed at all. A simulation confirmed it. In simulated people who keep the same habits all study, the same marker produced a large 'drop' (average odds ratio 0.37, against the real 0.12). Measured against that, the real drop was ordinary (calibrated p = .30, about −10 points after correction). Defining the marker as the first manual choice on a *clean* card produced the same kind of drop. H8 was replaced by tests whose anchor is not defined by the outcome (the first illogical card shown; the moment someone said it), each checked against a placebo. Those are the tests in this finding.

**Key numbers.**

- Pre-registered estimate OR 0.12; the same test on no-change simulations averages OR 0.37; calibrated p = .30 (risk difference)
- Placebo marker on clean cards: OR 0.22, p = .03, the same kind of 'drop'

**Method.** From docs/analysis/round1/h8_illogical_card.md. The null calibration redrew each person's choices 1,000 times from their own constant rate, reapplied the marker and refitted the same model.

**What participants said.** The move the H8 marker counts — going manual and leaving out the unneeded type — is described by several (P19, P13, P16, P17, P29). But noticing and going manual are not the same event. Some saw the excess and still took the card (P04, P28 'I still use its suggestion'). Others went manual only to trim an extra of a needed type (P10, P01). So a first manual 'fix' is a poor stand-in for the moment someone realised.

## 7.9 The Tactical Assistant is trusted more, and most people say so too

**Summary.** On the post-session questionnaire, operators trusted the Tactical Assistant more than the Strategic Assistant: 5.44 against 4.61 on a 1–7 scale, 26 of 33 people higher for tactical (p = .001). Most of the gap opens in the second session and in Strategic Heavy games, and people who said in the debrief that they preferred the tactical one also scored it higher. Tactical plans get only a few seconds of checking whatever their size, which fits "checking is costly, so it is skipped", but that mechanism test is fragile. Stated trust and actual use line up when people are ranked against each other, but not in size and not over time.

*Strength of evidence (register rating): moderate. Your verdict: keep.*

### 7.9.1 Trusted more: the basic result

[Pass 5: for the paper]

After every session operators rated each assistant on the same six statements ("is reliable", "I trust its recommendations", . . . ) from 1 to 7. Averaged over their two sessions, **26 of 33** rated the Tactical Assistant higher, 6 the Strategic Assistant, and 1 the same. The average gap is **+0.83 points** on the 7-point scale. That is a medium-sized effect (dz = .62), and every one of the six items points the same way. (Figure 33)

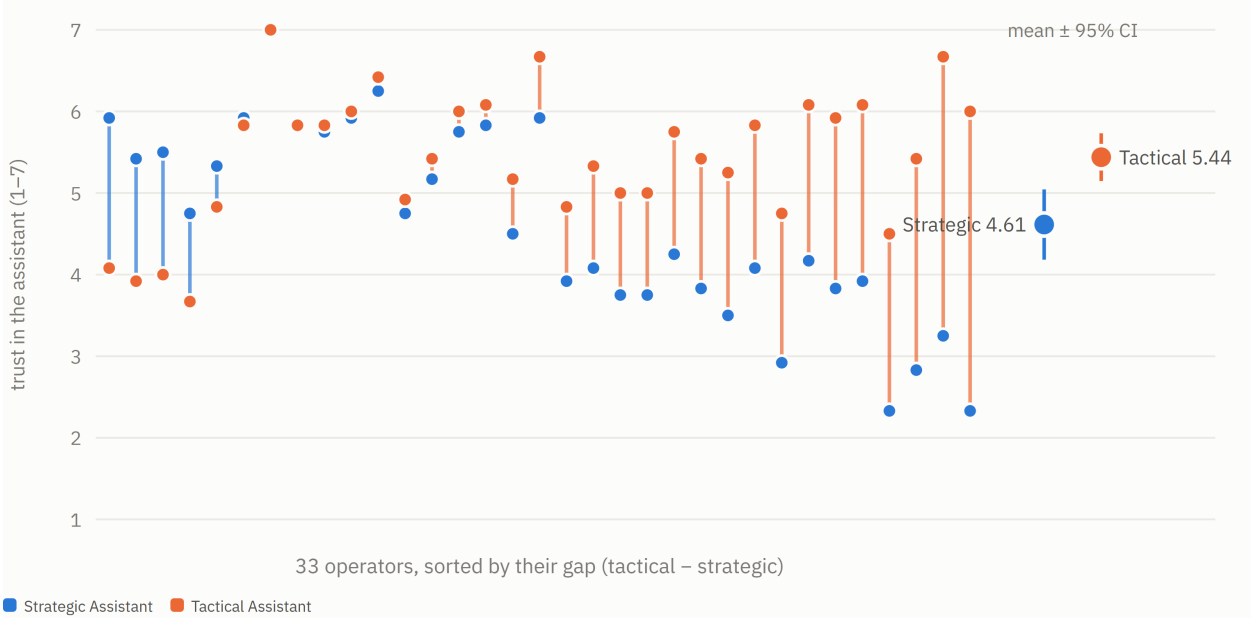



**Figure 33: Trusted more: the basic result**

**Key numbers.**

- Strategic 4.61 [4.18, 5.04], Tactical 5.44 [5.15, 5.73] (1–7, person means)
- Gap +0.83 [+0.35, +1.30]; paired t p = .0012; Wilcoxon p = .0010; dz = .62; n = 33
- 26 higher for tactical, 6 for strategic, 1 tied
- Item gaps: follow its recommendations +1.11, I trust it +1.01, reliable +0.94, performs well +0.74, useful +0.67 (all p < .01); confident in it +0.48 (p = .13)

**Method.** One value per person per assistant: the mean of the six items, averaged over that person's two sessions (33 people, P09 excluded). The test is a paired t-test on the difference tactical minus strategic, with a Wilcoxon signed-rank test as the distribution-free check. dz is the mean difference divided by the SD of the differences: .62 means the average person's gap is about six-tenths

of the person-to-person spread in gaps. The 95% CI says the true average gap is very probably between +0.35 and +1.30 points. Items are tested the same way, one at a time.

**What participants said.** In the round-2 preference coding, 17 of 33 favour the Tactical Assistant, 7 the Strategic, 3 both, 4 'it depends' and 2 no view. The usual reason is that the strategic cards visibly waste drones while the tactical plans 'seemed to be doing what I was doing'. Dissent comes from P10, P06 and P07 (trusted strategic more), P31 ('the tactical layer required a little bit more human intervention'), P12 (strategic is the 'more trivial calculation') and P21/P31, who saw tactical flaws (no backup drone, illogical task order).

**Quotes selected for this section.**

*"But I think in the [tactical level], it works very well." — P33 (debrief 0:28:24)*

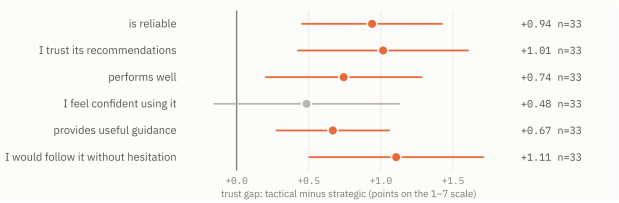
*"You can't blame the tactical for drone failures but you can blame the strategic for not having redundancy" — P03 (written interview)*

*"So the tactical one seemed to be doing what I was doing. So quite reliably. And obviously the strategic one wasn't." — P32 (debrief 0:23:52)*

### 7.9.2 Where the gap comes from: the second session and the Strategic Heavy scenario

[Pass 5: for the paper]

The gap is larger in Strategic Heavy sessions (+1.13) than in Tactical Heavy ones (+0.52, CI just reaches zero). It is also small after the first session (+0.37, CI crosses zero) and large after the second (+1.28). Both matter when taken together: holding session order fixed, Strategic Heavy adds about +0.6 to the gap and being the second session about +0.9. Tactical trust hardly moves (5.36 → 5.51 across sessions, and only +0.15 between scenarios); it is **strategic** trust that falls, and it is lowest in Strategic Heavy sessions. So "trusted more, especially in Strategic Heavy" largely means "the strategic one loses trust, especially in the scenario where its cards are used most". Four people said their trust *grew* with experience (C16). The chart opens on the six questionnaire items (every one favours tactical); **Scenario played** shows the two scenarios; **Also: first / second session** shows session order. (Figure 34)




**(a)** By questionnaire item.

**(b)** By scenario played.

**(c)** By session order as well.

**Figure 34: Where the gap comes from: the second session and the Strategic Heavy scenario**

**Key numbers.**

- Session 1 gap +0.37 [−0.15, +0.88]; session 2 +1.28 [+0.69, +1.87] (n = 33 each)
- Strategic Heavy sessions +1.13 [+0.53, +1.73]; Tactical Heavy sessions +0.52 [−0.01, +1.05] (n = 33 each)
- Same people, Strategic Heavy minus Tactical Heavy: +0.61 [−0.01, +1.23], paired t p = .054, Wilcoxon p = .038; strategic trust −0.46 (p = .087), tactical +0.15 (p = .25)
- Both in one model: Strategic Heavy +0.64 [+0.11, +1.17], p = .019; second session +0.93 [+0.40, +1.47], p < .001
- Strategic trust 5.00 → 4.23 from session 1 to 2; tactical 5.36 → 5.51

**Method.** The tactical-minus-strategic difference for each session (66 sessions). By session: averaged over the 33 first and the 33 second sessions. By scenario: both games of each scenario pooled (33 sessions each). Every person played one session of each scenario, so the scenario comparison is also done within person (paired t and Wilcoxon on each person's two gaps). Session order and scenario are partly tangled (about half played Strategic Heavy first), so a model with both (GEE, gap ~ Strategic Heavy + second session, clustered by person) separates them. Means have t-based 95% CIs.

**What participants said.** Those who name a session say it was the strategic tier that slipped in the second session: P19 ('you dropped the ball a little'), P33, P15 and P17. Tactical trust was steady or up (P19, P26). Against the pattern, P06 says strategic trust went up over time and P20 saw strategic issues only in the first run. Nobody tied the change to the Strategic Heavy scenario as such.

### 7.9.3 Why the tactical plan is kept: checking it is brief and does not grow with the plan

[Pass 5: for the paper]

Bigger missions mean bigger tactical plans. As plans grow, operators press Suggest more often (63% of plans with 3 or fewer drones → 93% with 10 or more) and keep the plan unmodified more often (63% → 86%). Once Deploy unlocks, they spend a median of about **2–4 seconds** on the plan whatever its size. Per drone, checking therefore falls steeply as plans grow (0.7 s → 0.2 s). A strategy card gets about 9 s. Almost every consulted plan is deployed as it comes: 468 of 486. The more there is to check, the less of it is checked proportionally, and the plan is taken. Operators describe exactly this. Nine say they check when it is cheap and skip it under pressure (C17). Three say the plan was too complex or too fast to check (C14). Five say they checked nearly everything, and they do decide more slowly (C18). Only 3 of 33 ever mention noticing a tactical flaw (C13).

**But** the pre-registered mechanism test (H2) is fragile. All three predicted parts point the right way (p = .019 uncorrected), but it does not survive correction for the round's several tests (.065). Among consulted plans, larger ones are actually modified a little *more*. Keeping goes up with size mainly because people consult more on big plans, not because they check consulted plans more lightly. The logs also cannot tell "too costly to check" apart from "nothing ever looked wrong". The tactical plan's flaws are hard to see, while the cards' over-allocation is visible. That half of the explanation belongs to *tier-flaw-visibility* in Section 7.8 and is not repeated here. (Figure 35)

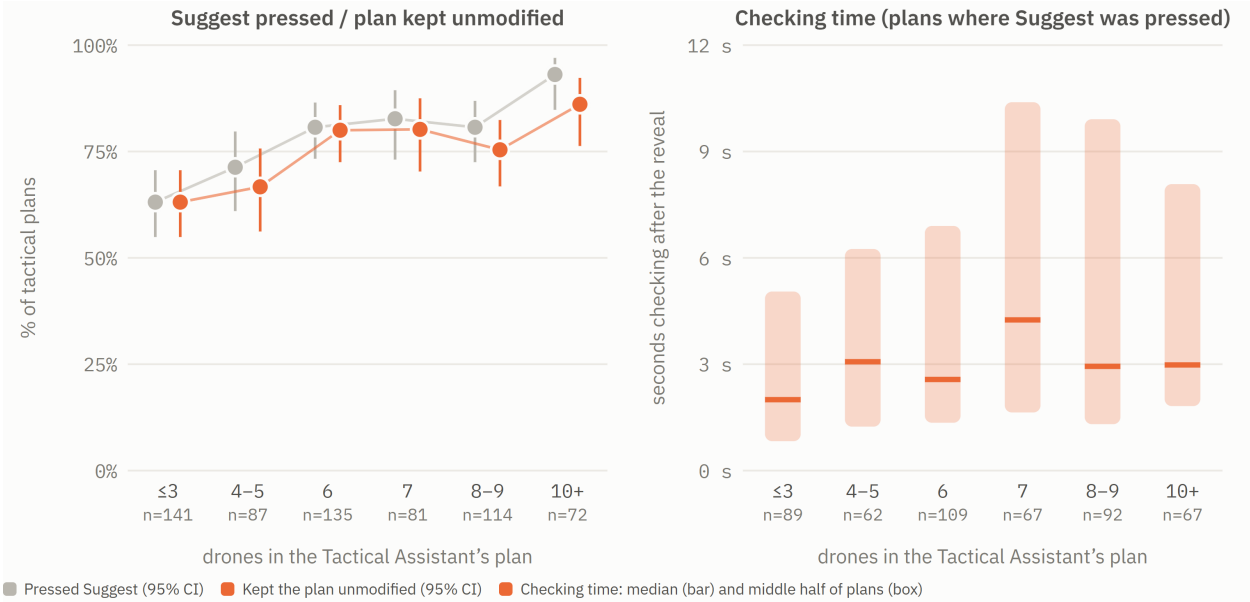



**Figure 35: Why the tactical plan is kept: checking it is brief and does not grow with the plan**

**Key numbers.**

- Kept unmodified rises with plan size: OR 1.76 per SD [1.25, 2.47], p = .001 (all 630 plans)
- Consulted rises: OR 2.52 [1.87, 3.38]; kept once consulted FALLS: OR 0.68 [0.50, 0.94], p = .017
- By bin (≤3, 4–5, 6, 7, 8–9, 10+ drones; n = 141, 87, 135, 81, 114, 72 plans): Suggest pressed 63%, 71%, 81%, 83%, 81%, 93%; kept unmodified 63%, 67%, 80%, 80%, 75%, 86%
- 10+ drones vs ≤3: Suggest pressed 93% vs 63%, OR 10.8 [3.2, 36.0]; kept 86% vs 63%, OR 5.2 [2.1, 12.7], both p < .001
- Median checking time by bin: 2.0, 3.1, 2.6, 4.2, 2.9, 3.0 s; per SD +0.60 s [−0.15, +1.35], p = .12; per drone 0.68 s at ≤3 drones → 0.23 s at 10+
- 468 of 486 consulted plans deployed unmodified (96%)

- Person-median check: tactical 2.85 s vs strategy card 8.80 s (n = 33, Wilcoxon $p < .0001$)
- H2 primary $p = .019$ uncorrected, .065 after Holm correction across round 1

**Method.** 630 confirmed tactical plans from 33 people (h2_verification_cost.load). They are grouped by the number of drones in the Tactical Assistant's plan into six bins with no gaps: $\leq$3, 4–5, 6, 7, 8–9 and 10+ drones. The bins were chosen so every bin has enough plans for a stable rate: at least 72 plans and at least 62 with a checking time in each (the smallest bins are 4–5 drones, 87 plans, and 10+, 72 plans). Every plan is used; none is dropped. "Consulted" means Suggest was pressed at least once. "Kept unmodified" means consulted and deployed with no change. "Checking time" runs from the last Suggest press to Deploy, minus the forced reveal: the planner adds one drone→task pair per second and Deploy is locked until it finishes, so that wait is not checking. Effects per SD of plan size (drones $\times$ tasks) come from GEE models on all 630 plans, clustered by person, adjusted for scenario and session; they do not depend on the bins. An odds ratio of 1.76 means each step up in plan size multiplies the odds of keeping the plan by 1.76. The two ends (10+ vs $\leq$3 drones) are also compared directly. The card comparison takes each person's median net decision time on strategic decisions: strategic_choice latency minus the card reveal, over all strategic decisions, manual ones included. The H2 numbers are from docs/analysis/round1/h2_verification_cost.md.

**What participants said.** Speaking to this point: 3 of 33 say the tactical plan was too complex or fast to check (C14), 9 that checking depended on time (C17), and 5 that they always checked (C18). The reasons match the finding: the plan is hard to check, so it is taken on trust ('I just took the AI's word for it'). Several say it is not more trust but the time to check (P04). P14 is the clear dissent: they checked tactical plans more, drone by drone, because cards can be read 'in one glance'.

**Quotes selected for this section.**

> *"as soon as it got to any sort of non-linear, difficult, high dimensional problem, I just took the AI's word for it, essentially, probably because I couldn't assess in the time" — P22 (debrief 0:30:06)*

> *"My reason is there are so many drones, so I don't have enough energy and attention to just the check them." — P27 (think-aloud s1 0:56)*

### 7.9.4 Stated vs revealed preference: they line up in rank, not in size or over time

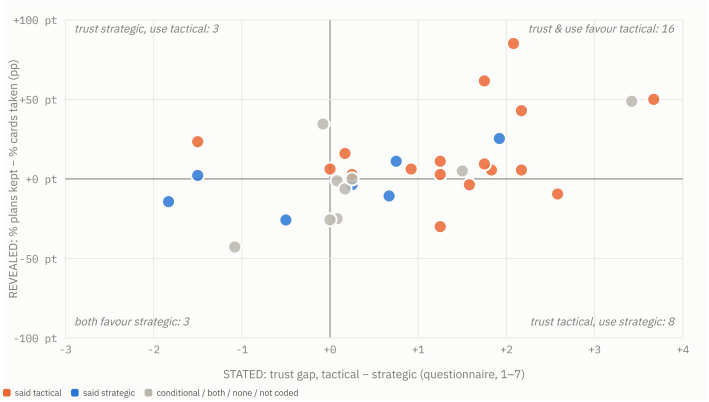
[Pass 5: numbers only]

Does the person who *says* (rates) the Tactical Assistant higher also *lean* on it more? Ranked against each other, yes, moderately. The person-level trust gap correlates $\rho = .47$ with the usage gap (share of tactical plans kept unmodified minus share of strategic decisions where a card was taken). 16 people favour tactical on both measures and 3 favour strategic on both. Three things do **not** line up:

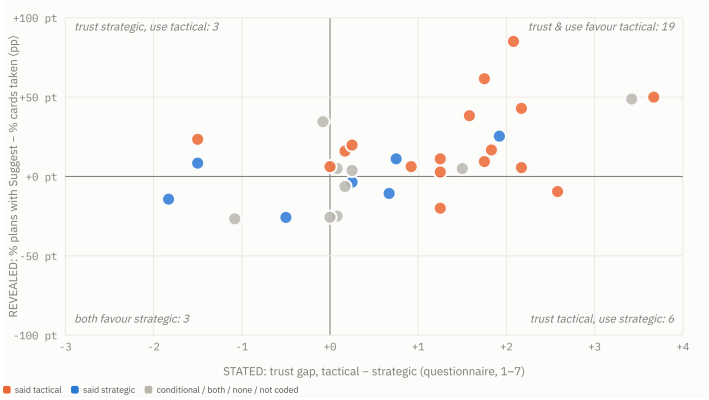
(1) **Size.** Stated trust favours tactical strongly (26 of 33), but use barely does. Cards are taken on 64% of strategic decisions and tactical plans kept on 72%, a gap of +8 points whose CI includes zero. Nine people rate tactical higher yet use the strategic cards more.

(2) **Change.** Within a person, a rise or fall in trust from session 1 to 2 does not go with a matching change in use ($\rho$ = .16 strategic, .13 tactical).

(3) **Lead.** Session-1 trust does not predict how card use changes in session 2 ($\rho = .07$). Trust ratings read like a stable opinion of the assistant, while use follows the moment: mission size, queue, time. The ratings describe the person; they do not forecast their next move. (Figure 36)




**(a)** Use measured as keeping the plan unmodified.




**(b)** Use measured as pressing Suggest at all.

**Figure 36: Stated vs revealed preference: they line up in rank, not in size or over time**

**Key numbers.**

- Trust gap vs usage gap (kept − took): $\rho = .47$ [.10, .73], p = .006, n = 33
- Quadrants: both favour tactical 16, both strategic 3, trust tactical/use strategic 9, trust strategic/use tactical 3 (2 on a zero line)

- Group use: took a card 64%, consulted the plan 76%, kept it 72%; usage gap +0.08 [−0.02, +0.17]
- Within person, change in strategic trust vs change in card use: $\rho$ = .16 [−.27, .52], p = .38
- Session-1 strategic trust vs change in card use s1→s2: $\rho$ = .07 [−.27, .40], p = .71; tactical .01
- Between people, strategic trust vs cards taken $\rho$ = .60 [.30, .78]; tactical trust vs plan kept $\rho$ = .31, p = .08

**Method.** One dot per person, both sessions pooled. Stated = mean tactical trust minus mean strategic trust (1–7). Revealed = share of their confirmed tactical plans kept unmodified minus share of their strategic decisions where they took a card (the alternative "pressed Suggest at all" gives the same picture, $\rho$ = .49). Spearman $\rho$ uses 4000-draw seeded bootstrap CIs. A caveat on (1): taking a card and keeping a plan are different acts with different costs, so the *level* of the usage gap is not a pure preference; the ranking is the cleaner comparison. "Change" correlates the session-2 minus session-1 differences within each person (33 people). "Lead" correlates session-1 trust with the later change in use. Session-level links are also shown (66 sessions): strategic trust ~ cards taken $\rho$ = .49, tactical trust ~ plan kept $\rho$ = .27. Note that the trust survey comes *after* the session it follows, so a same-session link can be use shaping trust as easily as the reverse.

**What participants said.** Several people say outright that how much they used an assistant was not the same thing as how much they trusted it. They took cards they rated lower because cards save time ('less reliable… but I still use its suggestion', P28). P20 lost trust in the strategic tier yet found it 'a bit more useful', and P11 would 'miss both equally' but use the strategic one more. This matches the gap between trust and use found in the data.

### 7.9.5 All the trust–use correlations in one view

[Pass 5: numbers only]

Between people, trust and use move together, most clearly for the strategic tier. Within a person over time, and as a predictor of the next session, the correlations are near zero. (Figure 37)

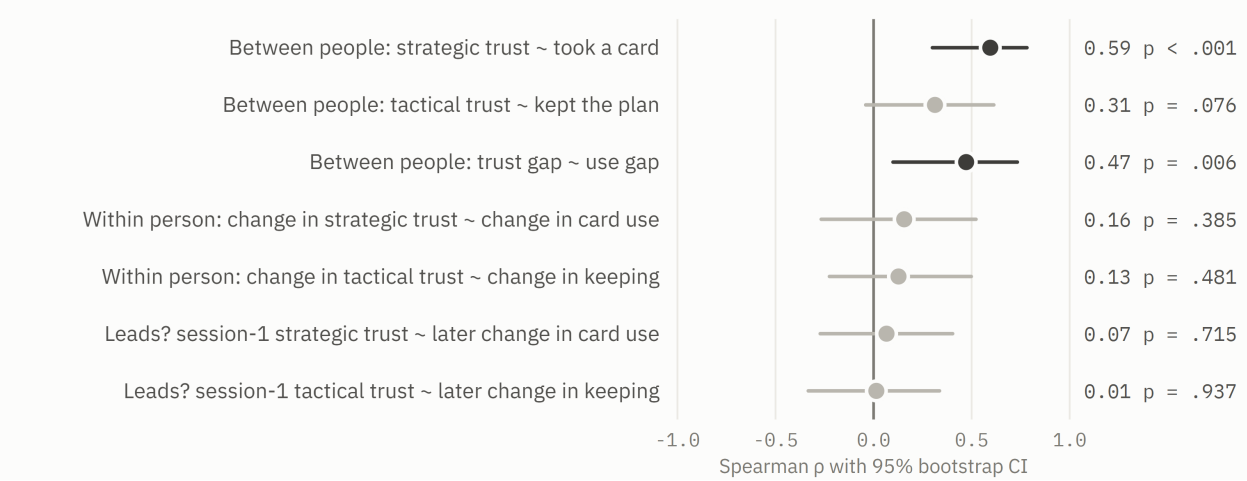



**Figure 37: All the trust–use correlations in one view**

**Method.** Each row is a Spearman $\rho$ with a 4000-draw seeded bootstrap 95% CI, at the person level (n = 33). A dark dot means $p < .05$. These are the same numbers as the section above, in one picture.

**What participants said.** 4 of 33 say their trust grew with experience or success (C16: P06, P10, P26, P32), and a few more describe a checking-then-trusting arc in play (P07, P06, P29). These accounts are all retrospective and smooth, while the data show no within-person link from trust to later use. Nobody describes trust falling and use following it within a session.

## 7.10 Beyond the findings: what operators said they want, believe and feel

**Summary.** Asked about real search and rescue, nobody rejected assistants like these, and 9 of 33 said a human must keep the final decision. What they asked for was mostly information: how the assistants decide (about 12) and failure risk or feedback on how they were doing (11). Some wanted help with a decision neither tier covers, choosing which mission to open next (P27, P32). How they pictured the assistants ranged from "a dozen lines of code" to something that learns or predicts, and several credited them with more than they had (learning from the operator, prediction, one shared plan across the tiers). Suggestive: one coder, and much of it unprompted talk during play.

*Strength of evidence (register rating): suggestive. Your verdict: keep.*

### 7.10.1 What they asked for

[Pass 5: numbers only]

The common thread is control plus information: keep the human in the decision and give them more to decide with (why the assistant suggested it, how likely drones are to fail, options in between). Only four wanted the assistant to do more. These are design requirements more than findings — and the "reasoning" request links to the tactical blind spot, where an explained refusal could have been checked. (Figure 38)

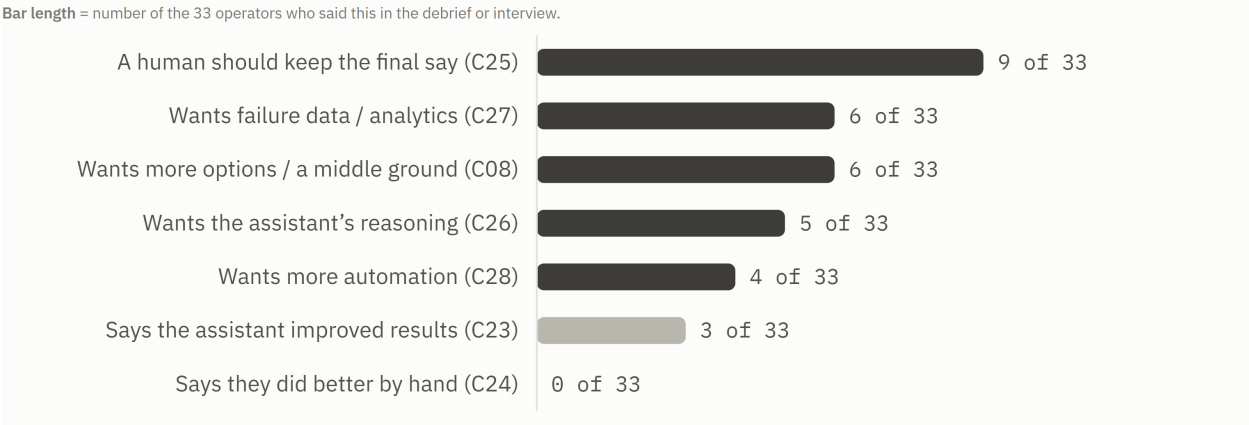



**Figure 38: What they asked for**

**Key numbers.**

- Human final say 9 of 33 (8 unprompted); failure data 6; more options 6; reasoning 5; more automation 4 (all unprompted)

**Method.** Debrief transcripts and interview notes coded against the codebook (CODEBOOK.md): C25 human final say, C27 failure data, C08 middle ground / more options, C26 reasoning, C28 more automation. "Unprompted" = at least one piece of evidence not given as a bare yes to a leading question.

**What participants said.** Across the coded material, 9 of 33 said a human should keep the final say (8 unprompted), 6 wanted failure probabilities or performance data, 6 wanted options between the two cards, 5 wanted the assistant's reasoning and 4 wanted more automation. Reading everything (including the in-play talk) raises the information asks: about 11 asked or wondered how the assistants decided, and 8 asked for failure or performance numbers. The dissent is small but sharp: P11 does not want to be 'reduced to pressing allocate, suggest, deploy'. The mini sections below take each strand in turn.

### 7.10.2 Real search and rescue: welcome, with a human deciding

[Pass 5: extra: may make it in]

What people said when asked how they would feel about systems like this in real search and rescue. 16 of 33 are quoted here (25 answered the question); nobody rejected it, and the common condition was a human who keeps the final decision.

**What participants said.** Asked the closing 'five years from now' question, at least 14 welcomed the idea, nearly always for speed ('a bad approximation is better than no response at all'), and 9 said a human must keep the final decision (8 unprompted). Conditions came with it: well regulated (P20), only where the job is understood and stakes low (P03, P05), only if better than today's AI (P15). Others raised a deeper worry: who is accountable when an AI fails to save someone (P31, P22), and military use of the research (P31). When pressed on whether a busy human could really check, P24 conceded they could not and still wanted one there.

### 7.10.3 Wanting to know how the assistants decide

[Pass 5: extra: may make it in]

Requests for the assistants' reasoning, and moments where people could not work out why a card or plan looked the way it did. 12 of 33 said something on this.

**What participants said.** 5 of 33 asked outright for explanation, auditability or understandable tools (P02, P12, P16, P25, P36); 6 more asked mid-play why a card looked as it did or said they could not tell (P01, P05, P06, P13, P25 in play, P32). P01 named the bind: more information to handle means relying more and trusting less. The requests point both ways: P05 found the strategic cards the hard ones to understand, while P19 found the card simple enough that its error was visible, and P22 said they trusted the tactical plan partly because they could not see how far from optimal it was. Nobody asked for an explanation of a single plan step; the asks were about how the assistant works in general.

### 7.10.4 Failure risk and feedback: wanting the numbers

[Pass 5: extra: may make it in]

Asks for drone-failure probabilities and for feedback on how well they (or the assistants) were doing, and misreadings of the random failures. 11 of 33 said something on this.

**What participants said.** 8 of 33 asked for failure probabilities or performance data (P10, P11, P12, P13, P25, P31, P32, P36); P10, P11 and P13 independently wanted the risk tied to place, terrain or drone type and fed into the allocation. Without it, 2 read patterns into failures that were in fact flat and random (P19: some tasks fail more; P01: the rate was 'cranked up'). 4 could not tell from the score how well they were doing (P10, P29, P31, P32).

### 7.10.5 More automation, and new jobs for an assistant

[Pass 5: extra: may make it in]

Wishes for the assistants to do more, or to do a job neither of them does now, and the few voices against. 8 of 33 said something on this.

**What participants said.** 4 of 33 asked in general for more automation (P13, P27, P30, P31) and 4 more proposed a specific new job: choose which mission to open next (P32, and P27's feasibility-then-criticality ordering), check the operator's own plan for slips (P11), build back-up routes into the tactical plan (P21), re-route spares after a failure (P30) and look ahead to missions not yet arrived (P30). The common thread is help with what neither assistant touches: choosing what to work on, and failures. Against: P11 does not want to be reduced to clicking.

### 7.10.6 What they thought the assistants were

[Pass 5: extra: may make it in]

How people pictured the assistants: simple rules or 'AI', whether they learn, predict or see beyond one mission. 13 of 33 said something on this.

**What participants said.** Two pictures sit side by side. 6 saw through to simple rules and judged them on that: 'a dozen lines of code' (P31), 'a matching algorithm' (P19), 'a little bit stupid', 'old computing' (P15), works but not optimal (P04); P32 and P04 said they would trust a provable optimiser over machine learning. Others credited the assistants with more than they had: P03 hoped it would learn from their choices, P27 thought it predicted, P13 wondered if it knew fast drones fail more, P21 assumed the two tiers share a plan. 2 noticed the cards look at one mission only (P24 had to ask, P30). 2 more took the illogical card for a bug or a planted error (P17, P27).

### 7.10.7 How the two tiers fit together

[Pass 5: extra: may make it in]

Remarks on the hierarchy itself: which level a human should hold, and how a strategic choice carries into the tactical one. 10 of 33 said something on this.

**What participants said.** The shared view is that the strategic choice sets up everything below it: a good allocation makes the tactical stage hard to fail, and drone failures can only be absorbed by drones sent at the start (P03, P19, P36). Several said the human belongs at the strategic level and the assistant at the tactical one, 'give it resources and allow it to optimise' (P05, P30, P32, P33 when asked, P12 reasoning that the strategic sum is the easier one), and P10 found an Aggressive card leaves only one sensible tactical plan. The cross-current: P21 builds the tactical plan themself because their strategic choice already carries a plan in their head.

### 7.10.8 Me or the assistant: who does it better?

[Pass 5: extra: may make it in]

Brought back from the sidelined 'who does better' finding: what people said about whether the assistant or they themselves got better results. 13 of 33 said something on this.

**What participants said.** Coded strictly, 3 of 33 said the assistant improved their results and none said they did better by hand. Read more widely, about 7 rated the assistant as good as or better than themselves (P06, P11, P24, P29, P31, P33, P35), nearly always because it was faster rather than wiser. 4 said they could do better given time or care (P04, P12, P17, P35) and P22 judged them about equal; P31 and P21 said they simply could not tell. Self-doubt shows through (P01 'I don't trust myself anymore', P35). This fits the data: the gain sits between people, not in what the assistant did for any one of them.

### 7.10.9 Interface and display asks

[Pass 5: extra: may make it in]

Concrete requests about the screens, plus the praise. 11 of 33 said something on this.

**What participants said.** 8 of 33 asked for something on screen: a total drone requirement per mission (P10, P13), a cue on one screen when the other needs attention (P10), drones-in-hub and feasibility in the queue (P27; the criticality sort exists but they had not found it), filtering by penalty (P19), confirmation before cancelling a task (P13), and keeping a Suggest plan when clicked away (P01). P06 said the task needs two monitors and P31 that the central map went unwatched. Praise came from 4 (P10, P14, P25, P07: easy, intuitive). Two people (P13, P14) were confused by the reversed NASA-TLX performance slider, which matters for that item's data.

### 7.10.10 It's a game: what real stakes, pace and training would change

[Pass 5: extra: may make it in]

How people saw the realism of the task: no lives at stake, the pace, what the simulation leaves out, and the training a real operator would need. 12 of 33 said something on this.

**What participants said.** 9 of 33 said real lives would change how they played: more stress (P01, P11, P15, P34), more care about efficiency (P07), while P01 expected to behave the same and saw that as the problem. Some rules read as unrealistic: abandoning a mission (P02, P13), partial rescue (P28: a drone that finds someone and leaves), task locations known in advance (P28, P36), the pace (P31, P36). 4 said real operators would need training or new skills (P29, P35, P36, P13).

### 7.10.11 Feelings: fun, stress, self-doubt, and talking to the assistant

[Pass 5: extra: may make it in]

The emotional side of the sessions: enjoyment, stress, doubts about their own ability, and treating the assistant like a person. 14 of 33 said something on this.

**What participants said.** 7 said it was fun or wanted to win (P01, P10, P11, P18, P24, P31, P36). Stress and drama ran alongside it, mostly around failures and empty reserves (P01 'I'm losing my job', P22 'everyone is going to die'). 5 voiced doubts about themselves (P01, P29, P33, P35, P27 not liking to decide fast), and for P27 and P35 it was a stated reason to rely more on the assistant. 4 talked to or personified the assistants: P19 scolded and apologised to the Strategic Assistant, P22 and P32 urged it on, P11 felt the pull to 'personify the computer and feel some sort of grudge', alongside pride when doing it themself.

# 8 In Their Own Words

[Generated (`qual.tex`): the qualitative findings built from long passages of the think-aloud, the questionnaire talk and the debriefs. Two findings you pruned this pass are left out ("Trust in the process, not the answer" and "What a real operation would need"; your note on the second: the limitations are implied and can be mentioned near the end). Participant P36 is quoted only in Section 9.]

## 8.1 You trust what you cannot check

**Summary.** The most articulate operators describe trusting the Tactical Assistant *because* they could not assess its plan, while the strategy card was simple enough to judge at a glance, and so was judged, often harshly. Under time pressure checking shrinks to a glance, and people say so with unusual candour: "if we look at the recording I was doing exactly that". A minority did check, and their trust was earned by comparing the assistant's plan with their own.

*Strength of evidence (register rating): qualitative. Your verdict: keep.*

### 8.1.1 Trusted because it cannot be checked

[Pass 5: for the paper] Your note: A nice few quotes that we can use to make a point.

The strategy card shows a few numbers per drone type beside the mission's own requirements, so its oddities are visible. The tactical plan is a web of routes on a map. P22 puts the consequence most clearly: they take the tactical plan "at its word" because they cannot see how far it is from optimal, and they suspect that is the only reason it seems so good. P19 says rating it would be "disingenuous" because they never really looked, and P01 that it is "a lot harder to mentally challenge" than drone numbers. This is the participants' own account of two quantitative results: the trust gap between the tiers (Section 7.9) and the visible versus invisible flaws (Section 7.8). It suggests the gap is partly a gap in *checkability*, not in quality.

**What participants said.** The clearest voices are P22, P19 and P01; P28 gives the same split from the other side (the strategic decision is the one you can analyse).

**Quotes selected for this section.**

> *"The tactical assistant, I never really saw it do anything wrong, but that's basically because I think the problem is so complex [that] seeing a more optimal solution is really hard. So I wonder if it's one of those things where these are based on the fact that I actually can't perceive the difference between its performance and the optimal." — P22 (talk during the questionnaire survey1 0:14:45)*

> *"I will be honest with you. I don't think I was paying as much attention to the suggestions of the tactical assistant as I could have been. Just purely because the tasks were often very multi-drone. I think I would have committed a lot of extra thought-space to assigning all of the drones individually to those. So I definitely inadvertently trusted the tactical [agent]. I think it's worth saying any assessment of the tactical system would probably be disingenuous just because of my lack of engagement with it. But the strategic assistant, I think on the most part, I have absolutely no problems with. There was that one issue, but that's okay because I noticed it because the display is straightforward and it's clear enough that you can as a human spot discrepancies if they arise." — P19 (talk during the questionnaire survey1 0:17:06)*

> *"i felt like i took a bigger pinch of salt for the strategic one because i felt like i could assess the quality of it more immediately whereas with the tactical assistant, I couldn't accurately and rapidly assess how well it was performing. So I just kind of said, OK, I have to take your word for it, essentially." — P22 (talk during the questionnaire survey2 0:26:53)*

### 8.1.2 Checking shrinks to a glance

[Pass 5: extra: may make it in] Your note: Less significant but might be worth a mention

Nobody describes careful checking under pressure. They describe a quick look ("does anything look ridiculous?"), knowing that a plan can look fine and not be, and accepting that trade because penalty accrues while they check. P31 gives the whole arc: over-checking at the start, under-checking by the end, less oversight as the session goes on, and a rating corrected against their own behaviour. P11 makes the expected-value argument explicitly: on a big mission even a dodgy plan pays, because of the time saved for the next mission. P28 is the exception, saying they checked every answer, though "a quick browse".

**What participants said.** P31 is the richest single source here; P29, P15 and P22 say it more briefly; P28 dissents.

**Quotes selected for this section.**

Researcher: Am I correct in saying you should be using the agents a little bit more often now?

*"Whatever lack of optimality that comes from using the AI agents is more than compensated for in their speed. Or at least in their speed of decision making. I also feel like the longer that this session has gone on, the less oversight that I've been giving it, and the more I've been focusing on the strategic layer and the incoming missions." — P31 (think-aloud s2 0:20:58)*

Researcher: So you say you especially trust it with a large mission. You mean you're happy to rely on it because of the speed?

*"Yeah. I think even if the tactical agent gave me a dodgy suggestion and the mission failed and I lost points, I probably would have gained more points from the time saved" — P11 (debrief 0:29:11, includes the researcher's words)*

*"Speed is the name of the game. So sometimes in these ones, I guess making a decision earlier is beneficial because at some point the other ones are going to pop up." — P22 (think-aloud s1 0:04:20, includes the researcher's words)*

### 8.1.3 Trust earned by comparison, and doubts that came with practice

[Pass 5: for the paper] Your note: Minimal but nice

Where trust was earned, it was by comparison. P32 trusted the tactical plan because it "seemed to be doing what I was doing" and the cards did not; P29 tested themself against the assistant early and found their own builds were the ones that failed; P11 decided they had been "too harsh" once Suggest matched their simple plan at the same speed. Practice cut both ways: P15 only saw the assistant's mistakes once they had time to look, and suspects they were there all along.

**What participants said.** P32, P29, P11 and P15.

**Quotes selected for this section.**

Researcher: So I would start with just how did you find that? Anything immediately that you would comment on?

*"I think I developed a stronger opinion on how much to trust each of them. By, I guess, comparing them to what I would have done. And I mean, I understood the task more as I went. So the tactical one seemed to be doing what I was doing. So quite reliably. And obviously the strategic one wasn't. So I guess it was very much comparing them to what I'd do. And then letting the tactical one get on with it." — P32 (debrief 0:23:45)*

Researcher: The degree to which you trusted the decisions the AI was making, uh, and how that progressed throughout the study?

*"I mostly trusted it, although, uh, I did kind of want to test it more at the beginning, like comparing it, like doing a few manual ones myself to see, like, whether it would be better, faster. I guess it was kind of hard to keep track of which tasks were being finished when, but I noticed that the ones that failed were normally cause I'd just done a manual one. So I guess that made me want to use the AI more often." — P29 (debrief 0:25:59)*

## 8.2 Who decides, and who answers for it

**Summary.** Asked about real search and rescue, nobody rejected assistants like these, but almost everyone wanted a human to keep the decision, and the reasons were moral as much as practical: accountability for a life not saved, a human "connected" to the operation, and the plan held in someone's head. People were candid that the human they want in the loop may not be able to check much. Against the grain, P13 wants enough automation that "the end users could be anyone".

*Strength of evidence (register rating): qualitative. Your verdict: keep.*

### 8.2.1 Accountability that just isn't there

[Pass 5: for the paper]

P31 sets out the dilemma: even if an AI saves more lives on average, when it fails to save one, "that accountability just isn't there". P22 weighs the same discomfort against saving five people sub-optimally rather than none, and settles on a human making the final decision. (P22's "a bad approximation is better than no response at all" refers back to an item of the pre-study questionnaire.) P31 also raised, unprompted, that this research "will inevitably lead to it being used in military".

**What participants said.** P31 and P22.

**Quotes selected for this section.**

> Researcher: But one of my questions is if in five years' time this was much more sophisticated, far more complicated, how would you feel about search and rescue scenarios being conducted with AI systems of this type in this kind of setup?
>
> *"Yeah, I feel like that's a very interesting question for more of a moral reason, because if you look at it entirely from the perspective of your option A, option B, and option A saves more lives, that always seems like a good thing, but taking it ... in a slightly less simplified way, you've got to make sort of moral judgments about how these resources are being allocated, and if someone's life was like failed to be saved, under a completely human system, you'd have some sort of like, not responsibility, what's the word I'm looking for? [interviewer suggests the word 'accountability']. Accountability, thank you! Of who was on the job that failed to save your life, whereas if it's an AI system, even if it performs on average better than a human, that accountability just isn't there." — P31 (debrief 0:27:53, includes the researcher's words)*
>
> Researcher: How do you feel about people using systems like this in, um, high risk scenarios?
>
> *"I think definitely with emergency scenarios, it's best to just get something going, even if it's not the most perfect plan. And AI lets you do that quite good, like quite quickly." — P29 (debrief 0:27:05, includes the researcher's words)*

### 8.2.2 A human in the loop, even one who cannot check

[Pass 5: for the paper]

Some want the human there regardless of what they can do: P24, pressed on whether a busy human could check, concedes they could not and still wants "a human stand here"; P28 says "humans should make that decision" because "we cannot guarantee what type of answers it will provide". P32 gives the organisational version: measure where people and agents add value and assign the tiers accordingly.

**What participants said.** P24, P28 and P32.

**Quotes selected for this section.**

> Researcher: If we took this five years in the future, and systems of this type were being used in real search and rescue scenarios with more complicated AIs and operators being assisted, how do you feel about that? And do you see any potential for good and for bad of doing that sort of thing?
>
> *"I think if I were being organisationally strategic about that, it seems like it's really helpful. Assuming that you're human constrained (if you've got loads of spare humans, I'd want to see if the AI was actually better) but assuming that you've got less humans than you ideally would have, it feels like a good accelerator. I'd want the analytics on, I guess, like drone utilisation and mission success. I feel like that would be a good place to be making these decisions. And should I be, for example, in this case where I think the strategic one was less good than the tactical, if I were being really strategic, I'd be saying: 'okay, people are going to do the strategic stuff, and we're just going to trust it for the tactical'. So I guess very much doing that optimisation around: 'is it the agents or the humans who are adding the most value, and where do they add the most value', and then have them allocate them... I'd want to understand, well, just because I like to understand everything, I'd want to understand empirically actually how it was performing, who should be doing what." — P32 (debrief 0:26:10, includes the researcher's words)*

### 8.2.3 Ownership, and the opposite view

[Pass 5: for the paper] Your note: This is one of the better ones

For some the reason to keep work is not safety but connection. P11 does not want to be "reduced to pressing allocate, suggest, deploy, because otherwise I might as well be in the next room", and admits to pride and "some sort of grudge" against the computer. P21 keeps both decisions because the allocation already carries a plan in their head that the assistant would "disrupt". P13 takes the opposite view: automate everything not ethically constrained, so that the operator's skill matters less.

**What participants said.** P11 and P21 against P13.

**Quotes selected for this section.**

> *"I don't want to be reduced to pressing allocate, suggest, deploy, because otherwise I might as well be in the next room. And I think the small missions give me the chance to do that more. And I'm like, 'oh, I can just do that. Don't worry. I'll do that'. Done. And I feel satisfaction." — P11 (debrief 0:30:55)*

> *"it would be a dreadful task without any of the tools, especially on the tactical side. It's just a lot of laborious dragging, holding shift at the right time.... And I guess I do feel I want to use (empahsis)my(/emphasis) solution, and I don't want to like, oh, you know, of course the computer thinks it's better. Or like personify the computer and feel some sort of grudge to hold it." — P11 (debrief 0:26:55, includes the researcher's words)*

## 8.3 Failure without feedback

**Summary.** Drone failures were random and uniform, and nothing on screen said so or said how likely they were. Operators felt the lack: P13 is "making blind decisions", P31 cannot tell whether their nudging helped, and several wanted failure rates tied to drone type or place built into the allocation. Without information they built their own theories (hostile places, "fires or something") and blamed themselves.

*Strength of evidence (register rating): qualitative. Your verdict: keep.*

### 8.3.1 Blind decisions, and failure data built into the plan

[Pass 5: for the paper]

P13 treats allocation as a decision about risk that cannot be made well without the failure rate, and later proposes feeding a terrain-based estimate into the allocation rule so that the reserve is sized to the risk rather than padded. P10 wants failure rates by drone type from historic data. (In P11's and P10's debriefs the idea of per-area risk was raised by the researcher; P13 arrived at it themselves.)

**What participants said.** P13, P10 and P11.

**Quotes selected for this section.**

> *"I think in a in a real world scenario where if I imagine I'm a drone swarm operator you will have some numbers presumably about 'so what's the likelihood for drone failure' right? Based on historic data, and that will presumably bias (or inform, not bias) but inform your selection if I'm like okay the lifter drone is much heavier therefore there's more wear and tear on it or whatever it's more likely to fail down that means that my redundancy strategy might change from 'send one additional drone for the lifter' towards 'send 20 percent more more lifters than you need" — P10 (debrief 0:28:53, includes the researcher's words)*

## 8.4 The hierarchy as one plan: how the two decisions connect

**Summary.** Operators did not experience two independent assistants. They describe one plan that starts at the strategic step: over-allocate at the top and the shortage cascades down; allocate generously and there is "really only one way" to assign the drones; build the allocation yourself and the tactical plan is already in your head. Several place responsibility for failures with the strategic tier, and use the cross-mission view the cards lack (who can wait, what to hold back). Whether one plan held across the tiers is the right ideal is the question the practitioner's account, at the end, takes up and answers differently.

*Strength of evidence (register rating): qualitative. Your verdict: core.*

### 8.4.1 The strategic step carries the risk

[Pass 5: for the paper]

P19 reaches the conclusion from the results: the tactical assistant "did everything", and the only remedy for failures is sending enough drones at the start, "the strategic assistant's prerogative". P28 weighs the same step from the other side: they hold drones back in the base because "I cannot predict when the next task will appear", and accept that a colour sent as a single drone may fail.

**What participants said.** P19 and P28.

**Quotes selected for this section.**

> *"Basically, it did everything. If failure rate of drones is unrelated to my assignment, so rate of failure as opposed to how many drones I send out, then all things considered, the tactical assistant is probably the best. The tactical assistant completed all the jobs and we didn't end up in any kind of a situation where we didn't have enough drones if it hadn't been for random drone failures. And the only way to remedy drone failures is to have sent enough drones at the beginning, which I think is the strategic assistant's prerogative." — P19 (talk during the questionnaire survey1 0:18:53)*

> Researcher: You found that one more demanding?

> *"I'm trying to be more defensive in the strategy, so I'm trying to reserve more and more drones in the base, rather than send them out to finish the task quickly, because I cannot predict when the next task will appear. So luckily, there's only one task is holding for a long time, only one there, and another problem is [that] such a choice will make the drones in the case becomes not sufficient for that case. So if I only send out one drone for one specific color, and it fails, then I have to abort that mission, and I only abort one of them, because that happened, and the rest is still able to finish, so I think this, such a deployment is still all right." — P28 (talk during the questionnaire survey2 0:28:05)*

### 8.4.2 One plan across both decisions

[Pass 5: for the paper]

The card-then-Suggest pattern in the logs (Section 7.3) is described from both sides. P10: a generous card means "there is really only one way of assigning them, which means if I just hit the suggest button then I can focus on something else". P21: building the allocation by hand lays out a plan in the mind, so switching to Suggest would disrupt it; their own plans chain drones through each other's tasks to get "redundancy for free". P21 also wonders whether the tactical assistant follows the strategic one's patterns. The practitioner's account at the end disagrees with this ideal: in staff planning each cell sees only its part, which is fine if each part has been through a sound process.

**What participants said.** P10 and P21.

**Quotes selected for this section.**

> *"it's better to use the autonomy on the tactical level on top of the strategic" — P10 (debrief 0:25:46, includes the researcher's words)*

> *"[Having manually set a strategic allocation] I prefer to [tactically] set the drones myself. Because I'm not sure how exactly it will be allocated [if I use the agent]. Because I have something in mind when I consider the allocation... When I make the selection for the drones, I kind of lay out a plan in my mind initially. So I feel like it's going to disrupt my thought process if I try to go around that." — P21 (think-aloud s1 0:14:16, includes the researcher's words)*

> *"Actually, they might work better if I use the AI agent for the selection and that as well, because. Yeah, because when I assume when it makes a selection, when it tries to make the drone allocation, it follows certain patterns, which could then be adopted by the tactical agent afterwards. And if I try to think of something else, so I'm going to split them like that and change them in a certain way. I'm not sure if the tactical agent will be able to match my thought process." — P21 (talk during the questionnaire between1 0:20:36)*

### 8.4.3 The cross-mission view the cards lack

The strategy cards look at one mission. Operators override them to act on what the cards cannot see: P22 strikes a "middle ground" to keep drones back for other tasks; P24 trims cards not because they are better than the AI but to save drones, and only then learns the cards see one mission; P19 reasons about who can afford to wait; P31 goes below even the Conservative card when other tasks are pending.

**What participants said.** P22, P24, P19 and P31.

**Quotes selected for this section.**

Researcher: Could you just summarize again, just in a sentence or two, how you were approaching each one, what scenarios you felt you were using each one?

*"I felt like sometimes I could find like a middle ground that had like maybe a little bit of gambling in there, essentially, that you wouldn't have the necessary coverage. But then also then that meant that you'd keep some more in reserve. So it wasn't just fully because it's cautious, it was also because it was actually to keep some in reserve for other tasks, because, yes, you're covering this task specifically very well. But that doesn't mean that then you actually have enough coverage for the, for the rest of your tasks. So I felt that often I could find, I could strike a middle ground. So I felt comfortable assessing and then overriding the strategic assistant." — P22 (debrief 0:28:28)*

*"And the problem is we don't know whether a extremely critical job is going to show up later." — P19 (think-aloud s1 0:09:07)*

Researcher: Now that I've had a little bit more time, I'm just thinking if my opinions have changed on this slightly. And for what reason?

*"If you wanted to send as few drones as possible, you could send four fast drones, one lifter, one camera. Whereas even the conservative recommendation recommended sending two of the lifters, and the cameras in case of drone failure. When I was in that situation, I had a lot of other pending tasks, so I went more conservative than the conservative recommendation." — P31 (talk during the questionnaire survey2 0:23:49, includes the researcher's words)*

## 8.5 New jobs for an assistant

**Summary.** Asked what the assistants should do, people rarely asked for more automation of the same kind. They asked for assistants that check: one that reviews the operator's own plan for slips rather than taking it over (P11). They asked for smarter rules (redundancy scaled to failure risk, backup routes in the tactical plan), for options between and beyond the two cards, and for an interface that knows what they have already planned. Two (P27, P32) asked for something neither assistant does: help deciding which waiting mission to open next.

*Strength of evidence (register rating): qualitative. Your verdict: keep.*

### 8.5.1 An assistant for what to open next

[Pass 5: for the paper]

Neither assistant helps with the first decision of every allocation: which waiting mission to open. P32 asked for it in play ("a suggestion of what to do next would be handy"), because a mission the reserve cannot cover sits at the top of the list anyway; P27 gave a rule, missions that can be done first, then by criticality, allowing for what is already running. The logs show the gap such an assistant would fill. With two or more missions waiting (411 decisions, 33 people), operators open the oldest (49%; chance 34%) and lean towards the more critical ($\times$1.45 per category above the waiting set's average, p = .02), but not towards missions they can actually cover: whether the reserve could supply a mission's drones makes no difference ($\times$1.04, p = .79), and at 41% of decisions the oldest waiting mission could not be covered. P27's order differs from what operators did at 59% of decisions. The queue had a "Sort: Score" toggle; the researcher told P27 only one participant had found it, and it is not logged.

**Key numbers.**

- Decisions with 2+ missions waiting: 411 (33 people). Opened the oldest 49% (chance 34%); when one waiting mission was uniquely the most critical, opened it 49% (chance 33%)
- Choice among the waiting missions (conditional logit, person bootstrap, 400 draws): oldest $\times$2.75 [1.97, 4.12], p < .001; newest $\times$1.64 [1.17, 2.32], p < .001; criticality, per category above the set's mean $\times$1.45 [1.09, 2.15], p = .02; coverable from the reserve $\times$1.04 [0.72, 1.58], p = .79; size per 3 drones $\times$0.95 [0.78, 1.11], p = .61
- The oldest waiting mission could not be covered from the reserve at 41% of these decisions; some waiting mission could not be covered at 57%. When some could not, the one opened was coverable 32% of the time (chance 39%)
- P27's order (coverable, then most critical, then oldest) picks a different mission from oldest-first at 59% of decisions; there operators opened P27's pick 35% of the time (chance 30%)

**Method.** Same decisions and waiting sets as the triage choice model (load.py), with two features added per waiting mission: its category (A–E, the penalty rate the operator sees), centred on the set; and whether the drones available in the reserve just before the decision covered its primary composition in every colour. Coverable is strict: tasks also have substitute compositions, and drones returning to the hub are not counted, so it understates what an operator could have done. scripts/register/pass2/queue_order.py; results in pass2/load.json, queueOrder.

**What participants said.** P27 and P32; both cuts include the researcher's words (flagged).

# 9 A Practitioner's Reading

[Generated (`practitioner.tex`). One participant with operational military experience, one point per passage you picked, each with its links back to the results above (agrees / qualifies / goes beyond) as cross-references; the last subsection gathers every link by kind. In the paper, describe them only as "a participant with operational military experience".]

## 9.1 A practitioner's reading: P36, a participant with operational military experience

**Summary.** One participant had operational military experience. They played like everyone else, and differed in who they are: the oldest participant, the only one with extensive command experience, and among the most positive about AI. Their sessions and debrief are the richest material in the study, and nearly every result here has a passage of theirs that agrees with it, qualifies it or goes past it. Each point below is one passage with its own links back; the last section gathers them. This is one person's expert commentary, not evidence about the cohort; in any paper, describe them only as "a participant with operational military experience".

*Strength of evidence (register rating): qualitative. Your verdict: core.*

### 9.1.1 How typical are they?

[Pass 5: numbers only] Your note: We will say that our expert was a good performer with a slightly higher workrate than others, prempting cards more, but was otherwise fairly typical (insert a couple numbers into here)

**Typical in how they played.** On every habit the findings measure (card use, how card use grows with mission size, Suggest use and how it grows with plan size, keeping Suggest plans unedited, the drones on a hand build, oldest-first) they sit in the middle half of the cohort. They were a little quicker on the strategic step (9.6 s per decision against a median of 11.4) and decided before the cards loaded a little more often (32% against 10%), at the 20th and 80th percentiles, and they scored a little above most (0.7 SD within game, 80th percentile). **Different in who they are.** They are the oldest participant (44; median 27), the only one with extensive command experience (24 of 33 report none), one of four with moderate drone experience, at the top of the AI-attitude scale (10 of 10, with one other), and among the least insistent that a human keep authority (3.0 against a median of 4.3). Their trust in the Strategic Assistant was higher than most (5.9; median 4.5). So the commentary that follows comes from someone whose play was ordinary: what they add is the reading, not a different way of playing. (Figure 39)

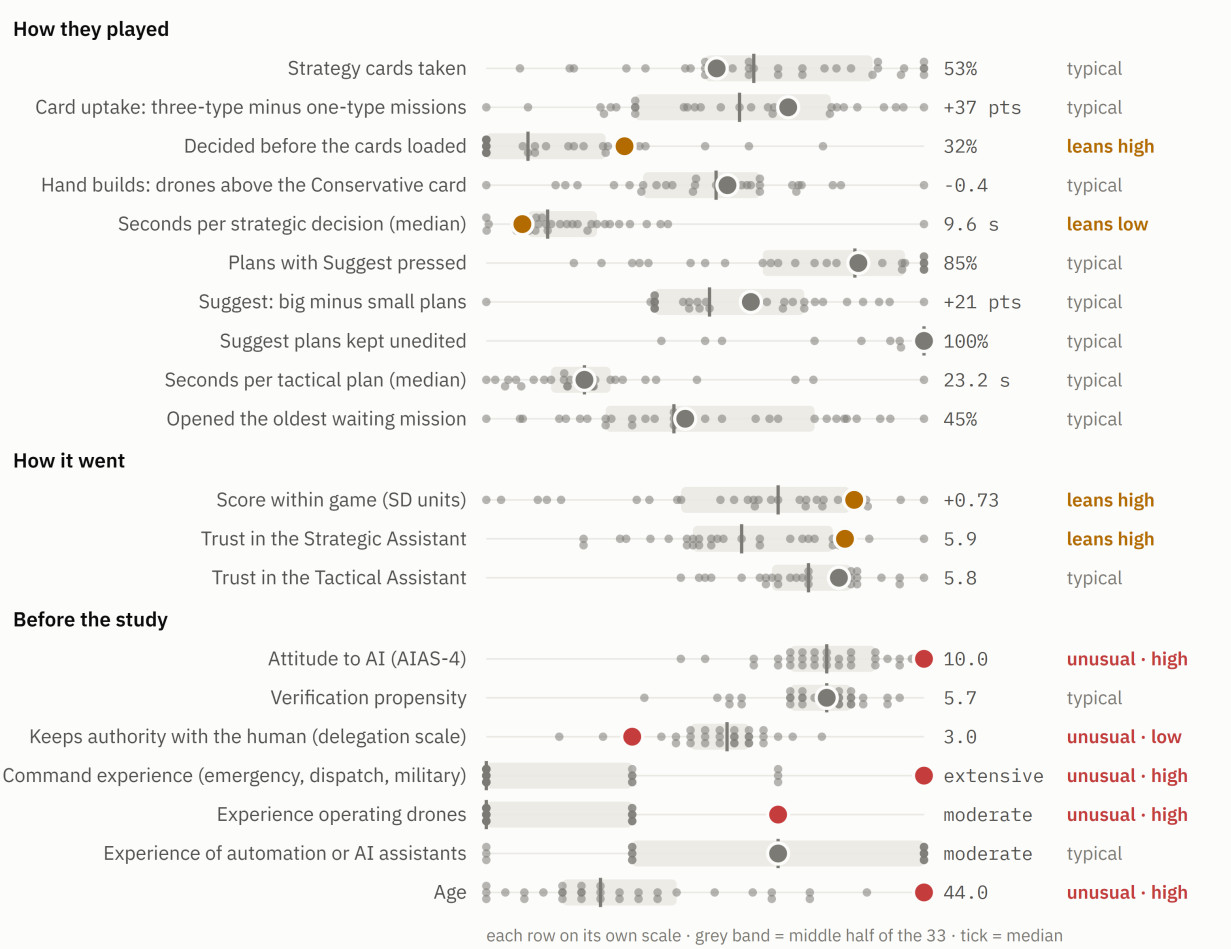



**Figure 39: How typical are they?**

**Method.** One value per person for each measure (33 people, P09 excluded): play habits from the same per-person table as the size and speed findings, oldest-first from the queue choice decisions, trust as the mean of the session ratings, and the pre-study questionnaire (experience items scored none 0 to extensive 3). P36's percentile rank among the 33 (ties share the middle). Typical = middle half; leans = 10th–25th or 75th–90th; unusual = beyond. With one person nothing is tested; this only places them. scripts/register/pass2/typical.py; results in pass2/typical.json.

### 9.1.2 How they played: for the whole board, not the mission

[Pass 5: for the paper] Your note: We'll briefly quote the second part I think

Five minutes in, P36 states the aim as throughput across every task while watching the reserve ("running a little bit low on blue"), and later gives the reason: "success is generated across all of the tasks rather than just an individual task". The two passages make one point, so they share it. This is the cross-mission view operators said the cards lack, and it fits the cohort's hand builds, which mostly sit at or below the cards.

#### Quotes selected for this section.

> *"so generally speaking all I'm trying to do here is get as many across as quickly as I can to complete all of the tasks managing my resources that I've got left here so I'm running a little bit low on blue over there and then just double checking on here so I've got the redundancy in there" — P36 (think-aloud s1 0:05:32)*
>
> *Your comment: Minor*
>
> *"so when I think it's going to be quicker because it's not necessarily about just that task it's about all of the tasks and success is generated across all of the tasks rather than just an individual task." — P36 (think-aloud s1 0:05:58)*
>
> *Your comment: Interesting*

#### Links back to the results.

- *Agrees with* Section 7.3.6, *Where the tactical time goes: hand-built and Suggest plans are close, with a small edge to Suggest in carrying the plan out*: Their aim is throughput across the board ("get as many across as quickly as I can to complete all of the tasks"), the same reason the cohort gives for starting Suggest and moving on.

#### Links back to the results.

- *Agrees with* Section 8.4.3, *The cross-mission view the cards lack*: Operators overrode the cards to act on what one mission's card cannot see. P36 says why in one line: "success is generated across all of the tasks rather than just an individual task".
- *Agrees with* Section 7.4.1, *Each manual decision against the Conservative card*: Playing for every task shows in the cohort's hand builds: on 44% of decisions they sit below both cards, mostly by leaving off the card's spare, which keeps drones for the next mission.

### 9.1.3 How they played: the card can be slower than doing it yourself

[Pass 5: extra: may make it in]

Three minutes into the first session, after taking a card on a small mission, P36 decides it would have been quicker by hand. It is the speed result in miniature: the card saves time only on big, three-type jobs.

#### Quotes selected for this section.

> *"Do you know what? Probably quicker just to hit three on there. But we'll go three on that one there. [. . . ] I probably should have done that one manually. It would have been quicker." — P36 (think-aloud s1 0:03:21)*

#### Links back to the results.

- *Agrees with* Section 7.3.5, *Why the rule makes sense: the card pays only on big strategic jobs; Suggest is not slower*: The card saves time only on big three-type jobs. Three minutes into their first session P36 finds it out: "Probably quicker just to hit three", then "I probably should have done that one manually. It would have been quicker."

### 9.1.4 How they played: simple by hand, complex to the assistant, and why

[Pass 5: for the paper] Your note: This one is a new idea that we can talk about in more depth in the paper: More complicated tasks can have a more branching failure space. This is not something we cover in this work, but could be an important factor in future works.

The fullest statement of the size rule in the study, with both halves of the reason. On a simple mission they already know the answer by eye, and they build in redundancy (every drone to every task) so that after a failure "you don't really need to change the story". On a complex one the options branch, so they let the assistant start and step in when needed. They also name the limit of the quick check that goes with handing over: whether it is right "is another question entirely". Their own log follows the rule.

#### Quotes selected for this section.

> *"sometimes it's going to be a little bit uh quicker to do these myself only because i know the answer based on what i'm seeing visually in front of me so i don't need to do that much [...] simple task you probably want to do yourself if you're going to get the solution quicker and the more complex tasks you're going to want somebody to give you that solution in front of you so you can give it a quick once over and go that looks about right um whether or not it is right of course it's another question entirely [...] the solutions it is coming up with uh i would have to go and do a much more thorough scan of the solution to say have you put redundancy into that plan or do i care like i've put redundancy into that plan so i don't have to come back to it i've put every turn on that last blue one every single drone is going to every single task so that if one of my drones fails my assumption is that my other drones are picking up the task whereas the more complex ones just literally looking have they allocated enough to that task started going because if it fails um think about like hypertextual branching paths so as soon as i get one failure that's going to change the entire story how much of that story do we need to change on the blue ones you don't really need to change the story you're just going to flick through whereas the other ones it's like okay no i've got more options so my option scope is broader and so it's going to need more human intervention to go into it so there's no point in me trying to think about all of that ahead of time i may as well let them start on that task and then intervene when i need to" — P36 (think-aloud s1 0:09:12, includes the researcher's words)*

#### Links back to the results.

- *Agrees with* Section 7.3.1, *Both tiers on one axis: the bigger the mission, the more it is handed over*: Small jobs by hand, big jobs to the assistant (cohort: card 56% to 75%, Suggest 54% to 91%). P36 gives both halves of the reason: on a simple job "i know the answer based on what i'm seeing visually", and a failure needs no re-plan ("you don't really need to change the story"); on a complex one "my option scope is broader", so "let them start on that task and then intervene when i need to". Their own log follows it (card 33% to 70%, Suggest 79% to 100%).
- *Agrees with* Section 7.9.3, *Why the tactical plan is kept: checking it is brief and does not grow with the plan*: "Give it a quick once over and go that looks about right ... whether or not it is right of course it's another question entirely": the brief check the timing data show, with its limit named.
- *Agrees with* Section 8.4.2, *One plan across both decisions*: P21 chained drones through each other's tasks for "redundancy for free". P36 does the same on simple missions ("every single drone is going to every single task so that if one of my drones fails ... my other drones are picking up the task"), which is why those plans need no checking afterwards.

### 9.1.5 How they played: the rule bends when the reserve is full

[Pass 5: extra: may make it in] Your note: minor, may make it in

A simple mission, so by their own rule one to do by hand; but the reserve is full and nothing is coming in, so they press Suggest anyway. The size rule is a default, not a law, and slack loosens it. It also cuts against the idea that people hand over because they are busy: here they hand over because they are not.

#### Quotes selected for this section.

> *"so again this one here that's relatively simple and I'm resource rich at the moment so we'll go nuts I'll start suggesting some stuff through because there's nothing else that's coming in" — P36 (think-aloud s2 0:20:33)*

#### Links back to the results.

- *Qualifies* Section 7.3.1, *Both tiers on one axis: the bigger the mission, the more it is handed over*: The size rule bends with slack: this mission is "relatively simple", but "I'm resource rich at the moment so we'll go nuts I'll start suggesting".
- *Agrees with* Section 7.6.1, *Queue depth against reliance — raw, and adjusted*: A busier moment does not push people onto the assistants. P36 presses Suggest at a quiet moment, "because there's nothing else that's coming in".

### 9.1.6 How they played: oldest first, and never without a reserve

[Pass 5: for the paper] Your note: Short note - they identified an classified a rule "never without a reserve". Links to the consistency factor in training.

Two of the cohort's habits in one sentence: open the mission that has waited longest unless something changes the priority, and build by hand with a spare ("minimum two, we're going to go three"). Their builds still land where the cohort's do relative to the cards.

#### Quotes selected for this section.

> *"this one's been there for the longest i've got nothing in there that's going to change my prioritization so i'm going to go into here and probably send manually i'm going to give you minimum two we're going to go three just never deploy without a reserve" — P36 (think-aloud s1 0:07:50)*

#### Links back to the results.

- *Agrees with* Section 7.7.1, *Order: still oldest-first well above chance*: Operators open the oldest waiting mission well above chance. P36: "this one's been there for the longest i've got nothing in there that's going to change my prioritization".
- *Qualifies* Section 7.4.1, *Each manual decision against the Conservative card*: Most of the cohort's hand builds sit at or below the cards (44% below both). P36 adds a spare on purpose ("minimum two we're going to go three just never deploy without a reserve"), yet their builds land where the cohort's do: on average 0.4 of a drone under the Conservative card, against a median of 0.6 under.

### 9.1.7 How they played: critical tasks over score

[Pass 5: for the paper] Your note: good

Out of Fast drones, they accept a lower score to get the critical tasks done, "if this is a real life scenario". The cohort leans the same way, opening the more critical waiting mission more often; P36 says the trade aloud and frames it as what a real operation would demand.

#### Quotes selected for this section.

> *"i've run out of blue so that's gonna hurt my score but at the same time i've got the critical tasks done so i'd rather do the critical tasks if this is a real life scenario i'd rather be focused on the critical tasks than getting those done to a higher value rather than moderate tasks of getting them done to a low" — P36 (think-aloud s1 0:07:50)*

#### Links back to the results.

- *Agrees with* Section 8.5.1, *An assistant for what to open next*: Operators lean towards the more critical waiting mission ($\times 1.45$ per category, on top of oldest-first). P36 goes further and spends score on it: "i've run out of blue so that's gonna hurt my score but ... i'd rather do the critical tasks if this is a real life scenario".

### 9.1.8 How they played: trusting the strategy card more, then seeing why that is a risk

[Pass 5: extra: may make it in]

One of the few who trusted the Strategic Assistant more than the Tactical, and their ratings say the same. The reason is that checking a card is easy (is there a reserve?), and a strategic mistake seemed less costly; then, mid-sentence, they reverse it: a bad allocation at the start leaves the tactical step "screwed". Easy to check is not the same as low stakes.

#### Quotes selected for this section.

> *"the strategic one i was less concerned about the solutions it presented so i hesitated less on the strategic than the tactical but only because all i was looking at there was have i got a reserve allocated whereas a tactical one it's 'have you covered all the tasks' and there is a reserve [. . . ] I wouldn't trust any of them without checking but I trust the strategic slightly more because the consequences are slightly - less they're probably not actually are they if you fuck up the allocation at the start of it then you're screwed going into the tactical" — P36 (talk during the questionnaire survey1 0:13:46)*

#### Links back to the results.

- *Qualifies* Section 7.9.1, *Trusted more: the basic result*: The cohort trusts the Tactical Assistant more; P36 is one of the exceptions, and their ratings match what they say ("I trust the strategic slightly more": 5.9 against 5.8 on the 1–7 scale, where the cohort's median strategic rating is 4.5). They hesitated less over the cards only because "all i was looking at there was have i got a reserve allocated".
- *Agrees with* Section 8.4.1, *The strategic step carries the risk*: P19 put the only remedy for failures with the strategic tier. P36 reaches the same point in mid-sentence, and it reverses their reason for trusting the cards: "they're probably not actually are they if you fuck up the allocation at the start of it then you're screwed going into the tactical".

### 9.1.9 The task: physics sets the pace, and the load moves to waiting

[Pass 5: for the paper]

The game's pace is not realistic: real drones are bounded by distance and speed, so decisions get faster but the operation does not. The load does not go away; it becomes waiting and contingency planning ("what am I going to do if this thing fails"), "loaded in a slightly different way".

#### Quotes selected for this section.

> Researcher: so I would ask in your professional opinion obviously this is clearly we're trying to get at something that we think is of interest in this kind of setting and stuff in reality I'd imagine the strategic and tactical thing would probably be done by two people but how relevant would you say this kind of set up is
>
> *"so in terms of delivery absolutely in terms of the pace probably not so much [...]if you think so the biggest thing for us that is always hard to convey to decision makers is: physics remains the bounding factor I've got to get from this point here to that point there and if the adversary can see me from 20k away then I'm doing that activity from 20k and if you're going at 3k an hour then that's going to take you a long time to get from here to over there so there's a level of that acceleration in terms of decision however on the flip side of it when the shit hits the fan you are probably going to be very frustrated; As I move this thing to here I've now got to wait for this thing to get into place to go and do this thing here hurry up and get there but at the same time as a commander you're sat there thinking okay what am I going to do if this thing fails what am I then going to do what am I going to do so your mental load is still loaded it's just loaded in a slightly different way so you're always trying to balance cognitive capacity against the time pressure effectively" — P36 (debrief 0:26:31)*

#### Links back to the results.

- *Qualifies* Section 7.6.1, *Queue depth against reliance — raw, and adjusted*: A busier queue did not push anyone onto the assistants. P36 puts the real pressure elsewhere: "physics remains the bounding factor", and the load becomes waiting and asking "what am I going to do if this thing fails". "Your mental load is still loaded it's just loaded in a slightly different way."
- *Agrees with* Section 7.3.6, *Where the tactical time goes: hand-built and Suggest plans are close, with a small edge to Suggest in carrying the plan out*: The time Suggest frees is spent elsewhere while a plan runs. P36 describes that dead time from the commander's side: "I've now got to wait for this thing to get into place", and uses it to plan for failure.
- *Agrees with* Section 7.10.10, *It's a game: what real stakes, pace and training would change*: Others said the pace was unreal; P36 agrees ("in terms of the pace probably not so much") and says what sets the real pace: "if you're going at 3k an hour then that's going to take you a long time".

### 9.1.10 The task: the pressure lands on the lower tier

[Pass 5: for the paper] Your note: maybe include

Higher up, decisions look further out and a new item is routine; at the lower tier everything arrives at once and must be sequenced. The game gave one operator both tiers; in a real hierarchy the time pressure the workload findings looked for would sit with the tactical level, and so would any assistant for what to do next.

#### Quotes selected for this section.

> Researcher: and I suppose I think it would be fair to say that a lot of traditional ways this kind of stuff is working I know people do incredibly long shifts and we're talking about watchkeeping over a wide area with increasing use of systems like this in a more I suppose the pace you can talk about that the distance context those distances are scaled down quite a lot within a tactical kind of small scale
>
> *"so it's how you're talking about when does the effect need to be delivered and coordinated in time and space then it's the higher up you are the further out and then the less so when you see something, 'cool I've now got to pass that information down but that's just like a routine routine task I'm going to move on to the next task' whereas if I see 10 things in front of me as it's all come down the chain I've got to do this thing now then that thing then that thing then that thing and then for it to stack back up the chain so yes there is a level of that is getting both faster" — P36 (debrief 0:28:21)*

#### Links back to the results.

- *Qualifies* Section C.3.2, *Said vs did: the people who said they hand over when busy*: Nine people said they lean on the assistant when busy, yet their reliance did not rise with the queue. P36 puts the busy in a real hierarchy: "the higher up you are the further out", where a new item is "just like a routine routine task"; at the lower tier "10 things in front of me as it's all come down the chain". The game gave one person both tiers.
- *Goes beyond* Section 8.5.1, *An assistant for what to open next*: P27 and P32 asked for an assistant that says what to open next. P36's lower tier is where that job would sit: "I've got to do this thing now then that thing then that thing then that thing".

### 9.1.11 Control: trust in the process, as with staff

[Pass 5: for the paper]

At battlegroup level and above (glossed in the quote as the study's strategic tier), a commander takes plans worked up by staff, because one person could not do the analysis in time, and trusts them because they have done those jobs and know how the answers are made. The link to the assistants is the gap: no operator has done the assistant's job, so the basis for that trust is missing.

#### Quotes selected for this section.

> Researcher: speaking specifically about the relationship between you and these two assistants these being autonomous systems yeah obviously that's not a million miles away from that being a human assistant who's helping you there that it's gonna be you and two people right do you have any feeling about the the difference there you spoke a lot about socializing socializing the decision-maker something like that how does that compare do you think to if you had two subordinate people here
>
> *"well it's kind of this how it works in military planning anyway so in anything above battlegroup level [our 'strategic'] there would be a staff there that would work up plans socialise those plans amongst themselves and back brief me as a commander. I would then go okay yeah I want you a bit of that plan and a bit of that plan give me this new plan. They'll go and write up by then sort of basically finish off that way. So it's a it is always a collective input into it but I'm just getting the synthesis of them work because I can't as an individual I couldn't work through the ground the enemy, the enemy's plan, the likelihood of my plan succeeding, the logistics behind it it would take me forever which is why your staff do it, so at that level you're already in a position where somebody's just coming back to you with the answers as long as you have trust in how they've come up with those answers then if it looks right: because you've gone through the process yourself [...] you'd have worked up in those levels so you kind of understand how those answers are generated so there's a level of trust in that process" — P36 (debrief 0:35:27, includes the researcher's words)*

#### Links back to the results.

- *Agrees with* Section 7.9.3, *Why the tactical plan is kept: checking it is brief and does not grow with the plan*: Plans are accepted after a brief look. P36 explains why that can be rational: answers from staff are taken "as long as you have trust in how they've come up with those answers", not re-derived.
- *Agrees with* Section 7.10.3, *Wanting to know how the assistants decide*: Several operators wanted to know how the assistants decide. For P36 that is the basis of trust: a commander can take the staff's answers because "you'd have worked up in those levels so you kind of understand how those answers are generated". No operator has done the assistant's job.
- *Agrees with* Section 7.3.5, *Why the rule makes sense: the card pays only on big strategic jobs; Suggest is not slower*: The card pays where a job is too big to do quickly by hand. P36 gives the same reason for having staff at all: "I couldn't work through the ground the enemy, the enemy's plan, the likelihood of my plan succeeding, the logistics behind it it would take me forever which is why your staff do it".

### 9.1.12 Control: verify by socialising the plan, and trust the maths

[Pass 5: for the paper]

Verifying, for P36, is a process around the plan (take it to those who will carry it out, keep it open to challenge), plus trust that the machine has "done the maths" and allocated enough. The second half is about what a two-second look at a tactical plan can confirm; the first is something the study's set-up has no place for.

#### Quotes selected for this section.

> Researcher: so you thought they say I might be worse
>
> *"I write a plan I will go and take it to the lowest tactical level which is something like this where you're moving different units around different people you'd probably speak to the platoon commander you'd speak to the platoon sergeant: 'I'm thinking of doing this', you'd bring your section commanders in you'd go through yours and that's the process 'guys this is what I'm thinking and this is what your likely mission is, go away and work out the plan'. [...] As humans we should be used to socialising our decision making process to an extent. How you do that is entirely up to the commander but at the same time I'm still going through a process where somebody is checking my thinking and has the option to challenge me so with AI it should always be open to challenge [...] but also trust in the fact it's done the maths behind it and has allocated enough platforms to do the task you're after" — P36 (think-aloud s2 0:17:40, includes the researcher's words)*

#### Links back to the results.

- *Qualifies* Section 7.9.3, *Why the tactical plan is kept: checking it is brief and does not grow with the plan*: This is from someone who kept every consulted tactical plan unedited. In P36's account verifying happens around the plan, not in a long look at it: take it to the people who will carry it out ("go away and work out the plan") and keep it "open to challenge". Beside that sits "trust in the fact it's done the maths behind it and has allocated enough platforms", which is about all the cohort's 2–4 seconds of checking can cover.
- *Goes beyond* Section 8.2.2, *A human in the loop, even one who cannot check*: Some operators want a human in the loop even one who cannot check. P36's check is not one person looking harder but a plan socialised so that "somebody is checking my thinking and has the option to challenge me".
- *Agrees with* Section 7.10.3, *Wanting to know how the assistants decide*: "With AI it should always be open to challenge": the same ask as the operators who wanted to know how the assistants decide.

### 9.1.13 Control: hand over the routine, keep the critical, and pay for it in skill

[Pass 5: for the paper]

Three ideas in one passage. Handing over lets them scan elsewhere and come back. They would take the critical thing themselves and leave the routine to the machine "as long as the resources are in place" (a criticality rule for handing over, which the study has not tested). And the cost of not building the plan is playing catch-up when it fails, so the key is not trusting the AI but the scarce, expensive skill of exploiting it.

#### Quotes selected for this section.

> Researcher: so the last scenario you said something about sort of speed accuracy and the main advantage of the agents being speed
>
> *"If I've sent that off it allows me to scan over here and then go back and check the answer over there. [...] That's what's going to be required if you're doing a drone swarm the whole point of it is to make sure that you have got, but the utility of it is that point of presence that point of intervention where I need something to check that and know I'm going to take over that critical thing myself because it's the most important thing whereas for the routine sort of activity of getting from A to B do you know what whatever you want to do as long as the resources are in place and I can see what resource I'll have to play with if something goes wrong whereas if I did all of that manually I would know all of that because I've built the plan I've got it in my head if I just said 'go nuts' when something goes wrong I've got to play catch up I've got to start thinking how do I now have to do that differently so the key thing for me in all of this is not necessarily trusting the AI it is the skills that a human is going to need to be able to exploit all of this [...] that's what air traffic controllers do and they're actually testing it for it and their pay reflects that because it's not a particularly common skill if you're going to start saying search and rescue teams you now need this skill set it's going to cost you this much to pay these people because they're a niche skill set" — P36 (think-aloud s2 0:20:56)*

#### Links back to the results.

- *Agrees with* Section 7.3.6, *Where the tactical time goes: hand-built and Suggest plans are close, with a small edge to Suggest in carrying the plan out*: Handing over frees the operator to work elsewhere: "If I've sent that off it allows me to scan over here and then go back and check the answer over there", which is where the cohort's Suggest time goes.
- *Agrees with* Section 7.9.3, *Why the tactical plan is kept: checking it is brief and does not grow with the plan*: Brief checking, whatever the plan size. P36 names its price: "if I just said 'go nuts' when something goes wrong I've got to play catch up", because the plan is not in your head.
- *Goes beyond* Section 7.2.3, *Who decides matters more than what is being decided*: Who decides predicts handover far better than the mission does. P36 draws the operational conclusion: "the key thing for me in all of this is not necessarily trusting the AI it is the skills that a human is going to need to be able to exploit all of this", a niche skill set like air traffic control, paid accordingly.

### 9.1.14 Risk: the failure rate of doing nothing is 100%

[Pass 5: for the paper]

Show failure rates, but with context, or people will refuse any risk: a bare "40%" reads as "no chance" when a person would fail as often and doing nothing always fails. Their answer to risk is to act and reduce it, and to train people to read the numbers.

#### Quotes selected for this section.

> Researcher: you would appreciate having a sort of a prior estimation of the failure chance and of what's coming in kind of thing
>
> *"but the interesting thing so that that always messes with people's heads doesn't it because we look at the failure 40% failure chance fucking no chance it's like okay but if you put a person into that scenario and it was all people doing it your failure chance would be this as well so there's got to be something where it shows a bit of context to it so just to understand that when you see that failure rate because everyone will try to push it up and up and up and up and up no one likes gambling particularly gambling in people's lives but particularly in the military environment or even in the search and rescue environment you're always gambling with someone's life like the failure rate of doing nothing is 100% so you've got to do something you're just trying to chip away at it and prioritise accordingly so having that information presented again great to see the information but it's then down to that person that you're training for them to be used to there's always going to be a failure rate I'm just trying to make it as low as reasonable" — P36 (debrief 0:32:25)*

#### Links back to the results.

- *Qualifies* Section 7.10.4, *Failure risk and feedback: wanting the numbers*: Several operators asked for failure numbers. P36 wants them with context ("there's got to be something where it shows a bit of context to it"), because a bare "40% failure chance" reads as "no chance" when "the failure rate of doing nothing is 100%".
- *Agrees with* Section 8.3.1, *Blind decisions, and failure data built into the plan*: Operators facing random failures with no numbers felt they were deciding blind. P36 agrees the numbers should be shown ("great to see the information") and adds that people then need training to act on them: "there's always going to be a failure rate I'm just trying to make it as low as reasonable".
- *Goes beyond* (not in the paper), *Lower-reliance operators attempt fewer recoveries*: Lower-reliance operators attempted fewer recoveries. P36's rule cuts the other way: in search and rescue "you're always gambling with someone's life", so you act and "chip away at it".

### 9.1.15 Risk: real stakes, experience, and the risk the commander no longer shares

[Pass 5: for the paper]

Their decisiveness came from planning experience and knowing it was a simulation. With real lives an inexperienced commander would be reckless or paralysed, and remote autonomy removes the shared risk that used to temper the order. They expect to be less decisive themselves if someone could die.

#### Quotes selected for this section.

> Researcher: I noticed that you're you were very decisive you were you didn't really wait around you just you just did stuff
>
> *"yes well and that's kind of a challenge within this is because I know it's a simulation there isn't someone's life on the line at the end of it and I've done enough planning activities to know and I'm very visual so looking at a glance yes it's allocated what I needed to allocate it cool let's start running with it whereas I can imagine if you've not had that experience that I've generated through non-use of drone swarming technology if you've not had that if you're you know young platoon commander who's literally just come out of Sandhurst and all is only a very experienced perhaps this is a simulation at Sandhurst and all of a sudden it's now you know the bullets are flying it's a two-way range they're gonna be much either they're gonna be foolhardy and just fucking go for it and people are gonna die or they'll be almost paralysed because it's like oh shit no if I do that someone is gonna die here's 40% failure rate on this task I'm gonna lose people but if I don't do that task this whole battle falls apart that's always been true regardless of whether or not technology was in there but before there was a level of interpersonal sort of contact I'm out there with my people I might die alongside them but I'm so I'm when I'm putting them under that risk I'm because I'm taking the same risk myself whereas something like this where you're not necessarily taking the risk yourself and you've put humans albeit they might be one removed from your autonomy but you've put humans on the way then that's going to then come back and impact on your decision-making so I imagine you're going to see me less decisive if someone's gonna die at the end" — P36 (debrief 0:33:43)*

#### Links back to the results.

- *Goes beyond* Section 7.10.10, *It's a game: what real stakes, pace and training would change*: "It's a game" was a common caveat. P36 says what changes with a life at stake ("I imagine you're going to see me less decisive if someone's gonna die at the end"), and that a commander who no longer shares the risk ("I might die alongside them") loses what used to temper the order.
- *Qualifies* Section 7.2.7, *Nothing measured beforehand predicts the style*: Prior experience did not predict style here. P36 expects it to matter most when the stakes are real: their own "looking at a glance yes it's allocated what I needed ... cool let's start running with it" comes from experience, where a new platoon commander would be "foolhardy" or "almost paralysed". A game with no lives on the line hides that difference.

### 9.1.16 Organising assistants: the top allocation sets the risk below

[Pass 5: for the paper]

Under-allocate at the top and the lower tier carries the risk; over-allocate and it is wasted elsewhere. The commander balances several tasks at once and puts more into the critical one. This is the coupling between the tiers that the cohort's card-then-Suggest pattern shows, said from the top.

#### Quotes selected for this section.

> Researcher: one thing that you did clearly is you very much immediately were happy to sort of automate the low level of detail the tactical level and spend more of your attention on the high level which makes a lot of sense to me and that's how probably about half my participants have behaved half the participants have said well I want to automate the high level and then I can fix it
>
> *"I would argue that if you've not got the resources available to you at the low level then or you've over allocated resources at the high level then you're running out of resources cascade down so you're it's a balance it's never perfectly weighted one way or the other so if you're let's say you split the task this dude here is doing this task and all of a sudden has got no resource left look at the task I need more resource I told you need more resource when I told you this is my plan and this is I've got no redundancy here you're carrying risk whereas if you've done both it's your risk to carry on the flip side of it if you've over allocated a resource that's just wasted resource now I can't complete you might want more resource but I've got five tasks that I need to do so I'm balancing these five do what you can with what you've got let me know where it all falls down because it doesn't matter if yours falls down because yours is a moderate task I've got a critical one over here that I'm allocating more to" — P36 (debrief 0:29:43)*

#### Links back to the results.

- *Agrees with* Section 7.3.3, *Tactical tier: Suggest rises with mission size, and goes with the strategic choice*: Handing over at one tier goes with handing over at the other (Suggest $\times 2.1$ after a card, same person, same size). P36 explains the coupling: the top allocation sets the lower tier's risk ("I told you need more resource when I told you this is my plan ... you're carrying risk"). Their own card-then-Suggest gap was close to zero, so the link is the cohort's, not theirs.
- *Agrees with* Section 7.4.1, *Each manual decision against the Conservative card*: Over-allocating at the top: "that's just wasted resource". In their own builds the cohort mostly trimmed the card's spare drone.
- *Agrees with* Section 8.4.3, *The cross-mission view the cards lack*: Operators overrode the cards to weigh other missions. P36 says it as a commander: "I've got five tasks that I need to do so I'm balancing these five", and "I've got a critical one over here that I'm allocating more to".

### 9.1.17 Organising assistants: narrow context is normal

[Pass 5: for the paper]

Staff cells each see only their part of the plan and first come together at the brief; that is fine if each part has been through a sound process. They add that this game's tasks are simple enough not to need more context. This is the clash with the operators who wanted one plan held across both decisions.

#### Quotes selected for this section.

> Researcher: both of them are myopic in terms of what they see so they see only the current thing so the strategic agent all it sees is the list of drones and it has a simple rule that says one drone for each dot plus one of each or something like that the fact that they see only that context how would you say that compares to the scenario you're talking about with humans planning across multiple layers this might be a six or seven layer command structure
>
> *"You kind of work in your own silo so if you're the g2'er [intelligence branch of general's staff planning team] you just focus on the enemy and you're delivering what the enemy is likely to do and you have a team working to you to do that so as a g2 officer you'll have like a one person will work on the enemy capability, one person will look at on the ground, one person will look on the sort of the human terrain and they'll work through their own analytical processes to come up with an answer then gets aggregated up and you go into it so going into that planning brief you as a g2'er are will only see your part of the plan and then the g3 [operations branch] guys will see their part of the plan and then that's kind of the first time you come together but generally speaking you're you're not getting all of the context okay so but you don't that you need that because as a person you need to be able to work through the context solution part of it yeah and this type of scenario here you wouldn't need that because of the simplicity of the task it's i've got to go and do these things and for those things i need these allocations so it's hard to sort of translate across the two as long as in each of them it's gone through the analytical process and spat out an answer" — P36 (debrief 0:38:36)*

#### Links back to the results.

- *Qualifies* Section 8.4.2, *One plan across both decisions*: Several operators wanted one plan held in mind across the tiers. P36 sees narrow context as normal ("you as a g2'er are will only see your part of the plan") and fine "as long as in each of them it's gone through the analytical process", and adds that this game's tasks are simple enough not to need more: a direct tension with those operators.
- *Goes beyond* Section 7.10.7, *How the two tiers fit together*: Operators wondered how the two assistants fit together. P36's model is staff cells that each see their part and first "come together" at the planning brief.

### 9.1.18 Organising assistants: one up, two down, and stay in your lane

[Pass 5: for the paper]

Each level is given the part of the level above that its task needs, and no more; an agent that reaches past its remit gives worse answers and is told "that's not your job". It pairs with the silos passage (both answer how much context a tier needs) but makes a separate point: what to give each level, rather than whether narrow context is acceptable.

#### Quotes selected for this section.

> Researcher: there's a philosophy of understanding what's going on one level above you and one level below you to have that or one or two levels above you or something for the context right so you you get told to go and i don't know um recce that road you need to know why you're wrecking that road so there's something above right and the idea that actually you need to have each of your agents in this in this scenario understanding the context a little above them i understand what you're saying that you don't want them to be thinking about the super wide context but is there an advantage to that linking up of understanding what's a little way up above you
>
> *"so it depends yeah so you could have your agent that's checking the colors might be the one that understands two up [the intent of those above you, and one about that] so you understand two up in the same way as you plan two down issue orders one down so you think about i'm going to be sure to report to my section commanders but my i'm thinking about what each of those individual soldiers doing with that scenario and that might be your agent that corrals all your thing but in a when we go through that staff process the person who's just doing a scenario they will understand the one up they will understand or part of the one up that's given to them as their task is there's an area of responsibility there's an area of interest there's a thing that the enemy is doing within this space i just need to kind of understand that that so it might be you might understand that your mission that's come to you within your organisation so that's technically your organisation's mission but you might consider that your one up so it's depending on on as long as the person doing the task is given the understanding they're required to do that task then it doesn't matter how it docks into everything else that's not their job that's the person above them to do that so for your agents yes it might be useful but no it might not be useful it depends and particularly it's not useful if they're giving shit answers like if they're coming out and going no i've allocated five over here because there's a thing over there it's like cool that's not your job your job is to do this thing over here and then there's other agents to do this thing over here and they don't need them to do both because that's where this agent is going to do both for them and that's quite often even humans doing it humans start thinking or g2 again they'll start thinking blue plan but your job is not to do the blue plan your job is to do the red plan stop thinking about or the blue is probably going to do this so i'm going to do this no no but i need to know that sir because i need to understand the context of what the red is going to do so you need to have an idea of what blue capabilities have but you don't have control over the blue plan and you're not there to anticipate the blue plan you're there to put it within this context just stick to your lane and do that output because someone else will do that for you" — P36 (debrief 0:42:47)*

#### Links back to the results.

- *Goes beyond* Section 7.10.7, *How the two tiers fit together*: How should the tiers fit? From doctrine: "you understand two up in the same way as you plan two down", each task comes with the part of the level above that it needs, and an agent reaching outside its lane is told "that's not your job".
- *Qualifies* Section 8.4.2, *One plan across both decisions*: The operators who wanted one shared plan are answered by a different design: each level gets "the understanding they're required to do that task", and "it doesn't matter how it docks into everything else".

### 9.1.19 Organising assistants: an agent that red-teams the others

[Pass 5: for the paper]

A checker that tests the data the other agents' answers rest on, so that a bad input is found before the answers built on it. It answers the hidden-flaw result without relying on the operator to notice.

#### Quotes selected for this section.

> *"One of the good things about ai is that you can have an agent whose only job is to go and red team [adversarially check] all of your outputs very quickly so if you've got that there is that hole in that data set and it go through and start all your job is to do is to check the data that they've based their answers on and then you can't go well this data is wrong so all these answers are now redundant we now need to look at this change this and then the answers will be right" — P36 (debrief 0:39:26)*

#### Links back to the results.

- *Goes beyond* Section C.4.1, *Strategic flaws are visible; tactical flaws are hidden*: The tactical flaw (a wrong "No workable plan") went almost unseen: only 4 people reacted to one aloud. P36's answer does not rely on the operator: "an agent whose only job is to go and red team [adversarially check] all of your outputs very quickly", checking "the data that they've based their answers on".
- *Agrees with* (not in the paper), *Assistants that check rather than decide*: P11 wanted an assistant that checks their own plan. P36's checker tests the other agents' data instead, so a bad input shows up before the answers built on it.

### 9.1.20 Read against the results: where a practitioner agrees, qualifies and goes beyond

Every link on the points above, in one place, grouped by how the passage bears on the result. **Agrees**: the size rule and its reasons, the card's speed, working across the board, oldest first, criticality, quick checking and its limit, trust in how answers are made, and the tiers coupled from the top. **Qualifies**: workload (the pressure in real operations is pace and the lower tier), the trust gap (easy to check is not low stakes), the size rule (it bends with slack), hand-build spares, checking (verifying is a process, not a look), failure data (it needs context), experience (it matters when the stakes are real), and one plan across the tiers (narrow context is normal). **Goes beyond**: skills and selection, verifying by challenge, always acting on risk, what real stakes do to decisiveness, staff cells and one-up two-down, a red-team agent, and where an assistant for what to do next would sit. This could close the paper.

**Agrees with the results (24).**

- Section 7.3.6, from the point in Section 9.1.2: Their aim is throughput across the board ("get as many across as quickly as I can to complete all of the tasks"), the same reason the cohort gives for starting Suggest and moving on.
- Section 8.4.3, from the point in Section 9.1.2: Operators overrode the cards to act on what one mission's card cannot see. P36 says why in one line: "success is generated across all of the tasks rather than just an individual task".
- Section 7.4.1, from the point in Section 9.1.2: Playing for every task shows in the cohort's hand builds: on 44% of decisions they sit below both cards, mostly by leaving off the card's spare, which keeps drones for the next mission.
- Section 7.3.5, from the point in Section 9.1.3: The card saves time only on big three-type jobs. Three minutes into their first session P36 finds it out: "Probably quicker just to hit three", then "I probably should have done that one manually. It would have been quicker."
- Section 7.3.1, from the point in Section 9.1.4: Small jobs by hand, big jobs to the assistant (cohort: card 56% to 75%, Suggest 54% to 91%). P36 gives both halves of the reason: on a simple job "i know the answer based on what i'm seeing visually", and a failure needs no re-plan ("you don't really need to change the story"); on a complex one "my option scope is broader", so "let them start on that task and then intervene when i need to". Their own log follows it (card 33% to 70%, Suggest 79% to 100%).
- Section 7.9.3, from the point in Section 9.1.4: "Give it a quick once over and go that looks about right ... whether or not it is right of course it's another question entirely": the brief check the timing data show, with its limit named.
- Section 8.4.2, from the point in Section 9.1.4: P21 chained drones through each other's tasks for "redundancy for free". P36 does the same on simple missions ("every single drone is going to every single task so that if one of my drones fails ... my other drones are picking up the task"), which is why those plans need no checking afterwards.
- Section 7.6.1, from the point in Section 9.1.5: A busier moment does not push people onto the assistants. P36 presses Suggest at a quiet moment, "because there's nothing else that's coming in".
- Section 7.7.1, from the point in Section 9.1.6: Operators open the oldest waiting mission well above chance. P36: "this one's been there for the longest i've got nothing in there that's going to change my prioritization".
- Section 8.5.1, from the point in Section 9.1.7: Operators lean towards the more critical waiting mission ($\times 1.45$ per category, on top of oldest-first). P36 goes further and spends score on it: "i've run out of blue so that's gonna hurt my score but ... i'd rather do the critical tasks if this is a real life scenario".
- Section 8.4.1, from the point in Section 9.1.8: P19 put the only remedy for failures with the strategic tier. P36 reaches the same point in mid-sentence, and it reverses their reason for trusting the cards: "they're probably not actually are they if you fuck up the allocation at the start of it then you're screwed going into the tactical".
- Section 7.3.6, from the point in Section 9.1.9: The time Suggest frees is spent elsewhere while a plan runs. P36 describes that dead time from the commander's side: "I've now got to wait for this thing to get into place", and uses it to plan for failure.
- Section 7.10.10, from the point in Section 9.1.9: Others said the pace was unreal; P36 agrees ("in terms of the pace probably not so much") and says what sets the real pace: "if you're going at 3k an hour then that's going to take you a long time".
- Section 7.9.3, from the point in Section 9.1.11: Plans are accepted after a brief look. P36 explains why that can be rational: answers from staff are taken "as long as you have trust in how they've come up with those answers", not re-derived.
- Section 7.10.3, from the point in Section 9.1.11: Several operators wanted to know how the assistants decide. For P36 that is the basis of trust: a commander can take the staff's answers because "you'd have worked up in those levels so you kind of understand how those answers are generated". No operator has done the assistant's job.

- Section 7.3.5, from the point in Section 9.1.11: The card pays where a job is too big to do quickly by hand. P36 gives the same reason for having staff at all: “I couldn’t work through the ground the enemy, the enemy’s plan, the likelihood of my plan succeeding, the logistics behind it it would take me forever which is why your staff do it”.
- Section 7.10.3, from the point in Section 9.1.12: “With AI it should always be open to challenge”: the same ask as the operators who wanted to know how the assistants decide.
- Section 7.3.6, from the point in Section 9.1.13: Handing over frees the operator to work elsewhere: “If I’ve sent that off it allows me to scan over here and then go back and check the answer over there”, which is where the cohort’s Suggest time goes.
- Section 7.9.3, from the point in Section 9.1.13: Brief checking, whatever the plan size. P36 names its price: “if I just said ’go nuts’ when something goes wrong I’ve got to play catch up”, because the plan is not in your head.
- Section 8.3.1, from the point in Section 9.1.14: Operators facing random failures with no numbers felt they were deciding blind. P36 agrees the numbers should be shown (“great to see the information”) and adds that people then need training to act on them: “there’s always going to be a failure rate I’m just trying to make it as low as reasonable”.
- Section 7.3.3, from the point in Section 9.1.16: Handing over at one tier goes with handing over at the other (Suggest $\times 2.1$ after a card, same person, same size). P36 explains the coupling: the top allocation sets the lower tier’s risk (“I told you need more resource when I told you this is my plan ... you’re carrying risk”). Their own card-then-Suggest gap was close to zero, so the link is the cohort’s, not theirs.
- Section 7.4.1, from the point in Section 9.1.16: Over-allocating at the top: “that’s just wasted resource”. In their own builds the cohort mostly trimmed the card’s spare drone.
- Section 8.4.3, from the point in Section 9.1.16: Operators overrode the cards to weigh other missions. P36 says it as a commander: “I’ve got five tasks that I need to do so I’m balancing these five”, and “I’ve got a critical one over here that I’m allocating more to”.
- (“Assistants that check rather than decide”, not in the paper), from the point in Section 9.1.19: P11 wanted an assistant that checks their own plan. P36’s checker tests the other agents’ data instead, so a bad input shows up before the answers built on it.

**Qualifies the results (10).**

- Section 7.3.1, from the point in Section 9.1.5: The size rule bends with slack: this mission is “relatively simple”, but “I’m resource rich at the moment so we’ll go nuts I’ll start suggesting”.
- Section 7.4.1, from the point in Section 9.1.6: Most of the cohort’s hand builds sit at or below the cards (44% below both). P36 adds a spare on purpose (“minimum two we’re going to go three just never deploy without a reserve”), yet their builds land where the cohort’s do: on average 0.4 of a drone under the Conservative card, against a median of 0.6 under.
- Section 7.9.1, from the point in Section 9.1.8: The cohort trusts the Tactical Assistant more; P36 is one of the exceptions, and their ratings match what they say (“I trust the strategic slightly more”: 5.9 against 5.8 on the 1–7 scale, where the cohort’s median strategic rating is 4.5). They hesitated less over the cards only because “all i was looking at there was have i got a reserve allocated”.
- Section 7.6.1, from the point in Section 9.1.9: A busier queue did not push anyone onto the assistants. P36 puts the real pressure elsewhere: “physics remains the bounding factor”, and the load becomes waiting and asking “what am I going to do if this thing fails”. “Your mental load is still loaded it’s just loaded in a slightly different way.”
- Section C.3.2, from the point in Section 9.1.10: Nine people said they lean on the assistant when busy, yet their reliance did not rise with the queue. P36 puts the busy in a real hierarchy: “the higher up you are the further out”, where a new item is “just like a routine routine task”; at the lower tier “10 things in front of me as it’s all come down the chain”. The game gave one person both tiers.
- Section 7.9.3, from the point in Section 9.1.12: This is from someone who kept every consulted tactical plan unedited. In P36’s account verifying happens around the plan, not in a long look at it: take it to the people who will carry it out (“go away and work out the plan”) and keep it “open to challenge”. Beside that sits “trust in the fact it’s done the maths behind it and has allocated enough platforms”, which is about all the cohort’s 2–4 seconds of checking can cover.
- Section 7.10.4, from the point in Section 9.1.14: Several operators asked for failure numbers. P36 wants them with context (“there’s got to be something where it shows a bit of context to it”), because a bare “40% failure chance” reads as “no chance” when “the failure rate of doing nothing is 100%”.

- Section 7.2.7, from the point in Section 9.1.15: Prior experience did not predict style here. P36 expects it to matter most when the stakes are real: their own "looking at a glance yes it's allocated what I needed ... cool let's start running with it" comes from experience, where a new platoon commander would be "foolhardy" or "almost paralysed". A game with no lives on the line hides that difference.
- Section 8.4.2, from the point in Section 9.1.17: Several operators wanted one plan held in mind across the tiers. P36 sees narrow context as normal ("you as a g2'er are will only see your part of the plan") and fine "as long as in each of them it's gone through the analytical process", and adds that this game's tasks are simple enough not to need more: a direct tension with those operators.
- Section 8.4.2, from the point in Section 9.1.18: The operators who wanted one shared plan are answered by a different design: each level gets "the understanding they're required to do that task", and "it doesn't matter how it docks into everything else".

**Goes beyond the results (8).**

- Section 8.5.1, from the point in Section 9.1.10: P27 and P32 asked for an assistant that says what to open next. P36's lower tier is where that job would sit: "I've got to do this thing now then that thing then that thing then that thing".
- Section 8.2.2, from the point in Section 9.1.12: Some operators want a human in the loop even one who cannot check. P36's check is not one person looking harder but a plan socialised so that "somebody is checking my thinking and has the option to challenge me".
- Section 7.2.3, from the point in Section 9.1.13: Who decides predicts handover far better than the mission does. P36 draws the operational conclusion: "the key thing for me in all of this is not necessarily trusting the AI it is the skills that a human is going to need to be able to exploit all of this", a niche skill set like air traffic control, paid accordingly.
- ("Lower-reliance operators attempt fewer recoveries", not in the paper), from the point in Section 9.1.14: Lower-reliance operators attempted fewer recoveries. P36's rule cuts the other way: in search and rescue "you're always gambling with someone's life", so you act and "chip away at it".
- Section 7.10.10, from the point in Section 9.1.15: "It's a game" was a common caveat. P36 says what changes with a life at stake ("I imagine you're going to see me less decisive if someone's gonna die at the end"), and that a commander who no longer shares the risk ("I might die alongside them") loses what used to temper the order.
- Section 7.10.7, from the point in Section 9.1.17: Operators wondered how the two assistants fit together. P36's model is staff cells that each see their part and first "come together" at the planning brief.
- Section 7.10.7, from the point in Section 9.1.18: How should the tiers fit? From doctrine: "you understand two up in the same way as you plan two down", each task comes with the part of the level above that it needs, and an agent reaching outside its lane is told "that's not your job".
- Section C.4.1, from the point in Section 9.1.19: The tactical flaw (a wrong "No workable plan") went almost unseen: only 4 people reacted to one aloud. P36's answer does not rely on the operator: "an agent whose only job is to go and red team [adversarially check] all of your outputs very quickly", checking "the data that they've based their answers on".

# 10 Discussion

[To write. Your pass-5 notes for it: the branching failure space of complex tasks as future work (Section 9.1.4); a rule like "never without a reserve" and the consistency of habits, for training (Section 9.1.6); the clash between one plan across the tiers and narrow context per tier (Sections 8.4.2 and 9.1.17). Material for it: the f-operator-wants finding (Section 7.10), the open questions recorded in the register, and the old draft's Step-4 benchmarking note: compare against adams2025supervise (workload at scale), wischnewski2023measuring/wildman2024trust (across-tier and multilevel trust), the variable-autonomy cluster including oberacker2026mosaic, and ramchurn2021trustwor for error propagation across a pipeline.]

## 10.1 Limitations

[Your pass-5 note: the real-operations limitations (pace, stakes, unknown task locations) are implied and can be mentioned here rather than as a finding. Draft list from the register: one sample of 33, almost all computing/engineering and strongly AI-positive (AIAS-4 mean 8.0/10); a game, not real stakes; four fixed games played in an order that depended on when a person took part; build changes during collection (Appendix B); single AI coder for the qualitative counts; the illogical card and the wrong refusal are side effects found after collection, not manipulations; trust measured only after each session, so a same-session trust–use link can run either way.]

# 11 Conclusion

[To write.]

# References


[1] Hussein Abbass, Jiangjun Tang, Rubai Amin, Mohamed Ellejmi, and Stephen Kirby. Augmented cognition using real-time EEG-based adaptive strategies for air traffic control. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 58:230–234, 2014. doi: 10.1177/1541931214581048.

[2] Julie A. Adams, Joshua Hamell, and Phillip Walker. Can a single human supervise a swarm of 100 heterogeneous robots? *IEEE Transactions on Field Robotics*, 2:46–80, 2025. doi: 10.1109/TFR.2024.3502316.

[3] Anduril Industries. Anduril's lattice: A trusted dual use platform for public safety and defense. https://www.anduril.com/news/anduril-s-lattice-a-trusted-dual-use-commercial-and-military-platform-for-public-safet July 2023. Accessed: 2026-05-07.

[4] Francesco Maria Delle Fave, Alessandro Farinelli, Alex Rogers, and Nick R Jennings. A methodology for deploying the max-sum algorithm and a case study on unmanned aerial vehicles. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 26, pages 2275–2280, 2012.

[5] Benjamin Hardin and Michael A Goodrich. On using mixed-initiative control: A perspective for managing large-scale robotic teams.

[6] Sandra G Hart and Lowell E Staveland. NASA task load index (TLX): Results of empirical and theoretical research. *Advances in Psychology*, 52:139–183, 1988.

[7] Amy Hocraffer and Chang S. Nam. A meta-analysis of human-system interfaces in unmanned aerial vehicle (UAV) swarm management. *Applied Ergonomics*, 58:66–80, 2017. ISSN 00036870. doi: 10.1016/j.apergo.2016.05.011. URL https://linkinghub.elsevier.com/retrieve/pii/S0003687016300989.

[8] Aya Hussein, Leo Ghignone, Tung Nguyen, Nima Salimi, Hung Nguyen, Min Wang, and Hussein A. Abbass. Characterization of indicators for adaptive human-swarm teaming. *Frontiers in Robotics and AI*, 9, 2022. ISSN 2296-9144. doi: 10.3389/frobt.2022.745958. URL https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2022.745958/full.

[9] Jih-Yuan Jian, Ann M Bisantz, and Colin G Drury. Foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, 4(1):53–71, 2000.

[10] S Kate Devitt. Trustworthiness of autonomous systems. In *Foundations of trusted autonomy*, pages 161–184. Springer, 2018.

[11] John D Lee and Katrina A See. Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1): 50–80, 2004.

[12] Joseph B. Lyons, Katia Sycara, Michael Lewis, and August Capiola. Human–autonomy teaming: Definitions, debates, and directions. *Frontiers in Psychology*, 12, 2021. ISSN 1664-1078. doi: 10.3389/fpsyg.2021.589585. URL https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2021.589585/full.

[13] Patricia L McDermott and Ronna N ten Brink. Practical guidance for evaluating calibrated trust. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, volume 63, pages 362–366. Sage Publications Sage CA: Los Angeles, CA, 2019.

[14] Christopher A Miller. Delegation management versus the swarm: A matchup with two winners.

[15] Kathleen L. Mosier and Linda J. Skitka. Automation use and automation bias. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 43(3):344–348, 1999. ISSN 1071-1813, 2169-5067. doi: 10.1177/154193129904300346. URL https://journals.sagepub.com/doi/10.1177/154193129904300346.

[16] Sasanka Nagavalli, Lingzhi Luo, Nilanjan Chakraborty, and Katia Sycara. Neglect benevolence in human control of robotic swarms. In *2014 IEEE international conference on robotics and automation (ICRA)*, pages 6047–6053. IEEE, 2014.

[17] Mohammad Naiseh, Dena Al-Thani, Nan Jiang, and Raian Ali. Explainable recommendation: when design meets trust calibration. *World Wide Web*, 24(5):1857–1884, 2021. ISSN 1386-145X, 1573-1413. doi: 10.1007/s11280-021-00916-0. URL https://link.springer.com/10.1007/s11280-021-00916-0.

[18] Changjoo Nam, Huao Li, Shen Li, Michael Lewis, and Katia Sycara. Trust of humans in supervisory control of swarm robots with varied levels of autonomy. In *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pages 825–830, 2018. doi: 10.1109/SMC.2018.00148. URL https://ieeexplore.ieee.org/document/8616144/. ISSN: 2577-1655.

[19] David Oberacker, Julia Richter, Philip Arm, Marvin Grosse Besselmann, Lennart Puck, William Talbot, Maximilian Schik, Sabine Bellmann, Tristan Schnell, Hendrik Kolvenbach, Rüdiger Dillmann, Marco Hutter, and Arne Roennau. MOSAIC: Modular supervised autonomy for intelligent coordination of heterogeneous robotic teams. arXiv preprint arXiv:2601.23038, 2026. Preprint (v3, July 2026); verify publication status before submission.

[20] Kazuo Okamura and Seiji Yamada. Adaptive trust calibration for human-AI collaboration. *PLOS ONE*, 15(2): e0229132, 2020. ISSN 1932-6203. doi: 10.1371/journal.pone.0229132. URL https://dx.plos.org/10.1371/journal.pone.0229132.

[21] Linda Onnasch and Eileen Roesler. A taxonomy to structure and analyze human–robot interaction. *International Journal of Social Robotics*, 13(4):833–849, 2021. ISSN 1875-4805. doi: 10.1007/s12369-020-00666-5. URL https://doi.org/10.1007/s12369-020-00666-5.

[22] Raja Parasuraman, Keryl Cosenzo, and Ewart de Visser. Adaptive automation for human supervision of multiple uninhabited vehicles: Effects on change detection, situation awareness, and mental workload. *Military Psychology - MIL PSYCHOL*, 21:270–297, 2009. doi: 10.1080/08995600902768800.

[23] Sarvapali D Ramchurn, Trung Dong Huynh, Feng Wu, Yukki Ikuno, Jack Flann, Luc Moreau, Joel E Fischer, Wenchao Jiang, Tom Rodden, Edwin Simpson, et al. A disaster response system based on human-agent collectives. *Journal of Artificial Intelligence Research*, 57:661–708, 2016.

[24] Tyler Reinmund, Pericle Salvini, Lars Kunze, Marina Jirotka, and Alan F. T. Winfield. Variable autonomy through responsible robotics: Design guidelines and research agenda. *ACM Transactions on Human-Robot Interaction*, 13(1), 2024. doi: 10.1145/3636432.

[25] Ericka Rovira, Kathleen McGarry, and Raja Parasuraman. Effects of imperfect automation on decision making in a simulated command and control task. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 49(1):76–87, 2007. ISSN 0018-7208, 1547-8181. doi: 10.1518/001872007779598082. URL `https://journals.sagepub.com/doi/10.1518/001872007779598082`.

[26] Thomas B Sheridan and William L Verplank. Human and computer control of undersea teleoperators. 1978.

[27] Mohammad D Soorati, Mohammad Naiseh, William Hunt, Katie Parnell, Jediah Clark, and Sarvapali D Ramchurn. Enabling trustworthiness in human-swarm systems through a digital twin. In *Putting AI in the Critical Loop*, pages 93–125. Elsevier, 2024.

[28] Andreas Theodorou, Manolis Chiou, Bruno Lacerda, and Simon Rothfuß. Editorial: Variable autonomy for human-robot teaming. *Frontiers in Robotics and AI*, 11, 2024. doi: 10.3389/frobt.2024.1465183.

[29] UK Ministry of Defence. Uk air power (jdp 0-30). Joint Doctrine Publication JDP 0-30, UK Ministry of Defence, November 2022. URL `https://www.gov.uk/government/publications/uk-air-power-jdp-0-30`. Accessed: 2026-05-07.

[30] Alberto Valero-Gomez, Paloma De La Puente, and Miguel Hernando. Impact of two adjustable-autonomy models on the scalability of single-human/multiple-robot teams for exploration missions. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 53(6):703–716, 2011. ISSN 0018-7208, 1547-8181. doi: 10.1177/0018720811420427. URL `https://journals.sagepub.com/doi/10.1177/0018720811420427`.

[31] Alan R. Wagner and Paul Robinette. An explanation is not an excuse: Trust calibration in an age of transparent robots. In *Trust in Human-Robot Interaction*, pages 197–208. Elsevier, 2021. ISBN 978-0-12-819472-0. doi: 10.1016/B978-0-12-819472-0.00009-5. URL `https://linkinghub.elsevier.com/retrieve/pii/B9780128194720000095`.

# A Simulation Parameters

# B Build Versions

[P09 (`v1.11`, errors on) excluded. Check the table against docs/STUDY_BUILD.md before submission; the register's sensitivity analyses on the `v1.12+` cohort are in the analysis scripts.]

**Table 4:** Simulation parameters (scenario parameter set v2.1). Everything except the arrival rate and the category weights is identical in the two scenarios.

| Parameter | Description | Value |
|---|---|---|
| *Fleet* | | |
| Fast (blue) | count; speed | 11; 11.0 u/s |
| Lifter (red) | count; speed | 11; 10.0 u/s |
| Camera (green) | count; speed | 11; 9.0 u/s |
| *Missions* | | |
| Category penalty rate | A / B / C / D / E | 0.05 / 0.10 / 0.15 / 0.25 / 0.40 pts/s |
| Archetype weights | Fast / Lifter / Camera / Mixed | 38 / 26 / 21 / 15 |
| Zone | radius; min. distance to hub; to other zones | 80; 150; 200 units |
| *Scenario-specific* | | |
| Strategic Heavy | mean inter-arrival; category weights A–E | 37 s; 40/38/16/5/1 |
| Tactical Heavy | mean inter-arrival; category weights A–E | 75 s; 5/13/28/38/16 |
| *Session, assistants, failures* | | |
| Session length | per session | 480 s (8 min) |
| Seed | all sessions | 42 (four games; Section 6.1) |
| Strategy card reveal | per card, independent | ≈4–5 s |
| Conservative card | spare per used type; top-up of remaining reserve | +1; 15% of every type |
| Suggest fill rate | tactical plan | 1 drone→task entry per second |
| Drone failure hazard | per deployed drone per second | $1/900$ (`v1.4`+) |
| Post-failure grace | no further failure on that mission | 30 s after recovery (`v1.5`+) |
| Penalty display step | | 15 s |
| Assistant error rates | $\varepsilon_S$, $\varepsilon_T$ | 0, 0 (all analysed sessions) |

**Table 5:** Platform builds used during data collection, with the changes that bear on the data. The build is recorded in every session's log; the full history is in `docs/STUDY_BUILD.md`.

| Build | Participants | Relevant to the data |
|---|---|---|
| `v1.0` | P01 | Task points half the later values (rescaled in analysis); failures from a per-mission schedule. |
| `v1.2` | P02 | As above; abandoned work re-queued as a residual mission from here on. |
| `v1.3`–`v1.4` | P03 | Points doubled; pre-study AI-attitude scales added; live failure hazard (1/900 per drone-second) on a fixed grid from `v1.4`; substitute compositions executable. |
| `v1.5`–`v1.6` | P04 | Recovery Suggest produces plans; 30 s post-failure grace; a half-built recovery plan survives a task completing. |
| `v1.7`–`v1.9` | P05–P07 | Strategy-card score bars corrected (Speed = best ETA / ETA; Reserve = share of reserve left); allocating clears the tactical screen; deploying closes the planner. |
| `v1.10`–`v1.11` | P10, P11 | Error-injection machinery added, switched off ($\varepsilon = 0$). |
| `v1.12` | P12–P30 | NASA-TLX performance slider carries a "runs backwards" note. |
| `v1.13` | P31–P36 | Games swapped: the same four mission streams played in the other session position. |

**Table 6:** The same measures by the four fixed games (mean (SD) over sessions). SH = Strategic Heavy, TH = Tactical Heavy; the number is the mission stream (plan index), not the session order. Sessions per game: SH1 17, SH2 16, TH1 16, TH2 17.

| **Measure** | **SH1** | **SH2** | **TH1** | **TH2** |
|---|---|---|---|---|
| Missions arrived (non-residual) | 13.0 (0.0) | 14.0 (0.0) | 6.0 (0.0) | 9.0 (0.0) |
| Strategic decisions made | 10.8 (2.3) | 12.1 (2.3) | 6.2 (1.4) | 7.6 (1.2) |
| Tactical plans confirmed | 10.6 (2.5) | 11.8 (2.4) | 6.1 (1.5) | 7.5 (1.1) |
| Tasks completed | 25.1 (4.4) | 28.0 (7.2) | 21.2 (6.0) | 32.8 (7.8) |
| Completion points[a] | 1266 (248) | 1288 (384) | 1169 (331) | 2058 (480) |
| Time penalty (points) | 288 (52) | 264 (66) | 232 (79) | 549 (92) |
| Score (points − penalty)[a] | 978 (284) | 1023 (443) | 940 (398) | 1509 (567) |
| Strategy card taken (% of decisions) | 58 (23) | 61 (31) | 78 (29) | 68 (26) |
| Manual before cards loaded (%) | 15 (19) | 22 (27) | 11 (26) | 11 (14) |
| Card edited, then deployed (%) | 10 (21) | 0 (2) | 1 (4) | 10 (19) |
| Suggest pressed (% of plans) | 75 (22) | 68 (30) | 75 (35) | 84 (21) |
| Suggest plan kept unmodified (% of plans) | 72 (24) | 66 (32) | 74 (35) | 76 (29) |
| Drone failures | 3.2 (1.7) | 4.9 (3.1) | 3.1 (2.0) | 4.5 (2.9) |
| Missions abandoned | 1.0 (1.1) | 1.4 (2.0) | 0.7 (0.9) | 0.9 (0.9) |
| NASA-TLX, raw mean (0–20) | 9.4 (3.1) | 8.2 (3.1) | 6.5 (2.9) | 10.6 (2.8) |
| Mental demand | 13.1 (3.8) | 10.2 (5.5) | 7.0 (5.2) | 14.7 (4.2) |
| Temporal demand | 13.5 (3.5) | 13.7 (4.4) | 11.2 (5.4) | 13.4 (4.5) |
| Performance (0 = perfect) | 8.1 (4.2) | 6.9 (5.6) | 7.8 (5.8) | 9.8 (3.8) |
| Effort | 10.2 (4.7) | 8.9 (4.7) | 5.9 (3.4) | 11.7 (4.0) |
| Frustration | 6.9 (4.8) | 5.1 (5.3) | 3.6 (3.9) | 8.7 (6.3) |
| Trust, Strategic Assistant (1–7) | 4.35 (1.29) | 4.42 (1.77) | 4.95 (1.43) | 4.75 (1.26) |
| Trust, Tactical Assistant (1–7) | 5.45 (1.05) | 5.58 (0.67) | 5.43 (0.95) | 5.30 (0.94) |

# C Supplementary Results

Sidelined analyses from the analysis register: robustness checks, alternative specifications and closer looks that support the main results but are not needed to follow them.

## C.1 Each operator has a personal style — individual, not a type (supplementary analyses)

Supplement to Section 7.2.

### C.1.1 After one session we can recognise a person's style in the next

[Pass 5: numbers only]

Each session is summarised as ten habits (below). Comparing everyone's session-1 profile with everyone's session-2 profile, the closest match is far more often the **same person** than chance allows. It is a "near neighbour" result, not exact identification: the own session is the single closest match for 7 of 33 people (chance: 1), but in the closest three for 41% (chance 9%), and on average ranks 8th of 33 (chance 17th). It survives taking the reliance axis out — so the style is more than "how much do you lean on the assistants". In the matrix, every row is one person's session 1, every column a session 2, darker = closer; the outlined diagonal is each person against themselves. (Figure 40)

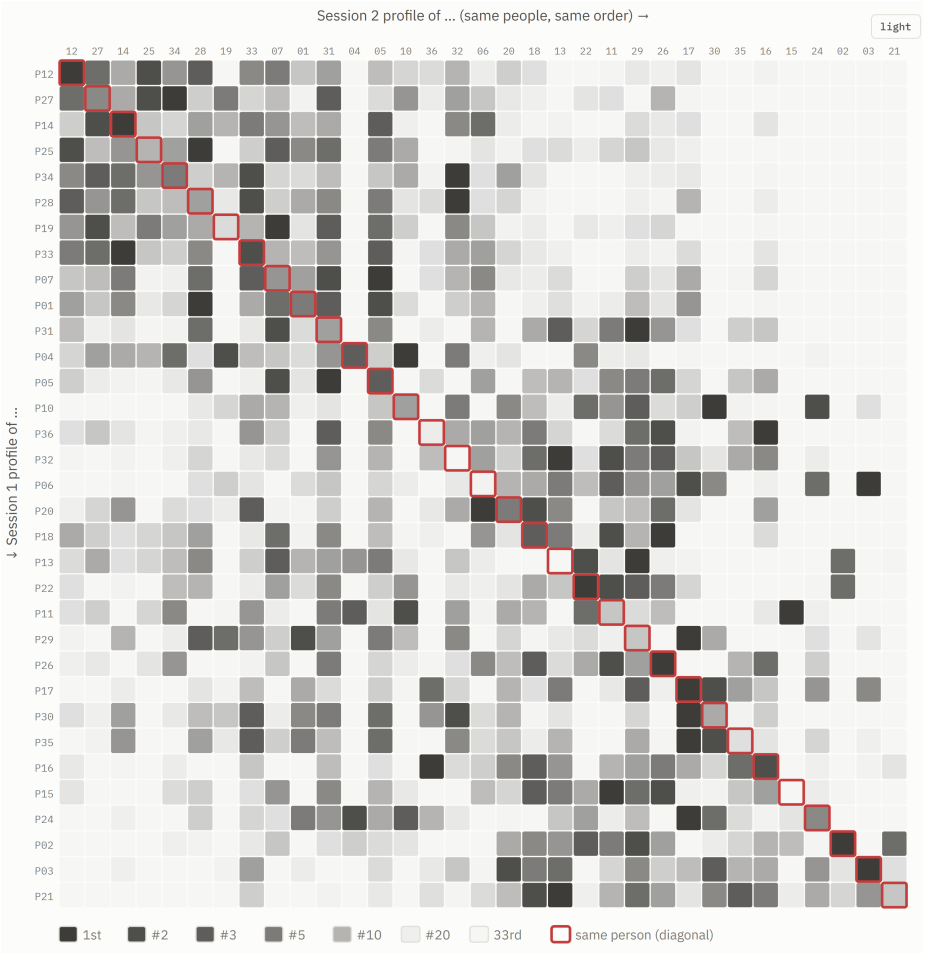



**Figure 40: After one session we can recognise a person's style in the next**

**Key numbers.**

- Own session nearest: 7 of 33 (session 1 $\rightarrow$ 2) and 5 of 33 (2 $\rightarrow$ 1); chance $\approx$ 1; permutation $p < .001$
- Own session in the closest 3: 41% vs 9% by chance; mean rank 8.2 of 33 vs 17.0 ($p < .001$)
- Reliance axis removed: 6 and 6 of 33, top-3 32%, mean rank 10.1 ($p < .001$) — the style is more than reliance
- Reliance axis alone: 4 and 4 of 33, top-3 27%, mean rank 11.1
- Profile correlation with own other session: median $r = .48$; with other people's: median $r = -.01$; own closer for 29 of 33 (sign test $p < .001$)

**Method.** Rows: 66 sessions (33 people $\times$ 2). Each session becomes a vector of ten habits: share of strategy cards taken; share of manual choices made before the cards loaded; share of cards edited; share of tactical plans where Suggest was pressed; share of Suggest plans kept unchanged; share of recovery episodes with Suggest pressed; time per strategic decision beyond the forced card delay (log); drones committed vs the Conservative card on single-type and on multi-type missions; card previews per decision. Each habit is turned into a z-score *within the game played* (so a hard game does not make everyone in it look alike; a habit with no data in a session, e.g. no recoveries, is set to that game's median). For each person's session-1 vector we measure the straight-line (Euclidean) distance to all 33 session-2 vectors and rank them; "rank 1" means their own session 2 is the nearest. Done both ways (session 1 looking among session 2s and vice versa). The p-value comes from shuffling who-is-who in session 2 10 000 times and counting how often chance does as well. "Reliance axis removed": the single strongest direction of variation (PC1, see below) is subtracted from every vector first. The profile correlation r is the Pearson correlation between a person's two ten-habit vectors (1 = same shape, 0 = unrelated). Exploratory and post hoc; the permutation test is clean. People are ordered by reliance in the matrix, so neighbours look alike — the claim is that the diagonal beats them.

### C.1.2 What a personal style looks like

[Pass 5: extra: may make it in]

Every person's ten habits in session 1 (light) and session 2 (dark). The top rows are people whose two sessions have almost the same shape (r up to .89); at the bottom are the few whose style changed (P13, P29, P32). Most people repeat themselves. (Figure 41)

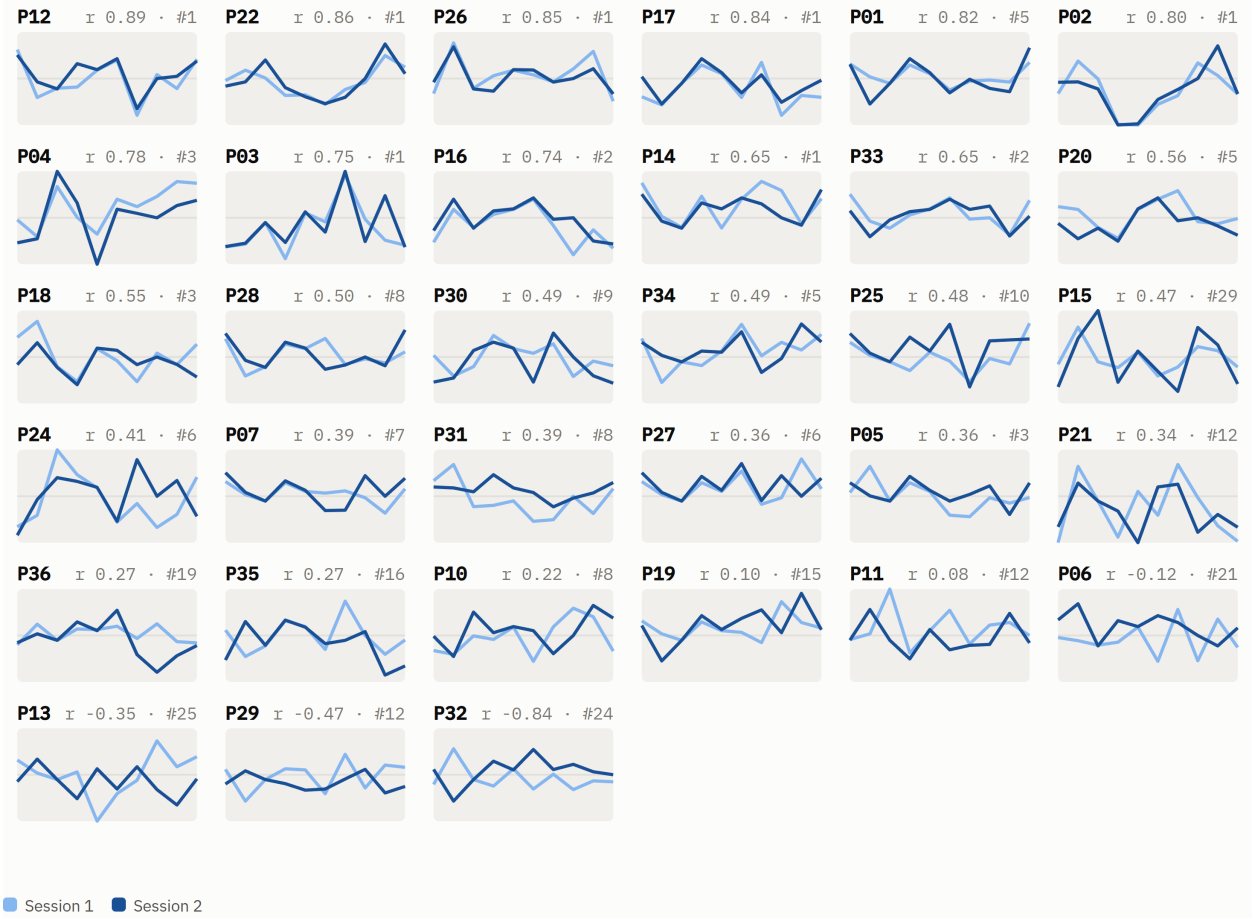



**Figure 41: What a personal style looks like**

**Key numbers.**

- 24 of 33 people have r $\geq$ .27 between their two sessions; 4 have r < 0

**Method.** Each line joins the ten habit z-scores (within game, clipped at $\pm$2.5 for drawing). r = correlation of the two lines; #k = where their own session 2 ranked among 33 (1 = nearest).

### C.1.3 Which habits carry over — and what each correlation means

[Pass 5: extra: may make it in]

Each small scatter is one habit: a dot is a person, x = that habit in session 1, y = the same habit in session 2. If people kept their place in the queue exactly, the dots would lie on the dashed line; if session 1 told us nothing, they would be a shapeless cloud. Six of the ten habits carry over clearly: keeping Suggest plans, pressing Suggest, taking cards, previewing cards, decision time, and Suggest in recovery. Topping up drones on single-type missions does not carry over at all. (Figure 42)

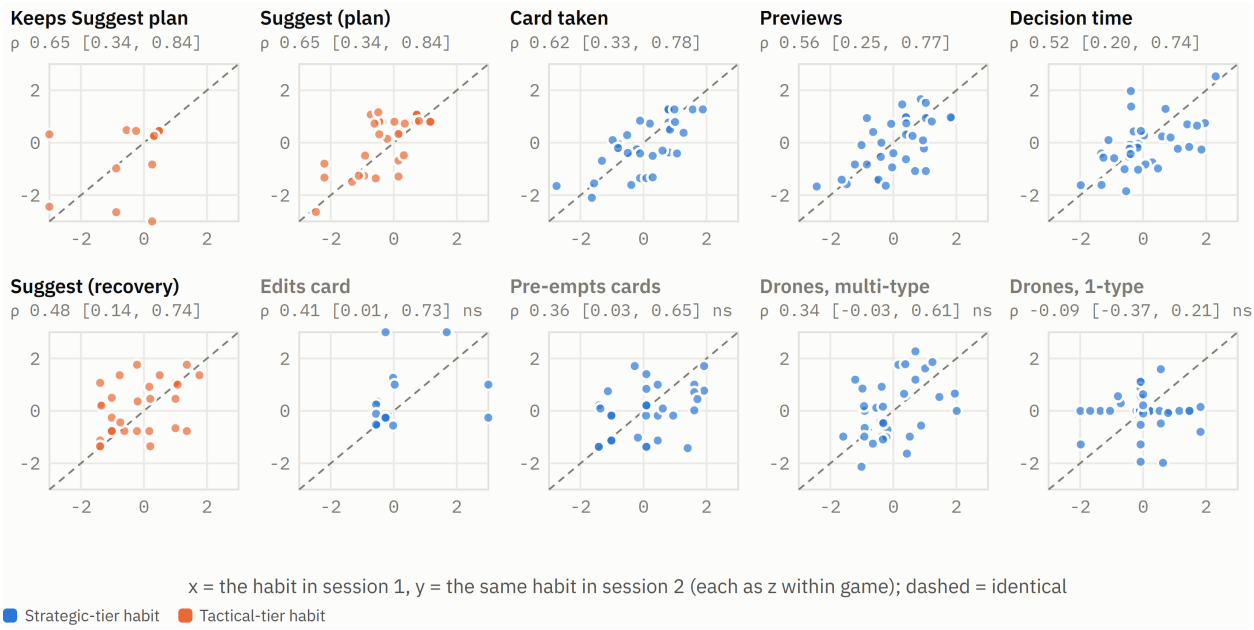



**Figure 42: Which habits carry over — and what each correlation means**

**Key numbers.**

- Survive Holm: keeps Suggest plan $\rho$ .65 [.34, .84]; presses Suggest .65 [.34, .84]; takes card .62 [.33, .78]; previews .56 [.25, .77]; decision time .52 [.20, .74]; Suggest in recovery .48 [.14, .74]
- Do not: edits a card .41; manual-before-cards (share of manual) .36; drones on multi-type .34; drones on single-type −.09
- Card uptake: ~2 distinct levels of people from one session, ~3 from two
- Score on the reliance axis, session 1 vs 2: $\rho$ = .65 [.38, .82]

**Method.** $\rho$ (Spearman) for habit X = the correlation, across the 33 people, between their rank on X in session 1 and their rank on X in session 2, each session first adjusted for the game played (z within game). $\rho$ = .62 for "takes the strategy card" means people keep roughly the same order: high card-takers in session 1 are mostly high in session 2, though not exactly. 95% intervals come from resampling people 2000 times; Holm corrects for testing ten habits. "Distinct levels" (separation strata) says how many groups of people the habit can tell apart reliably: one session of card uptake separates about 2 (high vs low), the average of two sessions about 3. Caution on "keeps the Suggest plan": 24 of 33 people kept every Suggest plan unchanged in both sessions (a ceiling), so its high $\rho$ mostly says that the nine who ever edit a plan tend to do so in both sessions. The pre-empting habit here is "share of *manual* choices made before the cards loaded" (undefined for 15 people who never chose manual in a session); the clearer per-decision version is in the next section.

### C.1.4 Not waiting for the cards is one of those habits

[Pass 5: extra: may make it in]

Committing a manual allocation before the strategy cards have even loaded is something some people do and others never do. People who did it in session 1 did it on a quarter of their decisions in session 2; people who did not, on 8%. Fifteen people never did it in either session. (Figure 43)

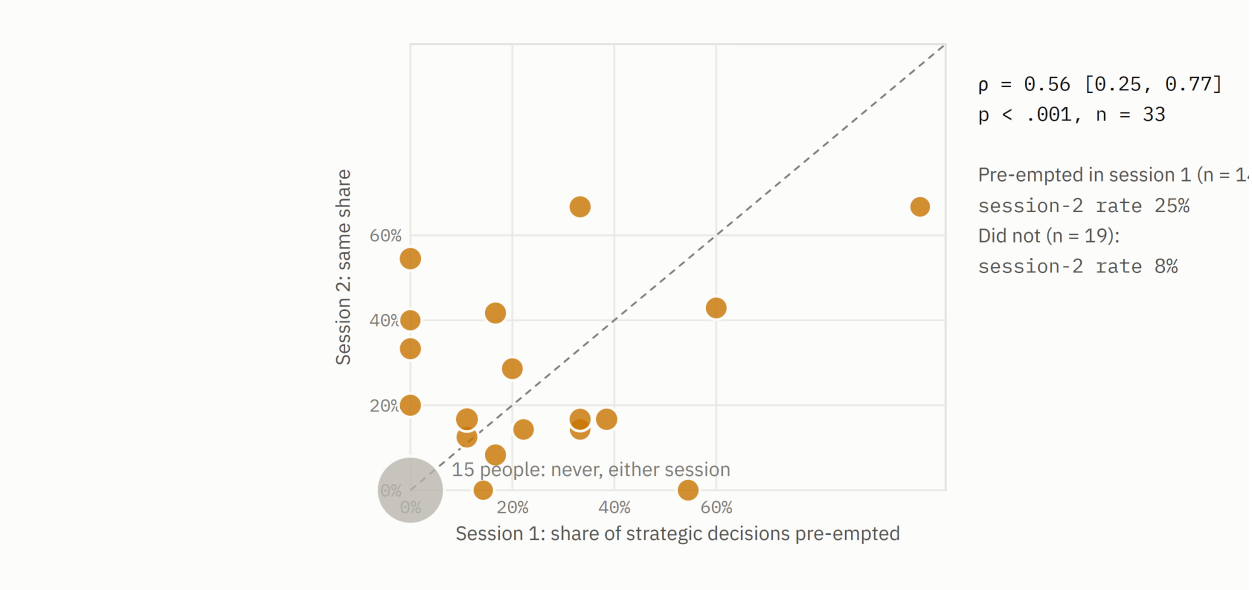



**Figure 43: Not waiting for the cards is one of those habits**

**Key numbers.**

- Session 1 vs 2: $\rho = .56$ [.25, .77], $p < .001$, $n = 33$
- Session-2 rate 25% for those who pre-empted in session 1 (n = 14) vs 8% for those who did not (n = 19)
- 86 of 564 decisions pre-empted; 46% of all manual choices; 15 of 33 never did it
- Round 2: person share of variance .24, mission .05

**Method.** Rows: 564 non-residual strategic decisions. Per person and session: share of decisions committed manually before the cards appeared (logged `manualBeforeCardsLoaded`). $\rho$ = Spearman of the session-1 share against the session-2 share across 33 people. Variance shares (person ~24%, mission ~5%) are from round 2 (F4.1, crossed random intercepts for person and the exact mission, which repeats across people because games are fixed). The 15 "never" people sit together at the origin; $\rho$ is driven both by them staying at zero and by the regular pre-empters staying high.

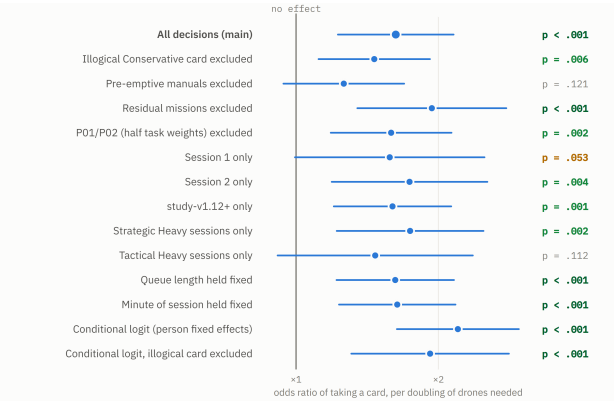
## C.2 Small jobs by hand, big jobs to the assistant — on both tiers, and it is where the assistant saves time (supplementary analyses)

Supplement to Section 7.3.

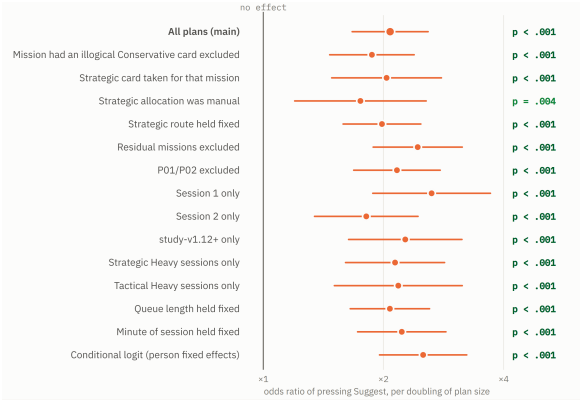
### C.2.1 Is it an artefact? Checks, all reported

[Pass 5: numbers only]

The worry: the illogical Conservative card (one that adds a drone of a type the mission does not need) appears only on one- and two-type missions — 71% of one-type decisions, none of the three-type ones — so "people avoid the card on small missions" could be "people avoid a visibly odd card". It is not: on one-type missions uptake is the same with an odd card (50%, 129 decisions) and without one (56%, 52 decisions), and with every odd-card decision removed the size effect is still there ($\times$1.46 per doubling; clean one-type vs three-type $\times$1.83). No $\varepsilon$-corrupted (deliberately wrong) cards were shown in any session, so card quality did not otherwise vary by size. The rule also holds with queue length held fixed, with the minute of session held fixed, per session, per scenario, without the half-weight participants and in a person-fixed-effects model. Two strategic rows do not reach $p < .05$: Tactical Heavy sessions alone (few one-type missions there) and cards-on-screen only (above). The tactical rule clears every row. (Figure 44)




**(a)** Strategic tier.




**(b)** Tactical tier.

**Figure 44: Is it an artefact? Checks, all reported**

**Key numbers.**

- Strategic, illogical card excluded: $\times$1.46 [1.12, 1.92], p = .006 (474 decisions); conditional logit $\times$1.92 [1.31, 2.82]
- One-type missions, odd card vs clean card: odds $\times$0.77 [0.33, 1.78], p = .55 — no difference

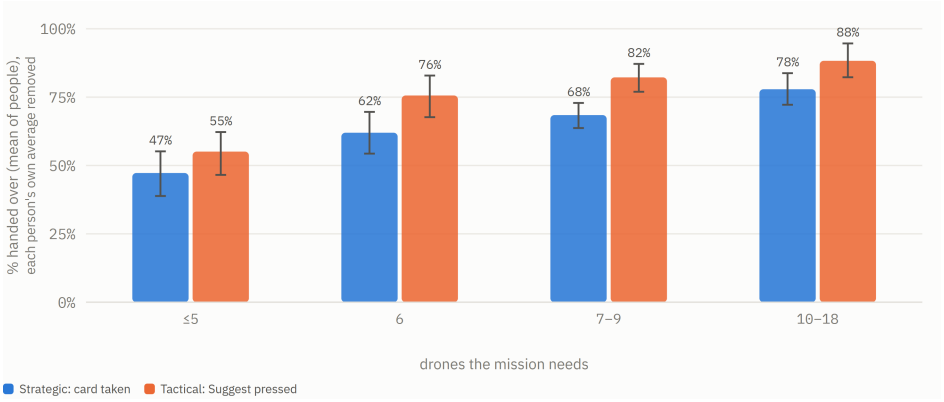
- Strategic range across 14 specifications: $\times 1.26$ to $\times 2.20$; tactical range across 15: $\times 1.75$ to $\times 2.64$
- Tactical, illogical-card missions excluded: $\times 1.87$ [1.47, 2.39]; strategic route held fixed $\times 1.98$

**Method.** Each row refits the main within-person model on a subset or with one more control. Illogical card: the *odd* flag from build_findings_synthesis.load_decisions (Conservative card carries a type the mission needs none of). Queue held fixed: the number of missions waiting at the decision, split within/between person (the queue story itself is Section 7.6's). Conditional logit: a different model family that compares each person only with themselves by construction (people who always or never hand over drop out, hence fewer people). The odd-card story itself is Section 7.8's — here it is only checked as a confound.

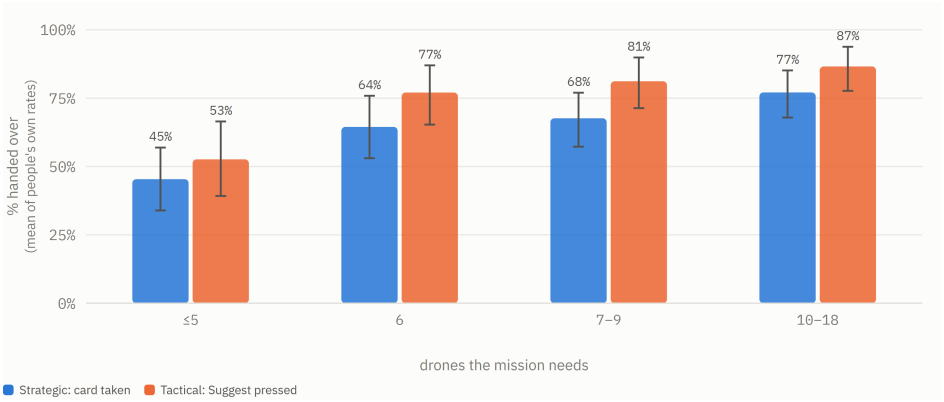
### C.2.2 Person by person

[Pass 5: extra: may make it in]

The same rule with each person counted once: for each band of mission size, each person's own hand-over rate, averaged over people. The default view removes each person's overall level first, so it shows how the *same* people shift between small and big missions. Both tiers rise band by band. 22 of 33 people hand over more on their biggest band than their smallest, on each tier; only 4 (strategic) and 2 (tactical) do the reverse. (Figure 45)




**(a)** Each person compared with their own average.




**(b)** Plain average of people.

**Figure 45: Person by person**

**Key numbers.**

- Strategic, own average removed: 47% (≤5 drones, 32 people) → 62% → 68% → 78% (10–18, 33 people)
- Tactical, own average removed: 55% → 76% → 82% → 88%
- Biggest band above smallest: strategic 22 people, below 4, equal 7 (sign test $p < .001$); tactical 22, 2, 9 ($p < .001$)

**Method.** Bands of drones needed: ≤5, 6, 7–9, 10–18 (both tiers; drones needed is the measure both tiers share). A person contributes to a band if they made at least 2 decisions in it. "Own average removed" = the person's rate in the band minus their rate over all bands, plus the overall rate, so bars stay on the percentage scale. Whiskers: 95% intervals from resampling people. Sign test: for each person, their biggest band against their smallest; are "more" more common than "less" than a coin flip would give (ties dropped)?

### C.2.3 The speed difference as estimates

[Pass 5: numbers only]

The same comparison as model estimates: negative means the assistant route was faster for the same person. (Figure 46)

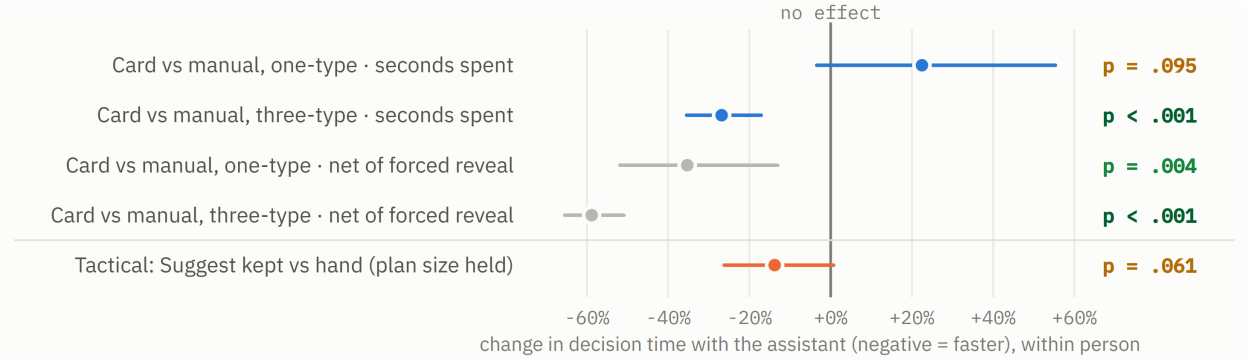



**Figure 46: The speed difference as estimates**

**Method.** As in the section above. Each row is a within-person % change in decision time with its 95% interval.

### C.2.4 The exception that proves the rule: the people who call the assistant slow

[Pass 5: extra: may make it in]

Six people complain that the assistant is slow (C06: P06, P15, P16, P26, P27, P29). On every measure the two groups' intervals overlap: none of the ten differences is reliable with six people. The directions are worth a sentence, not a claim. They take cards and press Suggest at the same rate as everyone else (58% vs 66%; 74% vs 76%), and they lean *towards* the size rule: they decide before the cards load more often (30% vs 13%, p = .055) and their three-type minus one-type gap is larger (+40 vs +19 points, p = .09). Read that way, the complaint and the rule are the same observation (the fixed wait is not worth it on small jobs), and P29 says as much. But this is a reading of six people, not a finding. (Figure 47)

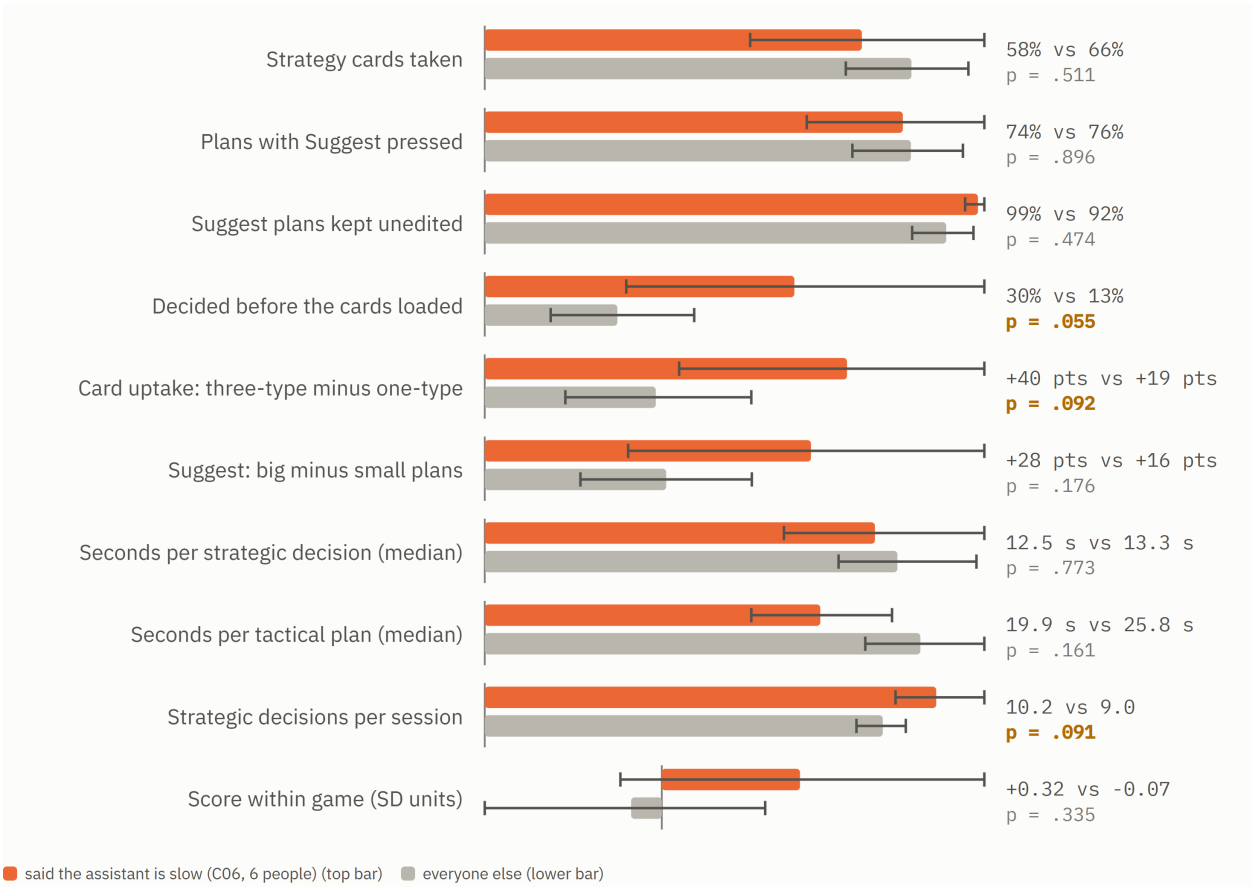



**Figure 47: The exception that proves the rule: the people who call the assistant slow**

**Key numbers.**

- Pre-emptive share 30% vs 13% (p = .055); three-type minus one-type card gap +40 vs +19 points (p = .09); decisions per session 10.2 vs 9.0 (p = .09)
- Tactical plan time 19.9 vs 25.8 s (p = .16); score +0.32 vs −0.07 SD (p = .34)
- Card uptake 58% vs 66% (p = .51); Suggest 74% vs 76% (p = .90) — reliance itself does not differ

**Method.** Per-person measures, averaged over the person's two sessions, for the six C06 people against the other 27. Bars are group means; whiskers are 95% intervals from resampling people within each group. Score = session score (P01/P02 rescaled for their half task weights) standardised within each of the four games. p = two-sided permutation test (group labels reshuffled 20,000 times); 10 comparisons, uncorrected, so expect about one p below .10 by chance.

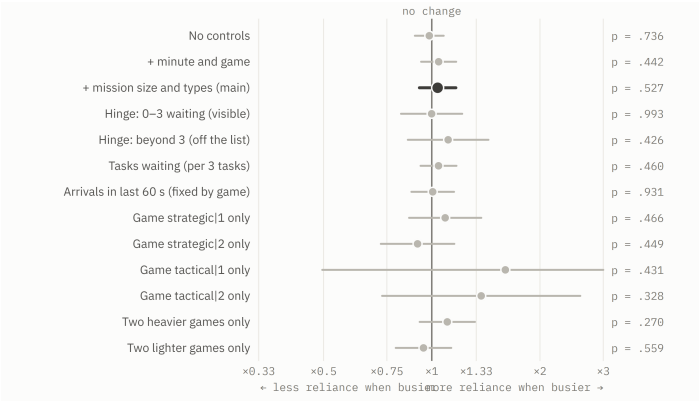
## C.3 A busier moment does not push people onto the assistants — even though they say it does (supplementary analyses)

Supplement to Section 7.6.

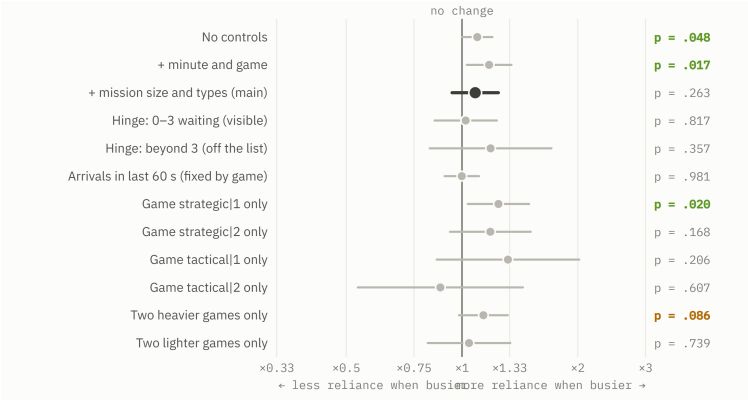
### C.3.1 Every way of asking it gives the same answer

[Pass 5: numbers only]

The within-person change in the odds of relying per extra mission waiting, under every specification tried — including each game on its own, the heavier vs the lighter pair of games, a 'hinge' that lets the effect differ beyond the visible list, tasks waiting instead of missions, and missions that had just arrived (which the operator cannot cause). Strategic: all 13 are compatible with no change. Tactical: the main model is flat, but two of twelve variants reach $p < .05$ (without mission size; Strategic Heavy game 1 alone) — a small upward lean that disappears once mission size is held fixed and is absent for the arrivals measure. (Figure 48)




**(a)** Strategic tier.




**(b)** Tactical tier.

**Figure 48: Every way of asking it gives the same answer**

**Key numbers.**

- Strategic main: OR 1.04 [0.92, 1.17], p = .53 per extra mission waiting; range across 13 specs 0.91–1.60, none $p < .05$.
- Strategic hinge: 0–3 waiting OR 1.00 [0.82, 1.22]; beyond 3 OR 1.11 [0.86, 1.44] — no change at the visible limit.
- Tactical main: OR 1.08 [0.94, 1.24], p = .26; + minute and game only: 1.18 [1.03, 1.34], p = .017; arrivals in last 60 s: 1.00 [0.90, 1.11].

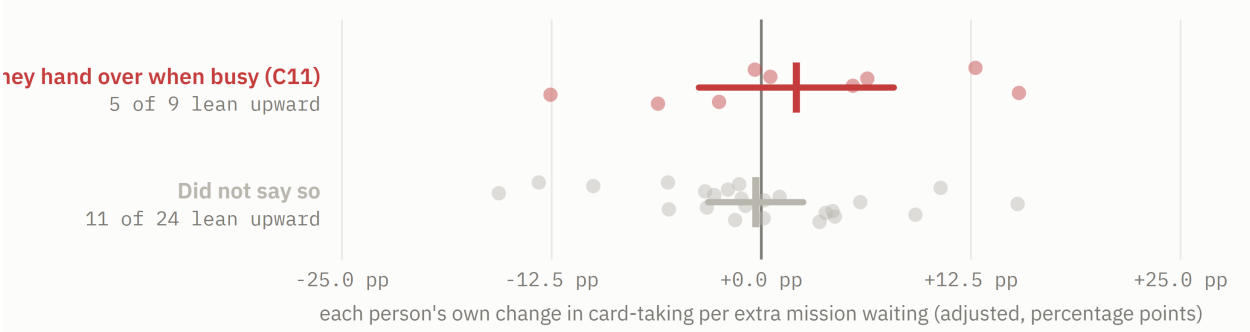
- Heavier games only (Strategic Heavy 1, Tactical Heavy 2): strategic 1.10 [0.93, 1.32], tactical 1.14 [0.98, 1.32].

**Method.** GEE logistic regression (decisions clustered within person, bias-reduced standard errors). 'Within person' means the queue is centred on that person's own average, so the estimate compares a person when busier with the same person when less busy; their average queue is kept as a separate term. 'Main' adds mission size, tasks, drone types, minute and game. An odds ratio of 1.04 means the odds of taking a card are 4% higher per extra mission waiting. To make the interval concrete: at the main model's upper limit (1.17) uptake would rise from 65% at 0 waiting to 80% at 5; at its lower limit (0.92) it would fall to 56%. So the data rule out a large effect but not a modest one either way.

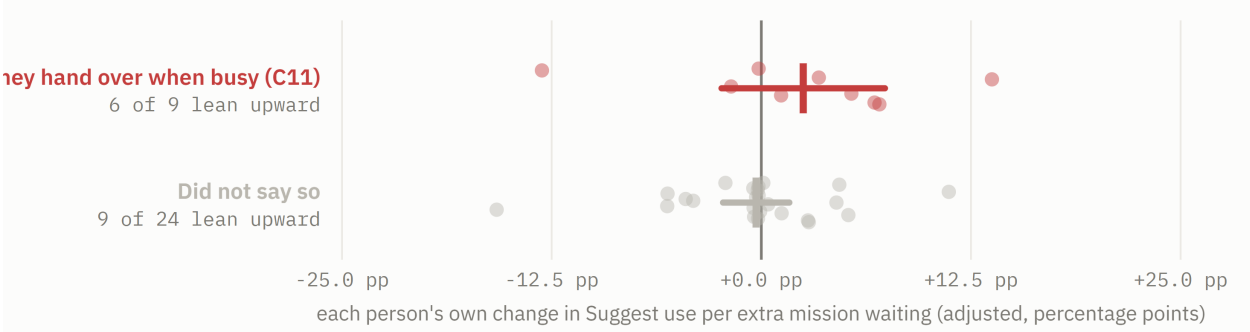
### C.3.2 Said vs did: the people who said they hand over when busy

[Pass 5: numbers only]

Nine operators (P05, P06, P11, P18, P20, P25, P32, P33, P35) said, unprompted, that they lean on the assistant more when busy, overwhelmed or rushed. Each dot is one person's own slope: how much their (adjusted) reliance changed per extra mission waiting. If what they said described what they did, the red group would sit clearly right of zero. It does not: on average they move about +2 points per extra mission, not distinguishable from zero or from the other 24 people. Their claim usually compared sessions ('the first one was so fast I used it a lot') — so the second view checks that version: reliance in their heavier game minus their lighter game. Again no difference. (Figure 49)




**(a)** Strategic tier, moment to moment.




**(b)** Tactical tier, moment to moment.

**Figure 49: Said vs did: the people who said they hand over when busy**

**Key numbers.**

- Strategic slopes: said +2.1 pp per extra mission (5 of 9 upward) vs not said −0.3 pp (11 of 24 upward); p = .41.
- Tactical slopes: said +2.5 pp (6 of 9 upward) vs −0.3 pp (9 of 24); p = .09 — the closest thing to support, and not significant.
- Queue × said interaction: strategic OR 1.13 [0.83, 1.53] p = .44; tactical 1.18 [0.91, 1.53] p = .20. Within the 'said' group alone the queue OR is 1.03 (strategic) and 0.97 (tactical).
- Heavier minus lighter game, adjusted: said +3.8 pp strategic, +1.5 pp tactical; everyone −1.1 / +1.2 pp; all intervals span zero.

**Method.** **Moment to moment:** for each person with 6+ decisions and some spread in queue depth, a straight line through their (observed − expected) values against queue depth; the slope is in percentage points per extra mission waiting. Groups compared with a Mann–Whitney test; also tested as a queue × 'said' interaction in the GEE above. **Busier session:** the four games' load measured by total drones demanded per session (Tactical Heavy 2: 115; Strategic Heavy 1: 89; Strategic Heavy 2: 88; Tactical Heavy 1: 63); for each person, their mean (observed − expected) in the heavier of their two games minus the lighter. With 9 people

in the 'said' group, only a large difference could have been detected — this is absence of evidence for the claim, and the point estimates are small.

### C.3.3 In play: overload talk comes at slightly busier moments, reliance does not follow

[Pass 5: extra: may make it in]

Using the narration recorded during play (30 people, 60 sessions), we found every moment someone said an overload-type phrase ('too many', 'can't keep up', 'oh no', 'struggling'…): 51 moments from 24 people. All nine of the 'said' group produced at least one. At those moments slightly more missions were waiting than at a random utterance (2.4 vs 2.1; p = .09) — so what people feel as overload is weakly tied to the queue. But in the minute either side, their card-taking and Suggest use were exactly what their missions and habits predict. The felt pressure is real enough; it just does not turn into handing work over. (Figure 50)

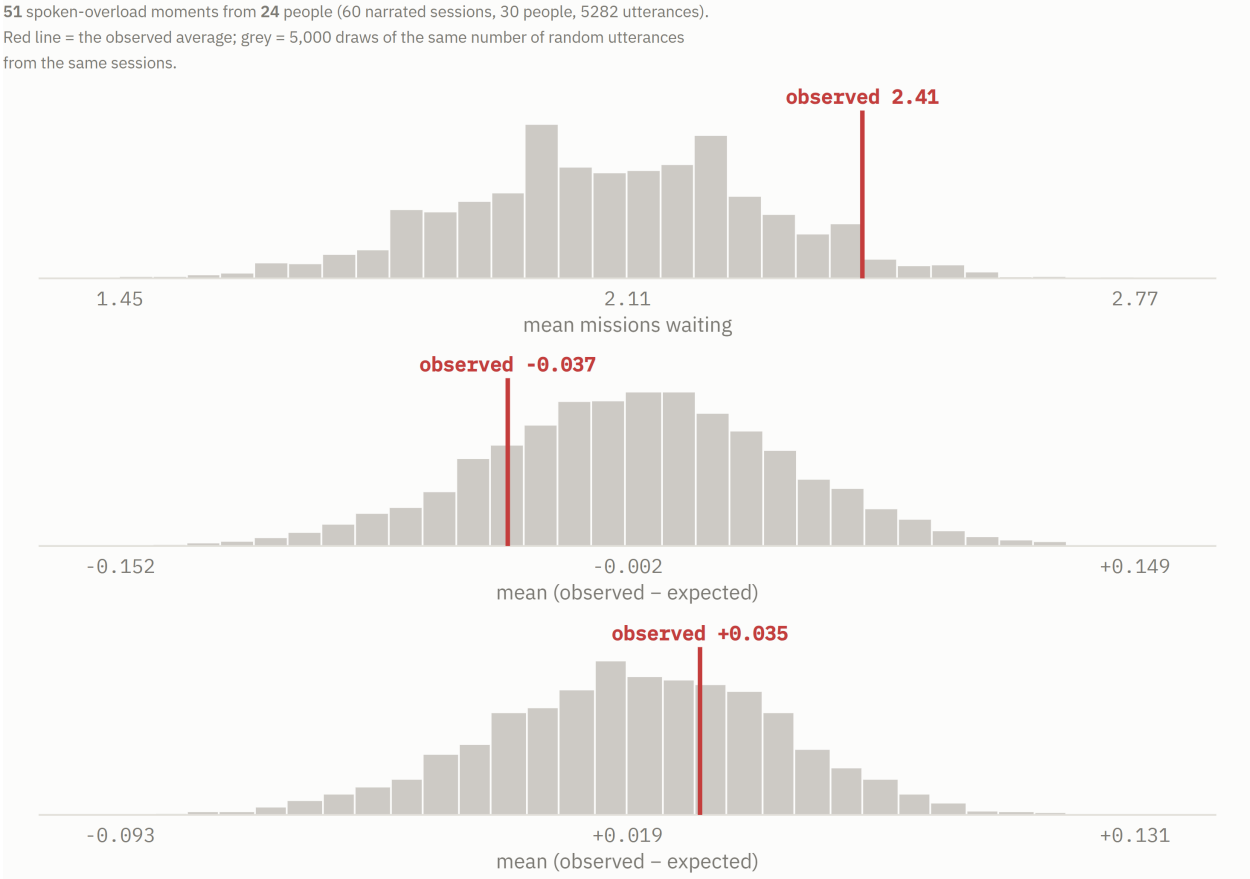



**Figure 50: In play: overload talk comes at slightly busier moments, reliance does not follow** Top: missions waiting at the overload moments; middle: strategy cards taken within ±60 s (observed − expected); bottom: Suggest pressed within ±60 s (observed − expected). Red line = observed mean; grey = 5,000 draws of the same number of random utterances from the same sessions.

**Key numbers.**

- Missions waiting at overload moments 2.41 vs 2.09 at random utterances (95% of random draws 1.71–2.45), p = .09.
- Card-taking within ±60 s: −3.7 pp vs expected (random utterances +0.1), p = .38. Suggest: +3.5 pp (random +2.1), p = .65.
- 16 of the 51 moments came from the 9 people coded C11; 35 from others.

**Method.** Phrase list fixed in round 2 (before any behaviour was looked at), applied to the transcript of each session window

(from segments.json; utterances 20 s apart merged into one moment). For each moment: missions waiting at that second (rebuilt from the log), and the mean (observed − expected) of strategic decisions and tactical plans started within ±60 s. Null: the same number of utterances per session drawn at random from ALL that session's speech, 5,000 times — so the comparison is 'overload talk' vs 'any talk', in the same people and sessions. Caveats: no speaker separation (researcher speech is in the transcripts); in round 2 about half of matched phrases were genuine stress on reading; alignment uses file times for 24 of 30 people (±2 s).

## C.4 The dip in strategy-card use belongs to the people who later said they noticed the illogical card, early in the session; it shows up whether or not they had met that card yet (supplementary analyses)

Supplement to Section 7.8.

### C.4.1 Strategic flaws are visible; tactical flaws are hidden

[Pass 5: numbers only]

Neither assistant had errors injected for anyone analysed (the $\varepsilon$ manipulation ran only for the excluded P09). Each tier still has a natural flaw. On the **strategic** side, it is the illogical extra drone type. It sits on the card face next to the mission's own requirements, everyone saw it, and 12 people mentioned it in debrief or interview (15 counting in-play remarks). On the **tactical** side, it is the wrong refusal: Suggest answers 'No workable plan' when a fix existed using a drone busy on another task. To the operator it looks exactly like a correct refusal. 14 people met one (23 wrong refusals out of 39). Four of them reacted aloud and accepted it, most saying they would have to abandon. Only P28 named the cause. Only 3 people ever described a tactical-plan flaw (C13), and just one of them had been wrongly refused. 'The strategic assistant is distrusted more' may partly be 'its mistakes are the ones you can see'. (Figure 51)

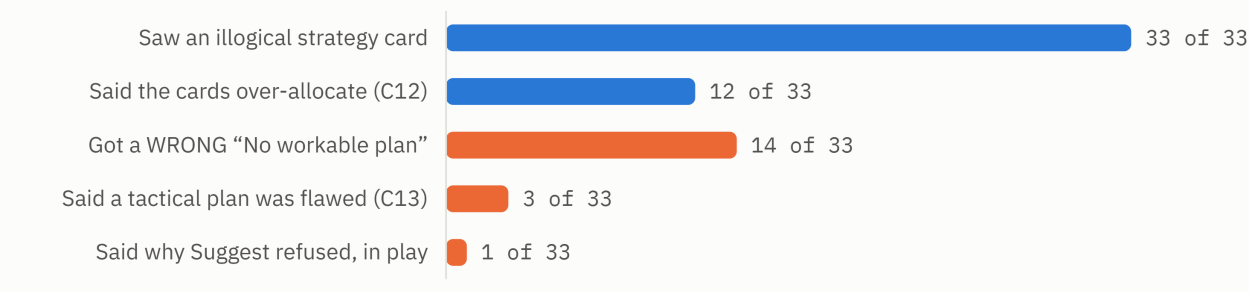



**Figure 51: Strategic flaws are visible; tactical flaws are hidden**

**Key numbers.**

- Strategic: 33 of 33 saw the illogical card (132 of 606 decisions); 12 coded C12; 15 incl. strict in-play remarks
- Tactical: 14 of 33 got $\geq$1 wrong refusal (23 of 39 refusals); 3 coded C13, 1 of them among the 14; 4 reacted aloud to a wrong refusal, 1 named why

**Method.** Strategic exposure: people with $\geq$1 illogical-card decision. Tactical exposure: people with $\geq$1 Suggest press in a recovery that returned 'No workable plan' or a partial plan while a fix using a busy drone existed (the 'blind' flag in h3_recovery). Noticing: qualitative codes C12 and C13 (debrief or interview), plus the hand-coded in-play remarks (extras.SPOKEN_STRAT / SPOKEN_TAC), each matched to the nearest preceding Suggest press.

### C.4.2 On the missions that carry the flaw, noticers stop using the cards

[Pass 5: extra: may make it in]

This is the cleanest behavioural trace, and it is where you would expect one. On their 1st illogical-card mission, noticers took a card 47% of the time and never-said people 56%. With each further illogical-card mission, noticers took a card about 14 points less often. By their 4th–6th they took one **0 times in 13** (7 people), going manual instead, mostly *before the cards had loaded* (62%). Non-noticers moved the other way: by their 4th–6th they took a card 74% of the time. Most of that was the illogical Conservative card itself (52%), which noticers took 0 times in 13. The handover analyst's check fits this: when people build a one-type mission by hand rather than take the Conservative card, nearly all of the 'trim' is dropping the unneeded-type spares (−1.2 drones overall, +0.05 on the types the mission needs). So hand-builders strip out exactly what is illogical; see Section 7.4. The manual-card-error card's version (people who mention a card error go manual more on one-type missions) holds in direction. But the same people also go manual more on three-type missions, where nothing is wrong (37% vs 22%). The one-type-*specific* part does not hold between people (interaction OR 1.12, p = .78). What is specific to the flawed missions is the *trend*: noticers learn to skip the cards there. (Figure 52)

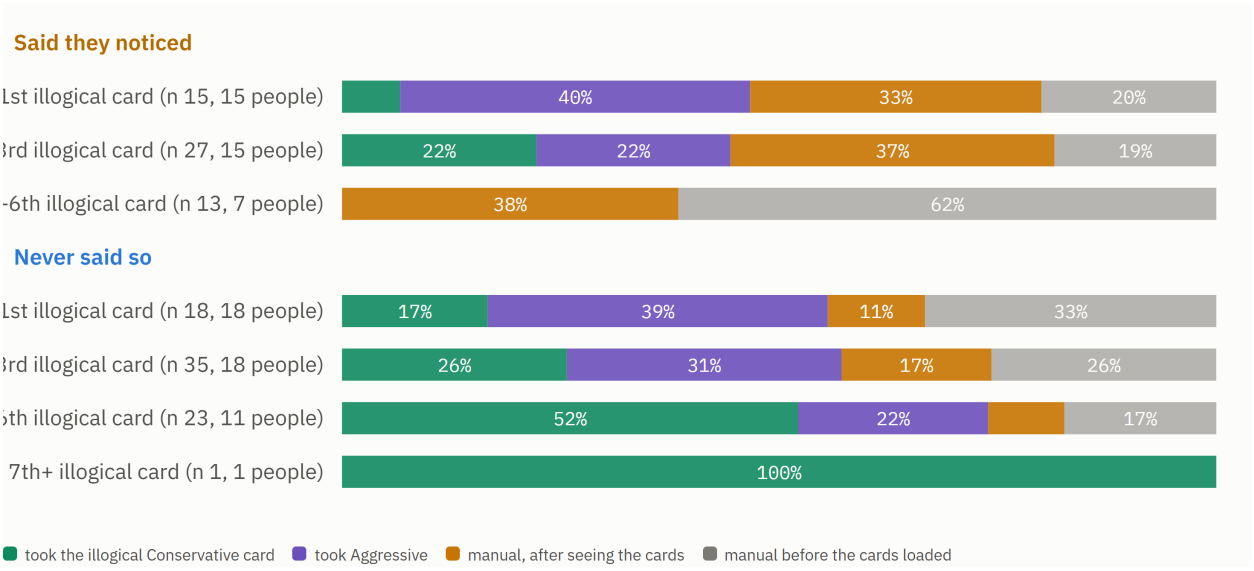



**Figure 52: On the missions that carry the flaw, noticers stop using the cards**

**Key numbers.**

- Took any card, per extra illogical card: noticers −13.9 points [−25, −3]; never-said +5.3 [−2, +12]; difference p < .001
- Took the illogical Conservative card: noticers −3.5 points per card; never-said +9.3 [+1, +18]; difference p = .001
- Manual before the cards loaded: noticers +15.4 points per card [+2, +29]; never-said −5.8; difference p = .012
- 4th–6th illogical card: noticers took a card 0 of 13 (7 people); never-said 74% of 23 (11 people), 52% the illogical card
- Manual on one-type missions: noticers 63% vs 37%, OR 2.41 [0.92, 6.32], p = .07; on three-type 37% vs 22%, OR 2.29, p = .06; one-type × noticed OR 1.12, p = .78

**Method.** Rows: the 132 illogical-card decisions, numbered per person (1st, 2nd, . . . ). Each is coded by what happened: took the Conservative card (the illogical one), took Aggressive, manual after the cards loaded, or manual before they loaded (a pre-emptive choice that never saw the cards). Trend = a linear model of each outcome on the card's number (capped at 7) × group, with game

held fixed and people clustered. The interaction p is whether the two groups' slopes differ. Between-person check (manual-card-error): manual ~ said + game on one-type missions, then on three-type, then the said × one-type interaction (logistic GEE, people clustered).

### C.4.3 What 'expected' and 'adjusted' mean

[Pass 5: numbers only]

**Expected** is the chance a model gives that this person takes a card on this decision. It knows only what the decision was (drones, tasks and drone types the mission needed), when it was (minute of the session, whether it was a re-queued mission, how many others were waiting), which of the four games was being played, and how often *this person* took cards on all their *other* decisions. **Adjusted** = the group's overall rate + the average of (observed − expected) at that position. In plain terms: 'how far above or below their usual self, given the mission in front of them, were people here?' The person term does most of the work (odds ×2.84 per unit; the model ranks takes above non-takes 81% of the time). So 'adjusted' mostly means 'compared with that person's normal habit'. Of the mission features, only the number of drone types matters (×1.51 per extra type). Because adjusted is a shifted average residual and not a probability, it can go above 100% at a small dot that sits far above its expectation. The old not-said +2 dot read 122% for this reason. (Figure 53)

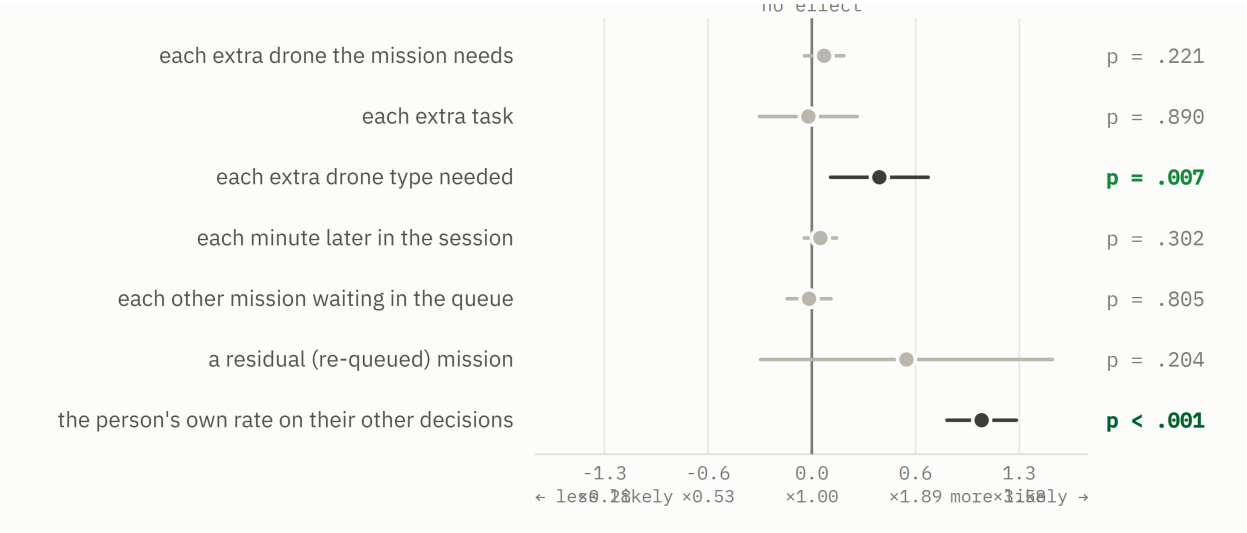



**Figure 53: What 'expected' and 'adjusted' mean**

**Key numbers.**

- Person's own rate elsewhere: odds ×2.84 [2.29, 3.51] per logit unit
- Each extra drone type needed: ×1.51 [1.12, 2.04]; drones ×1.08, tasks ×0.98, minute ×1.05, queue ×0.98 (all n.s.)
- Model ranks takes above non-takes 81% of the time (AUC 0.81)

**Method.** Logistic regression over all 606 decisions: took ~ drones + tasks + types + minute + game + queue length + residual flag + logit(person's leave-one-out rate, shrunk with +1/+2). The same specification as extras2.add_expected; the script checks the fitted values are identical. The chart shows each term's odds ratio with a 95% interval.

### C.4.4 Abandonment by what Suggest said

[Pass 5: numbers only]

Every drone failure that needed the operator is one recovery episode. Grouped by what Suggest answered: a **wrong refusal** ended in abandonment 53% of the time (9 of 17). When Suggest gave a plan it was 6% (3 of 51), and when operators fixed things by hand without Suggest it was 13% (9 of 69). A **correct refusal**, where no fix existed with the drones on the mission, ended in abandonment every time (11 of 11), as it should. The striking row is the first one. These were saveable missions, and the same 14 people abandoned only 6 of their 61 other attempted fixable failures. Abandonment was not their habit; the refusal is what is different. When nothing at all was tried, every failure was abandoned or left open (35 + 11). (Figure 54)

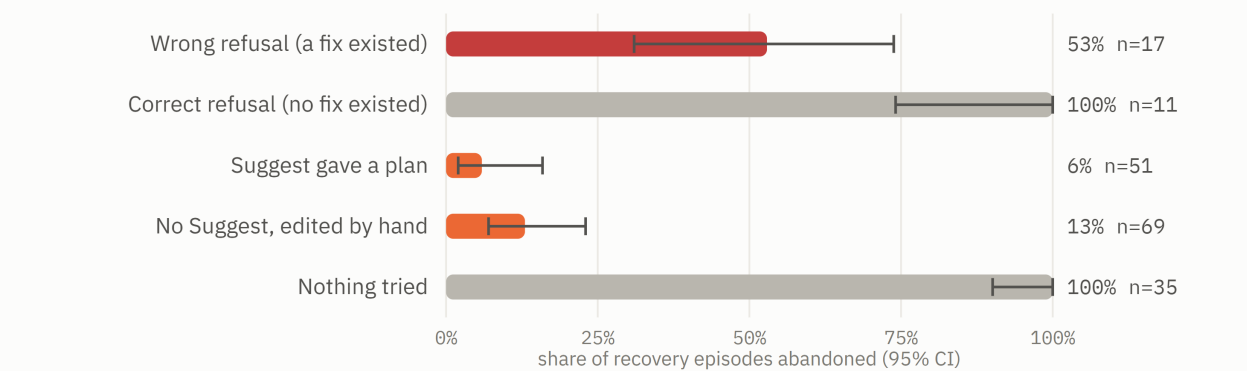



**Figure 54: Abandonment by what Suggest said**

**Key numbers.**

- Wrong refusal: 9 of 17 abandoned (53% [31%, 74%]), 14 people; all 17 fixable with drones on the mission
- Correct refusal: 11 of 11 abandoned (100% [74%, 100%]), 6 people
- Suggest gave a plan: 3 of 51 (6%); hand edits only: 9 of 69 (13%); nothing tried: 35 of 35 (+11 left open)
- Wrong refusal vs other attempted fixable failures: 9/17 vs 10/118; odds $\times$12.2, Fisher $p < .0001$; GEE $\times$12.3 [3.2, 47.6], $p = .0003$
- The same 14 people on their other attempted fixable failures: 6 of 61 abandoned

**Method.** Recovery episodes are rebuilt by h3_recovery.load_all (194 episodes, 33 people). It is a time-local rebuild of which drones were on the mission, idle or executing, at each Suggest press. A press is a **refusal** when Suggest could not cover every task the operator's live plan left short. A refusal is **wrong** when those tasks could have been covered by drones still on the mission, counting ones busy on another task (chaining them on). Suggest only offers drones that are not executing, which is its blind spot. An episode counts as a wrong refusal if it had at least one wrong-refusal press; as a correct refusal if it had refusals and none was wrong. Rates are abandoned $\div$ (abandoned + recovered), with Wilson 95% CIs. The comparison counts wrong-refusal episodes against every other fixable episode where something was attempted (Suggest gave a plan, or hand edits): Fisher exact test, plus a GEE logit clustered by person (odds ratio 12 means the odds of abandoning were twelve times higher).

### C.4.5 Do they check after a refusal? What the log shows between the refusal and the close

[Pass 5: extra: may make it in]

The timeline puts every refusal episode on one row, with 0 as the moment Suggest first said no. After a **wrong** refusal that ended in abandonment, not one operator dragged a drone in the planner. Three pressed Suggest again, got the same answer, and abandoned. None did anything on another mission in between, so they were not called away. 8 of the 9 abandoned within 10 s of the first refusal, at a median 3.8 s after the last one. That is time for a glance, not a re-plan. The 8 who **did not** take the refusal at its word look completely different: 7 of 8 started dragging drones, took a median 21 s, and saved the mission. After a correct refusal people behaved the same way as after a wrong one (abandoned in a median 4.4 s, no drags). That is right in those cases, and it shows operators get no cue to tell the two apart: the message is the same either way.

The verified case: P19, session 1, M003. Camera-2 failed. The Supply Drop needed a Camera. Two Lifters sat idle, and the only remaining Camera was busy on Search & Service, due to finish in about 45 s. Suggest said "No workable plan", and the operator abandoned 3 s later. Chaining that Camera on would have saved it (checked frame by frame on the recording). P19 later blamed themselves for failed missions (C19). No wrong-refusal abandoner is among the three who blamed the assistant (C20). (Figure 55)

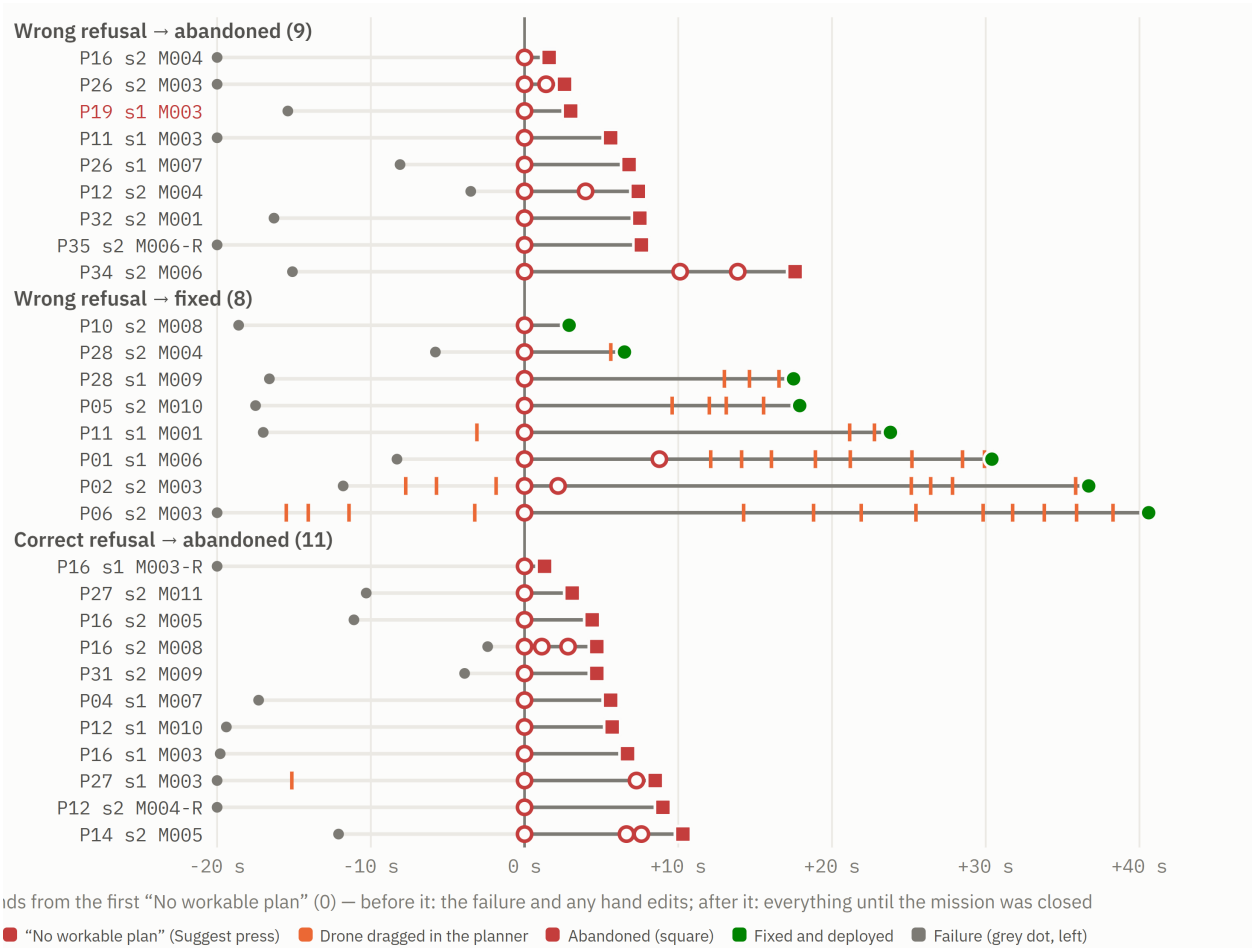



**Figure 55: Do they check after a refusal? What the log shows between the refusal and the close**

**Key numbers.**

- Wrong refusal → abandoned (9 episodes, 8 people): drags after the refusal 0 of 9; re-pressed Suggest 3 of 9; actions elsewhere 0 of 9; dragged before the refusal 0 of 9

- 8 of 9 abandoned within 10 s of the first refusal; median 6.8 s from the first refusal, 3.8 s from the last
- Wrong refusal → fixed (8 episodes, 7 people): 7 of 8 dragged drones after; median 20.9 s to deploy
- Correct refusal → abandoned (11, 6 people): 0 drags after, 3 retries, median 4.4 s from the last refusal
- Verified: P19 s1 M003, abandoned ~3 s after the refusal

**Method.** For each refusal episode the analysis takes the time of the first refusal and lists every logged operator action from then until the mission was abandoned or redeployed. That includes drags in the recovery planner (tactical_assignment_changed), further Suggest presses, and any action on another mission (strategic decisions, tactical confirms, recalls, other recoveries), from the raw event log. "Checked" here can only mean a *logged* action. Looking at the strategic map or the planner leaves no trace, so a fast abandonment could still hide a quick look, and the narration recordings are the place to check that. Seconds are session time; the session pauses if the primary window is hidden, so these are real play seconds.

## C.5 The Tactical Assistant is trusted more, and most people say so too (supplementary analyses)

Supplement to Section 7.9.

### C.5.1 Is the tactical plan given less attention in Strategic Heavy sessions?

[Pass 5: extra: may make it in]

A check of the idea that tactical trust is higher in Strategic Heavy sessions because the tactical plan gets less attention there. Raw, it does: in Strategic Heavy sessions people press Suggest less, edit less and spend less time in the planner. But Strategic Heavy plans are much smaller (5.3 vs 8.1 drones). Compared at the same plan size and session order, the planner is open about 12% less long and the checking is about 9% shorter, but neither is distinguishable from no difference, and editing is the same. The one clear difference is the other way round: on same-size plans people press Suggest *more* in Strategic Heavy (odds $\times 1.5$). People whose planner time drops more do not clearly trust the Tactical Assistant more ($\rho = -.24$, p = .19). And tactical trust itself barely differs between the scenarios (+0.15, p = .25); the gap comes from strategic trust. So the attention account is not supported, though a small effect cannot be ruled out. (Figure 56)

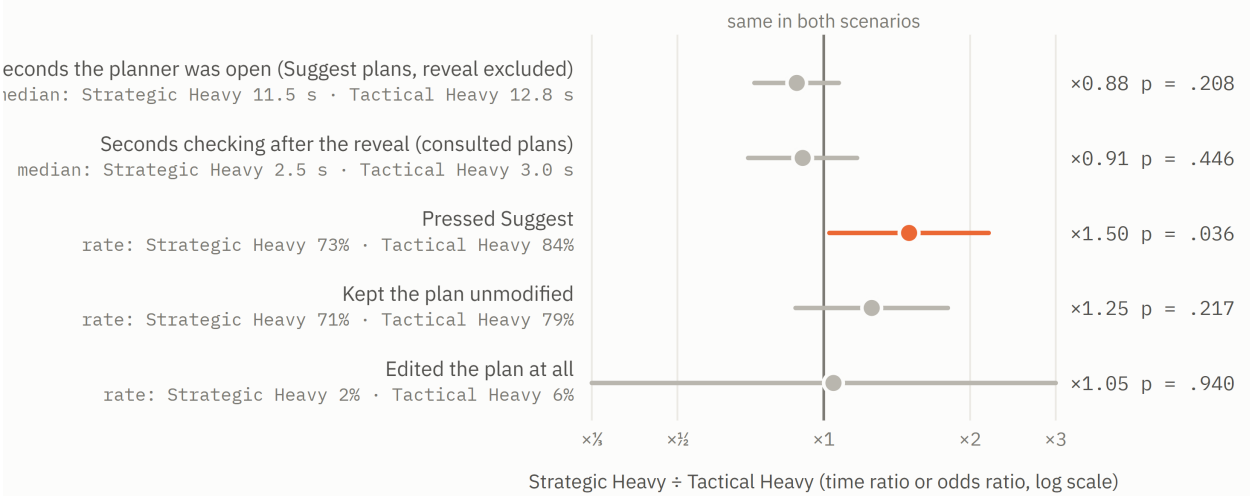



**Figure 56: Is the tactical plan given less attention in Strategic Heavy sessions?**

**Key numbers.**

- Planner time (Suggest plans): median 11.5 s Strategic Heavy vs 12.8 s Tactical Heavy; adjusted $\times 0.88$ [0.72, 1.07], p = .21
- Checking time after the reveal: median 2.5 s vs 3.0 s; adjusted $\times 0.91$ [0.70, 1.17], p = .45
- Suggest pressed: 73% vs 84% raw (OR 0.56, p = .005); same-size plans OR 1.50 [1.03, 2.18], p = .036
- Kept unmodified: 71% vs 79% raw (OR 0.67, p = .046); adjusted OR 1.26 [0.88, 1.80], p = .22
- Edited at all: 2% vs 6% raw (OR 0.44, p = .033); adjusted OR 1.05 [0.32, 3.47], p = .94
- Attention drop vs tactical-trust rise, person by person: $\rho = -.24$ [−.56, .12], p = .19, n = 31

**Method.** 630 tactical plans, 33 people (P09 excluded): 381 in Strategic Heavy sessions, 249 in Tactical Heavy. Five measures per plan. **Planner time**: from the planner opening to Deploy, minus the forced one-second-per-pair reveal, on plans where Suggest was pressed (486), so it is time spent with the assistant's plan. **Checking time**: last Suggest press to Deploy, minus the reveal (the "checking" measure of the section above). **Suggest pressed**, **kept unmodified** and **edited at all** on every plan. Each is compared

Strategic Heavy vs Tactical Heavy with a GEE model clustered by person: times on the log scale (so the result is a ratio of typical seconds), yes/no measures as odds ratios. **Raw** has scenario only; **adjusted** adds plan size (drones $\times$ tasks) and session order, so it compares like-for-like plans. The trust link: each person's average size-and-session-adjusted log planner time in each scenario, the Strategic Heavy minus Tactical Heavy difference, against their difference in tactical trust; Spearman $\rho$ with a bootstrap 95% CI (31 people with Suggest plans in both scenarios).

### C.5.2 A bad first session with the cards does not explain who loses strategic trust

[Pass 5: numbers only]

If strategic trust fell because people had bad experiences with the cards, then those who met more illogical cards in session 1 should lose more trust. They do not. The raw correlation even points the other way ($\rho = +.40$), but that comes entirely from scenario order: people who played Strategic Heavy first see the odd cards early and lose the least trust ($-0.29$ against $-1.26$ for those who played it second). Once order is held constant, the correlation is +.14 (p = .42). Round 1's pre-registered version (missions that failed after a card) was also null: $\rho = -.05$ [$-.39$, .30]. Read this as "no evidence that how much went wrong drives it", not as proof that experience doesn't matter.

**Key numbers.**

- Illogical cards seen in session 1 vs change in strategic trust: $\rho = +.40$ [+.03, +.70], p = .02 — order artefact
- Same, holding scenario order fixed: $\rho = +.14$, p = .42
- Mean change in strategic trust $-0.76$; Strategic Heavy first $-0.29$, Tactical Heavy first $-1.26$
- H5 primary (card failures in session 1): $\rho = -.05$ [$-.39$, .30]

**Method.** One row per person (33). The predictor is the count of decisions in session 1 where an illogical strategy card was on screen (from load_decisions' odd flag). The outcome is session-2 minus session-1 strategic trust. Spearman $\rho$ uses a 4000-draw seeded bootstrap CI. The partial version ranks both variables, removes the part explained by which scenario was played first, and correlates what is left. The $\rho = -.05$ is H5's primary test (docs/analysis/round1/h5_trust_drop.md).

### C.5.3 What they said agrees with what they rated

[Pass 5: numbers only]

In the debriefs and interviews, 17 operators named the Tactical Assistant as the one they preferred or trusted, 7 the Strategic one, and 9 gave a conditional, "both" or no answer. Those who said tactical had a questionnaire gap of +1.36 on average; those who said strategic were at −0.04. For 18 of the 24 who named one side, the questionnaire points the same way. The coding is from one coder (an AI model) and has not been second-coded, so treat the counts as suggestive. (Figure 57)

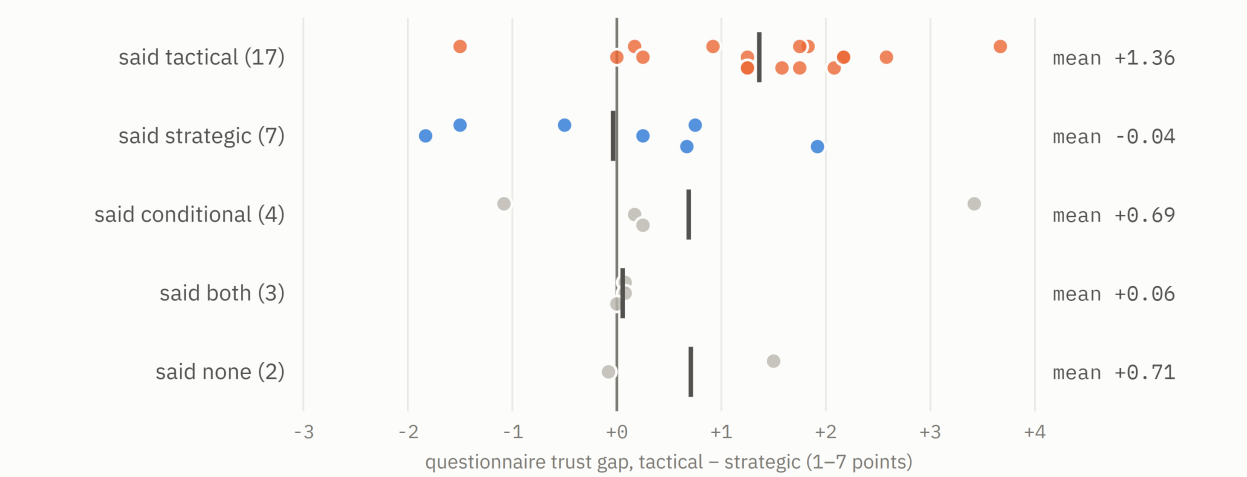



**Figure 57: What they said agrees with what they rated**

**Key numbers.**

- Said tactical 17, strategic 7, conditional 4, both 3, none 2
- Sign test 17 of 24: p = .064
- Questionnaire gap: tactical-sayers +1.36 [+0.76, +1.97] vs strategic-sayers −0.04 [−1.27, +1.19]; Mann–Whitney p = .033
- Questionnaire agrees with the stated side for 18 of 24

**Method.** Stated preference is the round-2 coding of each person's debrief and interview (docs/analysis/round2/wE_codes.csv, field pref). Its sign test asks whether 17 of 24 is more lopsided than a coin toss (exact binomial, two-sided, p = .064). The other test compares the questionnaire gap of tactical-sayers with strategic-sayers (Mann–Whitney, p = .033). Round 1's independent coding agreed poorly with it ($\kappa$ = .27), and round 2's $\kappa$ = .96 came from the same AI model. A human second coder is still needed (meth-qual-coder).

## C.6 A practitioner's reading: P36, a participant with operational military experience (supplementary analyses)

Supplement to Section 9.1.

### C.6.1 Passages not picked this pass

The passages from the same participant that were not picked for the main section, kept so nothing is lost. Each carries the links back it was first given; they were not re-read against a trimmed version.

# D Literature Suggestions

[Suggested additions keyed to the results, for the Related Work and Discussion. Not added to `refs.bib`: verify each reference before citing.]

- **Stable individual differences in reliance** (personal style, Section 7.2): Merritt & Ilgen (2008), "Not all trust is created equal: dispositional and history-based trust in human–automation interactions", *Human Factors* 50(2); Hoff & Bashir (2015), "Trust in automation: integrating empirical evidence on factors that influence trust", *Human Factors* 57(3).
- **Attitude scales not predicting behaviour**: Grassini (2023), "Development and validation of the AI Attitude Scale (AIAS-4)", *Frontiers in Psychology* 14 (the scale used); Schaefer et al. (2016), "A meta-analysis of factors influencing the development of trust in automation", *Human Factors* 58(3).
- **Task complexity and delegation** (mission size, Section 7.3): Lee & Moray (1994), "Trust, self-confidence, and operators' adaptation to automation", *International Journal of Human-Computer Studies* 40(1); Dzindolet et al. (2003), "The role of trust in automation reliance", *IJHCS* 58(6).
- **Anchoring on advice** (manual builds track the cards, Section 7.4): Tversky & Kahneman (1974), "Judgment under uncertainty: heuristics and biases", *Science* 185; Furnham & Boo (2011), "A literature review of the anchoring effect", *Journal of Socio-Economics* 40(1).
- **Workload and reliance** (the null, Section 7.6): Parasuraman & Manzey (2010) (already in the bib); Bailey & Scerbo (2007), "Automation-induced complacency for monitoring highly reliable systems", *Theoretical Issues in Ergonomics Science* 8(4).
- **Visible errors and reliance** (illogical card, Section 7.8): Dietvorst, Simmons & Massey (2015), "Algorithm aversion: people erroneously avoid algorithms after seeing them err", *Journal of Experimental Psychology: General* 144(1); Madhavan & Wiegmann (2007), "Similarities and differences between human–human and human–automation trust", *Theoretical Issues in Ergonomics Science* 8(4).
- **Automation bias and the cost of checking** (wrong refusals; tactical plans kept unchecked, Section 7.9): Bahner, Hüper & Manzey (2008), "Misuse of automated decision aids: complacency, automation bias and the impact of training experience", *IJHCS* 66(9); Goddard, Roudsari & Wyatt (2012), "Automation bias: a systematic review of frequency, effect mediators, and mitigators", *JAMIA* 19(1); Mosier & Skitka (already in the bib).
- **Measuring trust; stated vs revealed**: Kohn et al. (2021), "Measurement of trust in automation: a narrative review and reference guide", *Frontiers in Psychology* 12.
- **Methods**: Liang & Zeger (1986), GEE, *Biometrika* 73(1); Mancl & DeRouen (2001), bias-reduced sandwich SEs, *Biometrics* 57(1); Holm (1979), *Scandinavian Journal of Statistics* 6(2); McFadden (1979) on pseudo-$R^2$ (in Hensher & Stopher, eds., *Behavioural Travel Modelling*); Rousseeuw (1987), silhouettes, *Journal of Computational and Applied Mathematics* 20; Hubert & Arabie (1985), adjusted Rand index, *Journal of Classification* 2; Ericsson & Simon (1993), *Protocol Analysis* (think-aloud).